

Introduction to Information Technology — Study Notes

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About These Notes

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These notes were created for my personal study and revision. I am sharing them here because someone else might find them useful too.

Wishing you all success in your semester exams and IT learning journey. Stay disciplined, revise smartly, and keep confidence in your effort.

Duaon mein Yaad Rakhna.

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Block 1: Computer Fundamentals — Hardware, number systems, I/O devices, computer types

Block 2: Instructions, Memory and Storage — Memory hierarchy, instruction cycle, registers, secondary storage

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Block 4: Networks and Internet — LAN/MAN/WAN, topologies, WWW, connectivity, email

Block 5: Hypertext Markup Language — HTML tags, lists, tables, multimedia, linking

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(Detailed subsection numbers match SGOU SLM — see each unit heading in the notes.)

Block 1: Computer Fundamentals

Unit 1: Basic Hardware Concepts

1.1.1 Functions of a Computer

Theory

A digital computer is an electronic device that accepts user input, processes it, and generates the desired output. Like a juicer that takes fruit as input, processes it when the button is pressed, and produces juice as output, a computer transforms raw data into meaningful results through systematic processing. A computer performs four essential functions: accepting data (Input), performing operations on data as per instructions (Processing), providing the result to the user (Output), and retaining data for future use (Storage).

Important Points

- Four functions: **Input, Processing, Output, Storage**
- **Input** = data entered via keyboard, mouse, scanner, microphone, etc.
- **Processing** = operations performed on data according to program instructions
- **Output** = result displayed via monitor, printer, speakers, etc.
- **Storage** = saving processed data on memory devices (CD, USB, HDD, etc.)
- Alan Turing is considered the father of modern digital computers

For Exam

A digital computer is an electronic device that accepts input, processes it, and produces output. Its four functions are: (1) **Input** — accepting data through devices like keyboard and mouse; (2) **Processing** — performing operations on data per instructions; (3) **Output** — delivering results via monitor or printer; (4) **Storage** — saving data for future use. Computers require both hardware and software to function properly.

1.1.2 Basic Concepts

Theory

When dealing with computers, four key terminologies appear frequently: Hardware, Software, Firmware, and Liveware. Together they describe the complete computer ecosystem — physical components, programs, embedded essential programs, and the human users who operate the system. A computer without software is like a book with blank pages; firmware ensures essential startup programs run from ROM; liveware represents the human element that gives computers their purpose.

Important Points

Concept	Definition	Example
Hardware	Physical, tangible parts that can be seen and touched	Keyboard, mouse, processor, printer
Software	Set of programs bundled together; program = step-by-step instructions for a task	Windows, Linux, application programs
Firmware	Essential programs stored in separate memory; user cannot modify or delete	BIOS (in ROM) — runs POST at startup
Liveware	Human users who operate computers in daily activities	Students, professionals, operators

- Storybook analogy: cover/pages/ink = hardware; words/story = software
- POST (Power-on Self Test) checks hardware at startup using BIOS firmware

For Exam

Basic concepts of a computer system include Hardware (physical components like keyboard and processor), Software (set of programs that tell the computer what to do), Firmware (essential unmodifiable programs stored in ROM such as BIOS that runs POST at startup), and Liveware (human users). Hardware and software together make a computer useful, just as a book needs both physical pages and written content.

Diagram (refer SLM): Fig. 1.1.1 Computer System — keyboard, mouse, system unit, monitor, speaker, microphone.

1.1.3 Working of a Computer

Theory

The working of a computer can be compared to operating a television: switch on the TV, connect antenna/set-top box (input), change channel and adjust volume (processing), and view the programme on screen (output). Similarly, computers require both hardware and software to operate. The block diagram shows three functional units — Input unit, Central Processing Unit (CPU), and Output unit — with the CPU further divided into Control Unit (CU), Arithmetic and Logic Unit (ALU), and Memory Unit.

Important Points

Unit	Role
Input Unit	Receives data/instructions from outside world (keyboard, mouse, camera, mic)
Control Unit (CU)	Coordinates operations using timing signals from CPU clock; determines instruction execution order

Unit	Role
ALU	Performs arithmetic (add, subtract, multiply, divide) and logical (AND, OR, NOT) operations
Memory Unit	Stores data, instructions during processing, and results for future use
Output Unit	Delivers processed data/results to user in desired form

- TV analogy: hardware = TV components; channels = software
- CPU is the brain of the computer; all processing happens here

For Exam

A computer works through three functional units: Input unit (receives data from outside), CPU (processes data using Control Unit for coordination, ALU for calculations, and Memory Unit for temporary storage), and Output unit (delivers results to the user). Both hardware and software are required for proper functioning, similar to how a TV needs both physical components and broadcast channels.

Diagram (refer SLM): Fig. 1.1.2 Block diagram of a digital computer — Input unit, CPU (CU, ALU, Memory Unit), Output unit.

1.1.4 Von-Neumann Model

Theory

In 1945, John Von Neumann proposed a computer architecture concept known as Von-Neumann Architecture. This architecture consists of Control Unit, ALU, registers, memory unit, and input/output unit. Its key feature is the stored program concept — both data and programs reside in the same memory, enabling modern general-purpose computing where instructions and data are treated uniformly in memory.

Important Points

- Proposed by **John Von Neumann in 1945**
- Components: CU, ALU, registers, memory unit, I/O unit
- **Stored program concept:** data and programs in same memory
- Foundation of modern computer architecture
- Enables sequential instruction execution from memory

For Exam

Von-Neumann Architecture (1945, John Von Neumann) consists of Control Unit, ALU, registers, memory unit, and I/O unit. Its key feature is the stored program concept — both data and programs reside in the same memory, enabling modern general-purpose computing. This design became the foundation for virtually all modern computers.

1.1.5 Hardware Fundamentals

Theory

The major hardware components of a computer are the CPU (Processor), Motherboard, Memory, SMPS, Hard disk drive, Optical storage, Solid State Drive, and Registers. The processor is the most crucial component controlling all computer activities. Memory is classified into primary (RAM, ROM — directly accessed by CPU) and secondary (HDD, SSD, optical — permanent bulk storage). SMPS acts as the power supply unit, like the heart pumping power to all components.

Important Points

Component	Description
Processor (CPU)	Most crucial hardware; controls all activities; examples: Intel Core i7/i5, AMD Ryzen, Snapdragon
Motherboard	Printed circuit board (PCB); backbone connecting all devices
RAM	Volatile primary memory; read/write; contents lost when power off; reloaded from secondary at boot
ROM	Non-volatile primary memory; stores BIOS; types: PROM, EPROM (UV erase), EEPROM (electrical erase)
Secondary Memory	Non-volatile permanent storage — HDD, SSD, CD, DVD, USB
Registers	Small high-speed storage inside CPU; general-purpose and special-purpose
SMPS	Switched Mode Power Supply; PSU; better efficiency than linear supplies

- Memory address = unique number identifying each memory location (like house number)
- SSD uses NAND flash; faster, shock-resistant, lower power than HDD
- Processor and CPU terms are used interchangeably

For Exam

Key hardware components include the Processor (CPU — controls all computer activities), Motherboard (backbone PCB connecting all devices), Memory (Primary: RAM volatile read/write, ROM non-volatile with BIOS; Secondary: HDD, SSD, optical for permanent storage), Registers (fast temporary CPU storage), and SMPS (power supply unit). ROM types include PROM (non-erasable), EPROM (UV erasable), and EEPROM (electrically erasable).

Diagram (refer SLM): Fig. 1.1.3 Processors; Fig. 1.1.4 Motherboard; Fig. 1.1.5 RAM; Fig. 1.1.6 ROM; Fig. 1.1.7 HDD; Fig. 1.1.8 DVD Drive; Fig. 1.1.9 SMPS.

Unit Recap (from SLM)

- Computer receives data, processes it, and outputs the intended result
- Hardware = visible, touchable mechanical and electronic components
- Software = programs, OS, and data stored in memory/storage media
- Four essential components: input devices, output devices, memory, and CPU
- Input unit receives input; CPU processes (CU controls, ALU computes); memory stores during processing; output unit delivers results
- Von-Neumann architecture: ALU, Control Unit, Memory Unit, I/O — stored program concept
- Primary memory: RAM (volatile) and ROM (non-volatile); Secondary: HDD, SSD, optical
- Key hardware: Processor, Motherboard, Memory, SMPS, HDD, SSD, DVD drive, Registers

Unit 2: Number Systems and Codings

1.2.1 Number Systems

Theory

Computers understand only the language of numbers — machine language uses binary values 0 and 1. IC chips operate on two voltage states (presence = 1, absence = 0), so the base-2 binary number system is fundamental. A number system is a mathematical notation for representing numbers using symbols and digits consistently; the base or radix defines how many unique digits can occur in each position. Besides binary (base-2), programmers also use octal (base-8) and hexadecimal (base-16) to represent binary data more concisely.

Important Points

System	Base	Digits/Symbols	Key Terms
Binary	2	0, 1	Bit (1 digit), Nibble (4 bits), Byte (8 bits), MSB, LSB
Decimal	10	0–9	MSD (leftmost), LSD (rightmost); positional weights in powers of 10
Octal	8	0–7	Base 8 = 2^3 ; each octal digit = 3-bit binary group
Hexadecimal	16	0–9, A–F	Base 16 = 2^4 ; each hex digit = 4-bit binary group; compact binary representation

Decimal to Binary Conversion (Repeated Division by 2)

Decimal	Binary	Decimal	Binary
0	0	8	1000
1	1	9	1001
2	10	10	1010
3	11	15	1111
4	100	16	10000
5	101	50	110010
6	110	101	1100101
7	111	1000	111101000

Decimal ↔ Hex ↔ Binary (0–15)

Dec	Binary	Hex	Dec	Binary	Hex
0	0000	0	8	1000	8
1	0001	1	9	1001	9
2	0010	2	10	1010	A
3	0011	3	11	1011	B
4	0100	4	12	1100	C
5	0101	5	13	1101	D
6	0110	6	14	1110	E
7	0111	7	15	1111	F

- Binary advantages: simple arithmetic, easy chip design, two-level signals minimize errors
- Octal/Hex not directly understood by computers; used to shorten binary code for programmers
- Positional example: Decimal $1743 = 1 \times 10^3 + 7 \times 10^2 + 4 \times 10^1 + 3 \times 10^0 = 1000 + 700 + 40 + 3$

Worked Example 1: Decimal 25 → Binary (Division by 2)

Step	Division	Quotient	Remainder
1	$25 \div 2$	12	1
2	$12 \div 2$	6	0
3	$6 \div 2$	3	0
4	$3 \div 2$	1	1
5	$1 \div 2$	0	1

Read remainders **bottom to top** → $(25)_{10} = (11001)_2$

Verify: $1 \times 16 + 1 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1 = 16 + 8 + 1 = 25 \checkmark$

Worked Example 2: Decimal 58 → Binary

Step	Division	Quotient	Remainder
1	$58 \div 2$	29	0
2	$29 \div 2$	14	1
3	$14 \div 2$	7	0
4	$7 \div 2$	3	1
5	$3 \div 2$	1	1
6	$1 \div 2$	0	1

Read remainders bottom to top → $(58)_{10} = (111010)_2$

Verify: $32 + 16 + 8 + 0 + 2 + 0 = 58 \checkmark$

Worked Example 3: Decimal 13 → Binary (Sum of Weights Method)

Bit Position	2^3	2^2	2^1	2^0
Weight	8	4	2	1
Binary digit	1	1	0	1

$(13)_{10} = 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 1101_2$

Worked Example 4: Binary 11010110 → Hexadecimal

Group binary digits into **4-bit nibbles from the right** (pad left with 0 if needed):

```
1101 0110
  ↓    ↓
  D    6
```

$(11010110)_2 = \text{**}(D6)_{16}$

Worked Example 5: Binary 11101001 → Hexadecimal

```
1110 1001
  ↓    ↓
  E    9
```

$(11101001)_2 = \text{**}(E9)_{16}$

Worked Example 6: Binary 10110101 → Decimal

Bit	2^7	2^6	2^5	2^4	2^3
Value	1	0	1	1	0

$$= 128 + 0 + 32 + 16 + 0 + 4 + 0 + 1 = \mathbf{(181)_{10}}$$

Worked Example 7: Hex A3 → Binary → Decimal

$$A = 1010, 3 = 0011 \rightarrow (A3)_{16} = (10100011)_2$$

$$= 128 + 32 + 0 + 0 + 0 + 2 + 1 = \mathbf{(163)_{10}}$$

Worked Example 8: Decimal 357 → Octal

$$357 \div 8 = 44 \text{ R } 5; 44 \div 8 = 5 \text{ R } 4; 5 \div 8 = 0 \text{ R } 5$$

$$\text{Read remainders bottom to top} \rightarrow \mathbf{(357)_{10} = (545)_8}$$

$$\text{Verify: } 5 \times 64 + 4 \times 8 + 5 \times 1 = 320 + 32 + 5 = 357 \checkmark$$

Worked Example 9: Octal 545 → Binary

$$\text{Each octal digit maps to 3 bits: } 5=101, 4=100, 5=101 \rightarrow \mathbf{**(101100101)_2}$$

Worked Example 10: Hex 2F → Decimal

$$2 \times 16 + 15 \times 1 = 32 + 15 = \mathbf{**(47)_{10}}$$

Worked Example 11: Decimal 255 → Hex

$$255 \div 16 = 15 \text{ R } \mathbf{15 (F)}; 15 \div 16 = 0 \text{ R } \mathbf{15 (F)} \rightarrow \mathbf{**(255)_{10} = (FF)_{16}}$$

Quick Conversion Reference

From	To	Method
Decimal → Binary	—	Repeated division by 2; read remainders bottom-up
Binary → Decimal	—	Sum of (digit × 2^{position}) for each bit
Binary → Hex	—	Group into 4-bit nibbles from right; pad left with 0
Hex → Binary	—	Each hex digit → 4-bit binary (e.g., F → 1111)
Decimal → Octal	—	Repeated division by 8; read remainders bottom-up
Octal → Binary	—	Each octal digit → 3-bit binary

For Exam

Number systems used in computers include Binary (base-2, digits 0 and 1; bit, nibble, byte), Decimal (base-10), Octal (base-8, each digit = 3 bits), and Hexadecimal (base-16, digits 0–9 and A–F, each digit = 4 bits). Binary is the only language directly understood by computers because IC chips operate on two voltage states. Octal and hexadecimal provide compact representations of binary data for programmers.

Diagram (refer SLM): Fig. 1.2.1(a)(b) Binary system; Fig. 1.2.2(a)(b) Decimal; Fig. 1.2.3(a)(b) Octal; Fig. 1.2.4(a)(b) Hexadecimal; Fig. 1.2.5 Number System and Range.

1.2.2 Digital Codes

Theory

When key 'A' is pressed, it is mapped to decimal code 65, then converted to binary for the computer. Digital coding uses binary digits to represent letters, characters, and symbols. Common codes include BCD (each decimal digit in 4 bits), **Gray code (consecutive values differ by one bit)**, Excess-3 (add 3 to each decimal digit, express in 4-bit binary), and ASCII (7-bit character encoding). These codes bridge human-readable data and machine-level binary representation.

Important Points

Code	Description	Example
BCD	Binary-Coded Decimal; each decimal digit → 4 bits	$(357)_{10} = 0011\ 0101\ 0111$; $(123)_{10} = 0001\ 0010\ 0011$
Gray Code	Consecutive numbers differ by only 1 bit ; used in encoders, error detection	0 → 000, 1 → 001, 2 → 011, 3 → 010, 4 → 110, 5 → 111, 6 → 101, 7 → 100
Excess-3	Add 3 to each decimal digit, express in 4-bit binary; self-complementary	$(23)_{10}$: $2+3=5 \rightarrow 0101$, $3+3=6 \rightarrow 0110 \rightarrow \mathbf{0101\ 0110}$
ASCII	American Standard Code for Information Interchange ; 7-bit; 128 standard codes	A=65, a=97; Standard ASCII (English) + Extended ASCII (special chars)
EBCDIC	8-bit alphanumeric code by IBM; 256 symbols; used in mainframes	—

Digital Codes Comparison Table

Feature	BCD	Gray Code	Excess-3	ASCII
Full form	Binary-Coded Decimal	Gray Code (Frank Gray)	Excess-3 Code	American Standard Code for Information Interchange
Type	Weighted code	Non-weighted code	Unweighted, self-complementary	7-bit character encoding
Rule	Each decimal digit → 4 binary bits separately	Consecutive values differ by exactly 1 bit	Add 3 to each decimal digit, then 4-bit binary	Numeric code represents each character
Bit length	4 bits per decimal digit	Variable (n bits for n values)	4 bits per decimal digit	7 bits (128 standard codes)
Use	Calculators, digital clocks, decimal arithmetic	Optical encoders, error detection in communication	Simplified BCD arithmetic operations	Letters, numerals, punctuation in English

Feature	BCD	Gray Code	Excess-3	ASCII
Example (digit 5)	0101	0111 (decimal 5 in 3-bit Gray)	1000 (5+3=8 → 1000)	'5' = 53 (decimal)
Example (letter A)	Not used for letters	Not used for letters	Not used for letters	A = 65, a = 97
Limitation	Wastes bits (1010–1111 unused per digit)	Not suitable for arithmetic directly	Extra step of adding 3	Only 128 standard characters

Excess-3 Table (0–7)

Decimal	Binary	Excess-3	Decimal	Binary	Exces
0	0000	0011	4	0100	0111
1	0001	0100	5	0101	1000
2	0010	0101	6	0110	1001
3	0011	0110	7	0111	1010

Gray Code Table (0–7)

Decimal	Binary	Gray Code	Decimal	Binary	Gray (
0	000	000	4	100	110
1	001	001	5	101	111
2	010	011	6	110	101
3	011	010	7	111	100

ASCII Quick Reference

Character	ASCII (Dec)	Binary (7-bit)	Character	ASCII (Dec)	Binary
0	48	0110000	A	65	1000001
1	49	0110001	B	66	1000010
9	57	0111001	Z	90	1011010
Space	32	0100000	a	97	1100001

Worked Example: BCD encoding of 357

$(357)_{10} \rightarrow 3=0011, 5=0101, 7=0111 \rightarrow \mathbf{0011\ 0101\ 0111}$

Worked Example: Excess-3 encoding of 23

$2+3=5 \rightarrow 0101; 3+3=6 \rightarrow 0110 \rightarrow \mathbf{0101\ 0110}$

Worked Example: ASCII for "Hi"

H = 72 = 1001000; i = 105 = 1101001

For Exam

Digital codes represent characters and numbers in binary form. BCD encodes each decimal digit separately in 4 bits (e.g., 357 = 0011 0101 0111). Gray code ensures consecutive values differ by one bit, useful in encoders. Excess-3 adds 3 to each decimal digit before 4-bit binary encoding (23 → 0101 0110). ASCII uses 7-bit codes for characters (A=65, a=97). EBCDIC is IBM's 8-bit code with 256 symbols.

Unit Recap (from SLM)

- Machine language (binary) is the only language directly understood by computers
- Number system = set of values to represent quantity; base/radix = number of unique digits
- Binary: base-2; bit, nibble (4 bits), byte (8 bits); MSB, LSB
- Decimal: base-10; Octal: base-8; Hexadecimal: base-16
- BCD: each decimal digit in 4 bits; Gray code: consecutive numbers differ by one bit
- Excess-3: add 3 to each digit, express in 4-bit binary
- ASCII: 7-bit character encoding; A=65, a=97

Unit 3: Input/Output Devices

1.3.1 Input Devices

Theory

Input/output devices allow the computer to interact with the outside world. Input devices accept data and instructions from the user and convert them into a form the computer can process. Without I/O devices, a computer would be a “dumb terminal” with no external communication. Common input devices include keyboard, mouse, touchpad, scanner, and barcode reader — each designed for specific data entry needs.

Important Points

Device	Description
Keyboard	Most common input device; QWERTY layout; 101-key standard; primary device for data/instructions
Mouse	Pointing device; controls cursor; point, drag, drop; types: Mechanical, Optical, Wireless
Touchpad	Pointing device on laptops; touch-sensitive pad moves cursor
Graphic Tablet	Input for drawing images/graphics; like pencil on paper

Device	Description
Gamepad	Controller for computer/console gaming systems
Scanner	Reads text/graphics from paper; types: Flatbed, Handheld
Barcode Reader	Reads product barcodes; used in supermarkets, showrooms

- I/O devices are also called **peripheral devices** (surround CPU and memory)
- ATM example inputs: smart card, keyboard, touch screen
- OCR (Optical Character Recognition) reads printed/handwritten text
- MICR (Magnetic Ink Character Recognition) reads bank cheque codes
- Light pen: selects and displays objects on screen (used with graphics)

Input Device Selection Guide

Need	Best Device
Typing text/data	Keyboard
Pointing/clicking on screen	Mouse or Touchpad
Drawing/sketching	Graphic Tablet
Scanning printed document	Scanner (Flatbed/Handheld)
Reading product codes	Barcode Reader
Gaming controls	Gamepad
Touch-based input (ATM, kiosk)	Touch Screen

For Exam

Input devices accept data and instructions from the user. The keyboard is the most common standard input device with QWERTY layout. Mouse and touchpad are pointing devices for cursor control. Scanner converts printed text/graphics to digital form. Barcode reader reads product barcodes in retail. Other devices include graphic tablet, gamepad, light pen, and OCR/MICR devices.

Diagram (refer SLM): Fig. 1.3.1 Keyboard; Fig. 1.3.2 Mouse; Fig. 1.3.3 Touchpad; Fig. 1.3.4 Graphic Tablet; Fig. 1.3.5 Gamepad; Fig. 1.3.6 Scanner; Fig. 1.3.7 Barcode Reader.

1.3.2 Output Devices

Theory

Output devices communicate processed information back to the user. They are broadly classified into soft copy devices (monitor, speakers — visible/audible only while device is on, editable) and hard copy devices (printers, plotters — permanent physical output). Output types include text, graphics, audio, video, and tactile.

Printers are categorized as impact (touches paper: daisy wheel, dot matrix) and non-impact (no contact: laser, inkjet).

Important Points

Device	Type	Description
Monitor	Soft copy	Visual display; CRT replaced by LCD, LED, OLED, Plasma; TFT-LCD/LED flat panels
Multimedia Speaker	Soft copy (audio)	Sound output for movies, games, music
Printer	Hard copy	Text/graphics on paper; Impact: Daisy wheel (letter quality, noisy), Dot matrix (cheap, low quality); Non-impact: Laser (fast, high quality), Inkjet
Plotter	Hard copy	Large-format graphs, engineering drawings, maps; types: Drum, Flatbed

- Soft copy: available only when device is on; can be edited and saved
- Hard copy: permanent physical objects; cannot be edited after printing
- Dot matrix: serial and line dot-matrix types; laser printer uses rotating mirror
- Output types: Text, Graphics, Audio, Video, Tactile

For Exam

Output devices deliver processed results to users. Monitors provide soft copy visual output (LCD/LED/OLED). Speakers provide audio soft copy. Printers produce hard copy — impact printers (daisy wheel, dot matrix) touch paper; non-impact printers (laser, inkjet) do not. Plotters produce large-format engineering drawings and graphs. Soft copy is temporary and editable; hard copy is permanent.

Diagram (refer SLM): Fig. 1.3.8 Monitor; Fig. 1.3.9 Laser Printer; Fig. 1.3.10 Plotter.

Unit Recap (from SLM)

- Input devices give data to the computer; output devices return processed information
- Keyboard = most common input device (QWERTY layout)
- Mouse and touchpad = pointing devices (point, drag, drop)
- Scanner reads paper documents; barcode reader reads product barcodes
- Monitor = soft copy visual output; printer/plotter = hard copy output
- Speakers = audio soft copy output
- Printers: Impact (daisy wheel, dot matrix) vs Non-impact (laser, inkjet)

- Plotter = large-format graphs and engineering drawings

Unit 4: Different Types of Computing Systems

1.4.1 Evolution of Computers

Theory

In ancient times, people used fingers, pebbles, and simple calculators for computations. Key milestones include the Abacus (counting device), Napier's Bones (John Napier, multiplication aid), Pascaline (Blaise Pascal, 1642 — first mechanical adding machine), punched cards (Herman Hollerith, ~1880), and Charles Babbage's Difference Engine (1822) and Analytical Engine (1842 — father of computers). The digital era began with Claude Shannon's binary system; ABC (Atanasoff-Berry Computer, 1942) was the first electronic computer using vacuum tubes. Transistors (1947), FORTRAN (1957), IC (1959), mainframe with IC (1960), Intel 1KB memory chip (1970), and first microcomputer (1975, H. Edward Roberts) followed.

Important Points

- **Abacus** → **Napier's Bones** → **Pascaline** (1642) → **Punched cards** (1880–1970s)
- **Charles Babbage** = father of computers; Difference Engine + Analytical Engine
- **Claude Shannon** suggested binary system in digital era
- **ABC** (Atanasoff-Berry, 1942) = first electronic computer; vacuum tubes
- **1947**: Transistors; **1957**: FORTRAN; **1959**: IC; **1960**: Mainframe with IC
- **1970**: Intel 1KB memory chip; **1975**: First microcomputer (H. Edward Roberts)

Computer Generations Summary

Generation	Period	Technology	Key Development
Pre-electronic	Ancient–1800s	Manual/mechanical	Abacus, Napier's Bones, Pascaline
1st	1940s–50s	Vacuum tubes	ABC computer, ENIAC era
2nd	1950s–60s	Transistors (1947)	Faster, smaller; FORTRAN (1957)
3rd	1960s–70s	Integrated Circuits (1959)	Mainframe with IC (1960); Intel 1KB chip (1970)
4th	1970s–present	Microprocessors	First microcomputer (1975); personal computing era
5th (modern)	1980s–present	VLSI, AI, IoT	Smartphones, wearables, cloud computing

For Exam

Computer evolution progressed from Abacus and Napier's Bones to Pascaline (1642,

first mechanical adder). Charles Babbage is the father of computers (Difference Engine, Analytical Engine). The digital era introduced binary system (Claude Shannon) and ABC computer (1942). Transistors (1947), ICs (1959), and microcomputers (1975) led to modern computing.

Diagram (refer SLM): Fig. 1.4.2 Abacus; Fig. 1.4.3 Napier's Bones; Fig. 1.4.4 Pascaline; Fig. 1.4.5 Punched Cards.

1.4.2 Characteristics and Classification by Operating Principle

Theory

Computers possess key characteristics: Automatic (works without human intervention once started), Speed (completes in seconds what humans take years), Accuracy (consistent precision), Diligence (no fatigue, same accuracy for millions of calculations), Versatility (payroll, inventory, billing — many tasks), Power of Remembering (stores any amount of data), No IQ (follows instructions only), No Feeling (no emotions), and Storage (built-in + secondary). By operating principle, computers are classified as Analog (continuous physical quantities), Digital (digits/binary), or Hybrid (both — e.g., hospital ICU, heart beat measurement).

Important Points

- **Characteristics:** Automatic, Speed, Accuracy, Diligence, Versatility, Power of Remembering, No IQ, No Feeling, Storage
- Speed measured in ms → μ s → **nanoseconds** today; supercomputers in **MIPS** (Millions of Instructions Per Second)
- **Analog:** Continuous information; waves/physical quantities; e.g., thermometer, speedometer, petrol pump indicator
- **Digital:** Calculations with digits (binary); high accuracy; used in banking, education, insurance
- **Hybrid:** Speed of analog + accuracy of digital; e.g., hospital ICU, pacemakers, insulin meters

For Exam

Computer characteristics include automatic operation, high speed, accuracy, diligence (no fatigue), versatility, large storage capacity, no IQ (needs instructions), and no feelings. By operating principle: Analog computers use continuous physical phenomena (thermometer, speedometer); Digital computers use binary digits with high precision; Hybrid computers combine both qualities (hospital patient monitoring).

1.4.3 Types of Computers

Theory

Computers are classified by size, power, and usage into seven categories. From smallest processing power (wearables, handheld) to largest (supercomputers), each type serves specific needs — portability for laptops, computing power for workstations, multi-user support for mainframes, and massive parallel processing for supercomputers used in weather forecasting and scientific research.

Important Points

Type	Description	Use
Notebook/Laptop	Battery/AC powered; fits briefcase; keyboard, touchpad, screen	Travel, libraries, meetings, aircraft
Personal Computer (PC)	General-purpose, non-portable; fits office desk; single user	Home, office workspace
Workstation	Powerful desktop; more computing power, storage, graphics than PC	Engineers, architects, professionals
Mainframe	Large, expensive; hundreds to thousands of simultaneous users	Organizations, banks; below supercomputer in hierarchy
Supercomputer	Fastest; huge storage and computational power; MIPS	Weather analysis, scientific/numerical problems; e.g., IBM Deep Blue
Handheld (PDA)	Small, held in hand; Tablet PC, PDA, Smartphone	Gaming, presentations, word processing
Wearable	Very small; worn on body; low processing power	Medicine: pacemakers, insulin meters

For Exam

Computers by size and use include Notebook/Laptop (portable, battery-powered), Personal Computer (desktop, single user), Workstation (powerful desktop for professionals), Mainframe (supports hundreds/thousands of users), Supercomputer (fastest, MIPS, weather forecasting), Handheld/PDA (smartphone, tablet), and Wearable (pacemakers, insulin meters). IBM Deep Blue is an example of a supercomputer.

Diagram (refer SLM): Fig. 1.4.1 Different types of computers; Fig. 1.4.6 Notebook; Fig. 1.4.7 PC; Fig. 1.4.8 Workstation; Fig. 1.4.9 Mainframe; Fig. 1.4.10 Supercomputer; Fig. 1.4.11 Handheld; Fig. 1.4.12 Wearable.

Unit Recap (from SLM)

- Early devices (Abacus, Napier's Bones, Pascaline) led to modern computers

- Charles Babbage = father of computers; Claude Shannon = binary system
- Characteristics: Automatic, Speed, Accuracy, Diligence, Versatility, Storage, No IQ, No Feeling
- By operating principle: Analog (continuous), Digital (digits/binary), Hybrid (both)
- By size/use: Notebook, PC, Workstation, Mainframe, Supercomputer, Handheld, Wearable
- MIPS = Millions of Instructions Per Second (supercomputers)

Block 2: Instructions, Memory and Storage

Unit 1: Memory Representations and Hierarchy

2.1.1 Computer Memory

Theory

Computer memory is the storage space where data and instructions are stored — temporarily or permanently. Memory is physically organized as cells (each storing 1 bit) made of semiconducting materials, with unique addresses for access. Memory is classified as volatile (contents erased when power off — RAM) or non-volatile (retains contents — ROM, secondary). The memory hierarchy — Registers → Cache → Main Memory → Secondary → Offline (tape) — balances speed, size, and cost, with faster/smaller/expensive storage at the top.

Important Points

Topic	Details
Structure	Cells store 1 bit each; unique address per cell; bits grouped into memory words
Units	KB = 2^{10} = 1024 bytes; MB = 1024 KB; GB = 1024 MB; TB = 1024 GB; PB = 1024 TB
Operations	Read (fetch by address) and Write (store by address); Access time = time to read/write
Primary Memory	Directly accessed by CPU; RAM (volatile, read/write) + ROM (non-volatile); also called main/working memory
SRAM vs DRAM	SRAM: transistors only, no refresh, faster, expensive, used in cache; DRAM: transistor+capacitor, needs refresh, cheaper, used in main memory
Secondary Memory	Not directly accessed by CPU (needs interface); non-volatile, large, cheap; HDD, CD, DVD
Cache Memory	High-speed semiconductor between CPU and main memory; L1 (inside processor), L2, L3 (outside); hit/miss/penalty

Topic	Details
Memory Hierarchy	Registers → Cache → Main → Secondary → Offline; top=volatile/fast/small/expensive; bottom=non-volatile/slow/large/cheap

SRAM vs DRAM

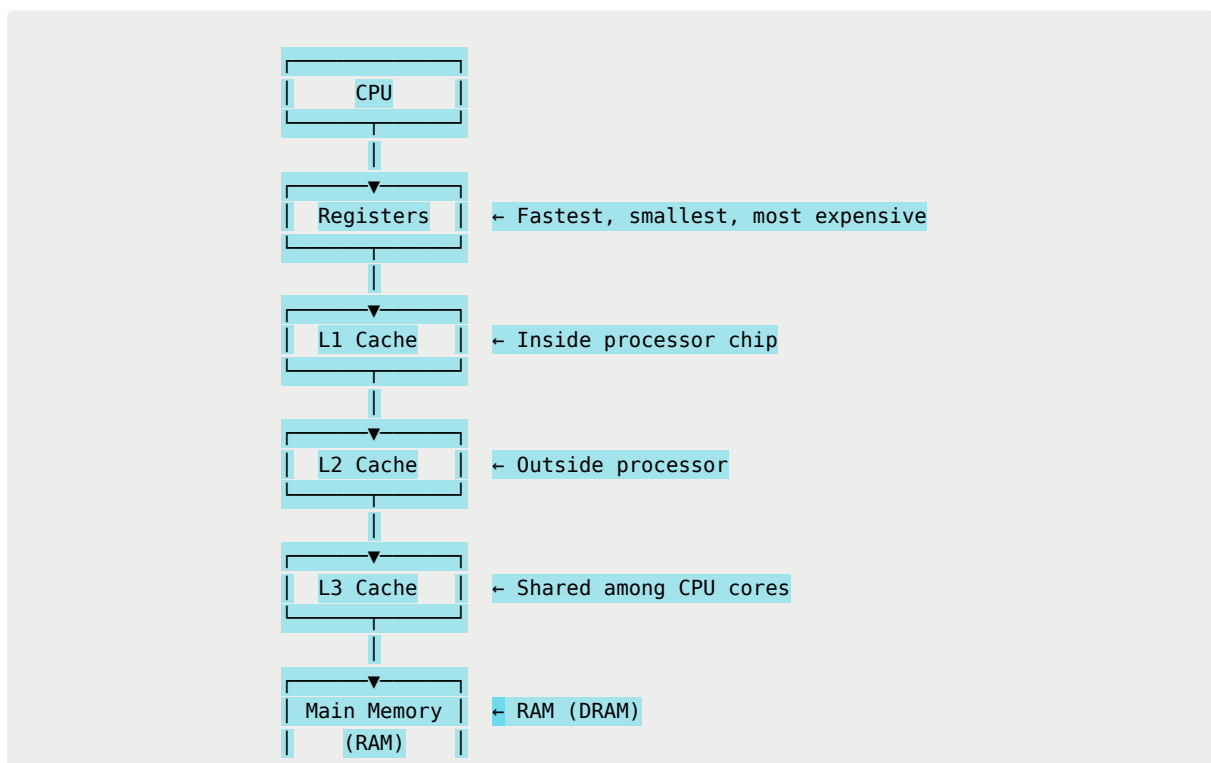
Feature	SRAM	DRAM
Components	Transistors only	Transistors + capacitors
Refresh	Not required	Required periodically
Speed	Faster	Slower
Cost	Expensive	Cheaper
Density	Low	High
Use	Cache memory	Main memory

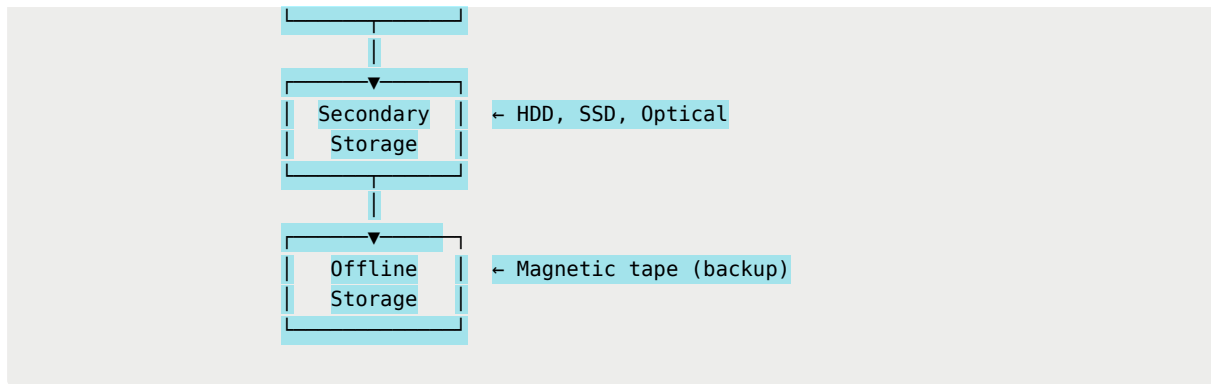
Cache Concepts

- **Cache hit:** data found in cache; **Cache miss:** data not in cache
- **Miss penalty:** extra time to fetch data from main memory into cache
- **Hit ratio** = hits / (hits + misses) = no. of hits / total accesses

Memory Hierarchy — Diagram Description (refer SLM Fig. 2.1.5)

The memory hierarchy is a pyramid structure with the CPU at the top. Data flows between levels — when the CPU needs data, it searches from the fastest level downward until found, then copies data upward into faster levels for future access.





Memory Hierarchy — Access Time and Characteristics

Level	Location	Volatile?	Relative Speed	Typical Size	Cost/
Registers	Inside CPU	Yes	Fastest	Bytes (tens)	Highe
L1 Cache	Inside processor chip	Yes	Very fast	32–64 KB	Very
L2 Cache	On/near CPU die	Yes	Fast	256 KB–1 MB	High
L3 Cache	Outside CPU cores	Yes	Fast	4–32 MB	High
Main Memory (RAM)	Motherboard	Yes	Moderate	4–64 GB	Mode
SSD	Internal/external	No	Faster than HDD	256 GB–4 TB	Medi
HDD	Internal/external	No	Slow	500 GB–16 TB	Low
Optical (CD/DVD/BD)	Removable drive	No	Slower	MB–GB	Very
Magnetic Tape	External backup	No	Slowest	TB+	Low

Hierarchy Rules (from SLM)

- **Top → Bottom:** speed decreases, storage size increases, cost per bit decreases
- **Bottom → Top:** access time decreases, cost per bit increases
- **Top three levels** (Registers, Cache, Main Memory) are **volatile**
- **Bottom two levels** (Secondary, Offline) are **non-volatile**
- College library analogy: study table = cache; department library = primary; central library = secondary

For Exam

Computer memory stores data and instructions in bit-level cells with unique

addresses. Units: KB through PB (each $\times 1024$). Primary memory (RAM volatile, ROM non-volatile) is directly accessed by CPU; secondary memory provides permanent bulk storage. Cache (L1/L2/L3) stores frequently used data between CPU and main memory. Memory hierarchy from Registers (fastest) to Offline tape (slowest) optimizes performance, capacity, and cost.

Diagram (refer SLM): Fig. 2.1.1 Different memories; Fig. 2.1.2 Structure of memory; Fig. 2.1.3 Memory cells analogy; Fig. 2.1.4 Cache Memory; Fig. 2.1.5 Memory Hierarchy.

Unit Recap (from SLM)

- Memory cells store 1 bit; addressed by unique numbers
- Read and Write are main operations; access time = read/write duration
- Units: KB, MB, GB, TB, PB (each $\times 1024$)
- Primary (RAM/ROM), Cache, Secondary — three memory types
- SRAM (cache) vs DRAM (main memory)
- Cache hit/miss/penalty; L1/L2/L3 cache levels
- Memory hierarchy: Registers $\rightarrow$ Cache $\rightarrow$ Main $\rightarrow$ Secondary $\rightarrow$ Offline

Unit 2: Instruction Set and Instruction Cycle

2.2.1 Instructions and Instruction Set

Theory

Instructions are the words of a computer's language — one step in a program for performing a specific task. A set of instructions forms a program; programs bundled together form software. The instruction set is the complete vocabulary of instructions a microprocessor supports (e.g., 8086 has ~117 basic instructions). Instruction Set Architecture (ISA) is the interface between hardware and software — the only way to interact with hardware. General instruction format has two fields: Opcode (operation to perform) and Address (operands).

Important Points

Sample Instructions (8086)	Operation
ADD	Adding two numbers
COMPARE	Compare two numbers
IN	Input from device (keyboard)
LOAD	Load from RAM to register
OUT	Output to device (monitor)

Sample Instructions (8086)	Operation
STORE	Store from register to RAM
JUMP	Jump to designated address

- Instruction format: **Opcode field + Address field** (e.g., ADD A, B — ADD=operator, A,B=operands)
- Address types: One-address, Two-address, Three-address instructions
- **ISA** = only way to interact with hardware

For Exam

Instructions are single steps in a program; instruction set is the complete group of instructions a microprocessor supports. ISA is the hardware-software interface. Instruction format contains Opcode field (operation) and Address field (operands). Example: ADD A, B adds operands A and B. Programs form software when bundled together.

2.2.2 Instruction Cycle

Theory

A program in memory has a sequence of instructions executed through a cycle for each instruction. The instruction cycle has four phases: (1) **Instruction Fetch** — processor takes instruction from memory; (2) **Instruction Decode** — identifies operation and operands; (3) **Operand Fetch** — retrieves operand values from memory addresses; (4) **Execution** — performs the specified operation. For ADD A, B: fetch instruction, decode as addition with operands A and B, fetch values of A and B, execute A+B, store result.

Instruction Cycle — Step-by-Step Explanation

Phase	Step	What Happens	Registers Involved
1. Instruction Fetch	1a	CPU reads memory address stored in PC (Program Counter)	PC holds address of next instruction
	1b	Instruction at that address is copied into IR (Instruction Register)	IR ← instruction from memory
	1c	PC is incremented to point to the following instruction	PC = PC + 1
2. Instruction Decode	2a	Control Unit (CU) interprets the opcode field in IR	CU reads IR
	2b	CU identifies the operation to perform (e.g., ADD, LOAD, JUMP)	—

Phase	Step	What Happens	Registers Involved
	2c	CU identifies operand locations (registers or memory addresses)	—
3. Operand Fetch	3a	CPU retrieves operand values from memory using addresses in instruction	MAR ← memory address
	3b	Data at that address is read into MDR (Memory Data Register)	MDR ← data from memory
	3c	Operands loaded into appropriate registers for ALU use	GPR, Accumulator
4. Execution	4a	ALU performs the operation specified by decoded instruction	Accumulator
	4b	Result stored in destination (accumulator, register, or memory)	Result → Accumulator or MDR
	4c	Condition flags updated if applicable (zero flag, carry flag)	Condition Code Register

Complete Worked Example: ADD A, B

Step	Phase	Action
1	Load program	Program containing ADD A, B is loaded into main memory
2	Instruction Fetch	PC points to ADD A,B; instruction fetched from memory into IR
3	Instruction Decode	CU identifies: Operation = ADD , Operands = A and B
4	Operand Fetch	Values of A and B fetched from memory addresses into registers
5	Execution	ALU computes A + B; result stored in Accumulator
6	Store	Result written back to memory location A (destination operand)
7	Repeat	PC points to next instruction; cycle repeats for entire program

Important Points

- **Four phases:** Fetch → Decode → Operand Fetch → Execute
- Program must be loaded into main memory before execution begins
- PC (Program Counter) always holds address of the **next** instruction to fetch

- IR holds the instruction **currently being** decoded/executed
- Each instruction in the program goes through the complete cycle sequentially
- When not running, programs are stored in **secondary memory**

For Exam

The instruction cycle executes each program instruction in four phases: Instruction Fetch (CPU reads instruction from memory address in PC into IR, then increments PC), Instruction Decode (CU interprets opcode and identifies operands), Operand Fetch (values retrieved from memory via MAR/MDR into registers), and Execution (ALU performs operation, result stored). For ADD A, B: fetch instruction, decode as addition, fetch values of A and B, execute A+B, store result in A.

Diagram (refer SLM): Fig. 2.2.1 Instruction Cycle.

2.2.3 RISC and CISC

Theory

RISC (Reduced Instruction Set Computer) uses a simple, limited instruction set with simple hardware; instructions like LOAD/STORE are independent; each instruction takes a single clock cycle; examples: PowerPC, SPARC. CISC (Complex Instruction Set Computer) uses complex instructions that can be larger than one word; reduces program instruction count but increases cycles per instruction; examples: VAX, Intel x86, AMD. RISC suits high-end applications (telecom, image/video processing); CISC suits low-end applications (home automation, security).

Important Points

Feature	RISC	CISC
Instruction set	Simple, limited	Complex, large
Decoding	Simple	Complex
Execution time	Short (single clock cycle)	Longer (multiple cycles)
Instruction format	Fixed	Variable
Memory interaction	Register-to-register (LOAD/STORE)	Memory-to-memory
Code length	Longer	Shorter (less RAM)
Examples	PowerPC, SPARC	VAX, Intel x86, AMD
Applications	High-end: telecom, image processing	Low-end: home automation

For Exam

RISC uses simple instructions executed in single clock cycles with register-to-register operations (PowerPC, SPARC). CISC uses complex instructions reducing program size but requiring more cycles per instruction (Intel x86). RISC has simple decoding and fixed format; CISC has complex decoding and variable format. RISC

suits high-performance applications; CISC reduces memory requirements with compact code.

Unit Recap (from SLM)

- Instructions are words of computer language; programs form software
- Instruction set = all instructions a microprocessor supports
- Instruction format: Opcode field + Address field
- Instruction cycle: Fetch → Decode → Operand Fetch → Execute
- RISC: simple, fast, single cycle; CISC: complex, compact code
- ISA is the only way to interact with hardware

Unit 3: Registers, Cache Memory and Virtual Memory

2.3.1 CPU Registers

Theory

Registers are the fastest storage in the memory hierarchy — small temporary storage inside or near the CPU for fast data retrieval during instruction execution. They are classified by purpose into general types (Data, Address, Status/Flag registers) and specific registers (Accumulator, MAR, MDR, GPR, PC, IR, Condition Code). Registers are accessed at higher speed than conventional memory and are essential for CPU performance.

Important Points

Register	Full Name	Function
Data Register	—	Holds data for an operation
Address Register	—	Holds address of data or instructions
Status/Flag Register	—	Indicates processor status or result of arithmetic operation
Accumulator	ACC	Stores intermediate results after ALU arithmetic/logical operations
MAR	Memory Address Register	Holds address of memory location to be accessed
MDR	Memory Data Register	Holds data to be read from or written to memory
GPR	General Purpose Register	R0, R1...Rn-1; temporary data during operations
PC	Program Counter	Contains address of next instruction to be executed

Register	Full Name	Function
IR	Instruction Register	Stores instruction about to be executed (fetched from PC)
Condition Code Register	—	Contains 1-bit flags (0 or 1) indicating operation status (e.g., zero flag)

- Registers = top of memory hierarchy; fastest storage mechanism
- Internal registers (inside processor) vs external registers (outside processor)

For Exam

CPU registers are the fastest temporary storage at the top of memory hierarchy. Key registers: MAR (memory address), MDR (memory data), Accumulator (ALU results), PC (next instruction address), IR (current instruction), GPR (temporary data), and Condition Code Register (1-bit flags). Registers enable fast data retrieval for instruction execution.

Diagram (refer SLM): Fig. 2.3.1 Memory hierarchy; Fig. 2.3.3 Registers in CPU.

2.3.2 Cache Memory

Theory

Cache memory is very high-speed memory lying just below registers in the hierarchy, between the processor and main memory. Frequently used instructions and data are placed in cache for faster access. Three levels exist: L1 (primary, inside processor chip, smallest/fastest), L2 (secondary, outside processor, larger), L3 (shared among CPU cores in multi-core processors). Cache hit means data found in cache; cache miss requires fetching from main memory (miss penalty). Hit ratio = hits / (hits + misses) measures cache performance.

Important Points

- L1 = inside processor; L2/L3 = outside processor; L1 has lowest latency
- **Hit ratio** = hits / (hits + misses)
- **Advantages:** faster than main memory, less access time, stores frequently executed programs
- **Disadvantages:** limited capacity, high cost
- On miss: fetch from main memory → if not there, fetch from secondary → copy to main → copy to cache

For Exam

Cache memory is high-speed storage between CPU and main memory storing frequently used data. Three levels: L1 (inside processor, fastest), L2, L3 (outside, shared in multi-core). Cache hit = data in cache; cache miss = not found (miss)

penalty applies). Hit ratio = hits/(hits+misses) measures performance. Cache improves data retrieval efficiency despite limited capacity and high cost.

Diagram (refer SLM): Fig. 2.3.2 Cache memory; Fig. 2.3.4 Cache Memory.

2.3.3 Virtual Memory

Theory

Virtual memory is an OS feature that compensates for physical RAM shortages by moving data pages between RAM and disk storage, allowing programs to believe they have more memory than physically available. It separates user logical memory from physical memory — programs run smoothly even with relatively small RAM. Physical addresses are hardware addresses in main memory; logical addresses are generated by the processor. Virtual address space is the virtual storage assigned to a process (program in execution).

Important Points

- Extends physical memory using disk; provides memory protection
- **Physical address:** hardware address in main memory
- **Logical address:** generated by processor
- **Virtual address:** allocated to virtual memory location accessed as if in main memory
- **Virtual address space:** virtual storage assigned to a process
- **Real address:** address of storage location in main memory
- OS swaps required program parts between main and secondary memory

Virtual Memory — How It Works (Step by Step)

1. Program requests more memory than physically available in RAM
2. OS divides program into **pages** (fixed-size blocks)
3. Required pages loaded into RAM; rest kept on disk (secondary storage)
4. When CPU needs a page not in RAM, a **page fault** occurs
5. OS swaps out less-used pages from RAM to disk, loads needed page into RAM
6. User/program sees a continuous large address space — unaware of swapping

Address Types Summary

Address Type	Generated By	Refers To	Example Use
Logical Address	Processor (CPU)	Virtual memory location	Program's view of memory
Virtual Address	OS/MMU	Address in virtual address space	Mapped to physical or disk

Address Type	Generated By	Refers To	Example Use
Physical Address	Hardware (actual RAM)	Real location in main memory	Actual RAM chip location
Real Address	—	Storage location in main memory	Same as physical address

For Exam

Virtual memory allows running large programs on computers with limited RAM by using disk as extension of main memory. The OS manages swapping between primary and secondary storage. Physical addresses are actual hardware memory locations; logical addresses are processor-generated. Virtual address space is assigned per process. This separation of logical and physical memory enables efficient memory management and protection.

Diagram (refer SLM): Fig. 2.3.5 Applications (TV app store analogy for virtual memory).

Unit Recap (from SLM)

- Registers: fastest storage; Data, Address, Status types
- MAR (address), MDR (data), Accumulator (ALU results), PC (next instruction), IR (current instruction)
- Condition Code Register with 1-bit flags
- Cache L1/L2/L3; hit ratio = $\text{hits}/(\text{hits}+\text{misses})$
- Virtual memory extends RAM using disk; virtual vs physical/logical addresses

Unit 4: Secondary Storage Devices

2.4.1 Secondary Storage and Magnetic Storage

Theory

Secondary storage provides cheap, non-volatile, large-capacity permanent storage since primary memory is expensive, limited, and volatile. The processor needs primary memory for fast intermediate results during processing. Secondary storage characteristics: non-volatile, large size, cheaper than primary memory. Devices are fixed (hard disk drive) or removable (pen drive, optical discs). Magnetic storage stores data on magnetic medium using magnetization patterns — either sequential access (magnetic tape) or direct/random access (magnetic disk).

Important Points

Type	Access	Description	Capacity/Speed
Magnetic Tape	Sequential	Older; backup; high recording density; low cost per bit	Large data; portable; sequential only
Hard Disk (HDD)	Random	Main data storage; metal/plastic disks with iron oxide coating	4500–7200 rpm; GB–TB; ms access time
Floppy Disk	Random	Oldest portable; obsolete	1.44 MB standard

- Secondary storage: non-volatile, large, cheap permanent storage
- Primary memory still needed for speed during processing
- HDD stores data using magnetization patterns on disk

For Exam

Secondary storage provides non-volatile, large, inexpensive permanent storage. Magnetic storage uses magnetization on magnetic medium. Magnetic tape is sequential access, used for backup. Hard disk (HDD) is the main random-access storage device with rpm 4500–7200 and capacity in GB–TB. Floppy disk (1.44 MB) is obsolete. Fixed devices include HDD; removable include pen drives and optical discs.

Diagram (refer SLM): Fig. 2.4.1 Secondary Storage Devices; Fig. 2.4.2 Magnetic Tape; Fig. 2.4.3 Hard Disk; Fig. 2.4.4 Floppy Disk.

2.4.2 Optical Storage

Theory

Optical discs store data read and written by lasers, with much higher capacity than floppies. CD (Compact Disc, invented by James Russell, first produced 1982) holds 650–700 MB; types include CD-R (write once) and CD-RW (rewritable). DVD (Digital Versatile/Video Disc) holds 4.7–17 GB depending on layers/sides. Blu-ray uses blue laser with shorter wavelength for high-definition content — single layer 25 GB, dual layer 50 GB.

Important Points

Media	Capacity	Key Features
CD	650–700 MB (80-min)	CD-R (record once), CD-RW (rewritable); Nero software for burning
DVD	4.7 GB (single-layer) to 17.08 GB (double-sided double-layer)	DVD-R (write once), DVD-RW (erasable)
Blu-ray (BD)	25 GB (single layer), 50 GB (dual layer)	Blue laser; HD content; 3D movies

For Exam

Optical storage uses lasers to read/write data. CD holds 650–700 MB (CD-R write

once, CD-RW rewritable). DVD stores 4.7–17 GB for movies and data. Blu-ray uses blue laser for HD content — 25 GB single layer, 50 GB dual layer. Optical discs replaced floppies due to higher capacity.

Secondary Storage — Complete Capacity Comparison Table

Device	Type	Capacity	Access	Media/Technology	Storage
Floppy Disk	Removable	1.44 MB	Random	Magnetic disk	Obsolete
CD (Compact Disc)	Removable	650–700 MB	Random	Optical (laser)	Common
DVD (single-sided, single-layer)	Removable	4.7 GB	Random	Optical (laser)	Medium
DVD (single-sided, dual-layer)	Removable	8.5–8.7 GB	Random	Optical (laser)	Excellent
DVD (double-sided, single-layer)	Removable	9.4 GB	Random	Optical (laser)	Best
DVD (double-sided, dual-layer)	Removable	up to 17.08 GB	Random	Optical (laser)	Maximum
Blu-ray (single layer)	Removable	25 GB (~2 hr HD video)	Random	Optical (blue laser)	High speed
Blu-ray (dual layer)	Removable	50 GB (~4.5 hr HD video)	Random	Optical (blue laser)	Long
HDD (Hard Disk Drive)	Fixed/Removable	GB – TB (typ. 500 GB–16 TB)	Random	Magnetic disk	4 million
SSD (Solid State Drive)	Fixed/Removable	GB – TB (typ. 256 GB–4 TB)	Random	NAND flash memory	Fast
Pen Drive (USB Flash)	Removable	8 GB – 512 GB+	Random	USB flash (NAND)	Portable
External HDD	Removable	TB (typ. 1–8 TB)	Random	Magnetic disk	Best
Memory Stick	Removable	Varies (MB–GB)	Random	Flash (Sony)	Common

Device	Type	Capacity	Access	Media/Technology	S
					m
Magnetic Tape	Removable	TB+	Sequential	Magnetic tape	B lc

DVD Capacity Breakdown (from SLM)

DVD Structure	Capacity
Single-sided, single-layer	4.7 GB
Single-sided, double-layer	8.5 – 8.7 GB
Double-sided, single-layer	9.4 GB
Double-sided, double-layer	up to 17.08 GB

Blu-ray Capacity Breakdown (from SLM)

Blu-ray Type	Capacity	Content Equivalent
Single layer	25 GB	~2 hours HD video or ~13 hours standard video
Dual layer	50 GB	~4.5 hours HD video or ~20 hours standard video

SSD vs HDD Quick Comparison

Feature	HDD	SSD
Technology	Magnetic disk (spinning platters)	NAND flash (no moving parts)
Speed	Slower (ms access)	Much faster (µs access)
Capacity	Higher (up to 16+ TB)	Lower per cost (typ. up to 4 TB)
Power	Higher consumption	Lower consumption
Reliability	Sensitive to shock	Shock resistant
Cost per GB	Lower	Higher (decreasing)
Best for	Bulk storage, backup	OS drive, laptops, performance

2.4.3 USB, Pen Drive, External HDD, Memory Stick, and SSD

Theory

USB (Universal Serial Bus) is an industry standard interface with Type A, B, and C ports connecting peripherals like cameras, smartphones, and printers. Pen drives are USB flash storage devices for documents, photos, music, and video. External HDDs are portable high-capacity (terabytes) backup devices connected via USB. Memory sticks (introduced by Sony for cameras) are compact removable flash media. SSDs (Solid State Drives) use NAND flash for faster, lower-power, more reliable storage than traditional HDDs.

Important Points

Device	Description	Advantages
USB	Universal Serial Bus; industry standard port interface	Type A, B, C ports; connects peripherals
Pen Drive	USB flash storage; pen-shaped	Scratch/dust/magnetic resistant; low power; portable; affordable
External HDD	Outside computer with own enclosure; USB connected	High capacity (TB); backup; portable; easy to use
Memory Stick	Sony portable flash media for cameras	Compact; large capacity; no moving parts; durable
SSD	NAND flash secondary storage	Faster, less power, more reliable than HDD; portable or internal

For Exam

USB is the standard interface for connecting devices. Pen drives are USB flash storage for files and media. External HDDs provide terabyte backup capacity via USB. Memory sticks are Sony's compact flash media. SSDs use NAND flash, offering faster access, lower power consumption, and greater reliability than HDDs, though at higher cost per gigabyte.

Diagram (refer SLM): Fig. 2.4.5 Pen Drive; Fig. 2.4.6 Memory Stick.

Unit Recap (from SLM)

- Secondary storage: non-volatile, large, cheap permanent storage
- Magnetic: tape (sequential, backup), disk/HDD (random, main storage), floppy (1.44 MB, obsolete)
- Optical: CD (650–700 MB), DVD (4.7–17 GB), Blu-ray (25–50 GB, blue laser)
- **USB = Universal Serial Bus**; pen drive = USB flash storage
- External HDD = backup; Memory stick = Sony portable media
- **SSD = NAND flash**, faster and more reliable than HDD

Block 3: Software

Unit 1: System Boot up and Software Layers

3.1.1 Software — System and Application

Theory

Software is a collection of computer programs, data, instructions, and documentation that enables a computer to perform specific tasks. **An instruction is a step-by-step procedure**; a **program is a set of instructions**; **hardware is all physical parts**;

programs bundled together form **software**. The OS acts as the interface between user and hardware — the most important software on any computer. **Software** divides into **System Software** (manages hardware, provides platform) and **Application Software** (helps users perform specific tasks). Early systems used DOS (command-based); modern systems use GUI (graphical icons and mouse).

Important Points

#	System Software	Application Software
1	Interface between apps and hardware	Designed per user requirements
2	Often low-level; also C, Python, Java	High-level languages
3	Operates computer hardware	Performs specific user tasks
4	Installed with OS	Installed per user need
5	Less/no user interaction	Full user interface
6	Runs independently; provides platform	Needs system software to run
7	OS, compilers, assemblers, debuggers, drivers	Word processors, browsers, media players

- System software features: communicates directly with hardware, smaller size, complex, high speed, versatile
- **OS** = core system software; first layer loaded at startup; intermediary between user, apps, hardware
- **Device drivers** translate between OS and hardware (printers, graphics cards, network adapters)
- **Language processors:** assembler, compiler, interpreter — convert programming languages to machine code
- **Utility software:** compression tools, disk defragmenter, backup, antivirus
- Application: general purpose (word processor, spreadsheet, presentation, database, multimedia) or specific/customised (payroll, airline reservation, Tally, inventory/HRM)

For Exam

Software is a collection of programs, data, instructions, and documentation enabling computers to perform tasks. It divides into system software (manages hardware, provides platform — OS, device drivers, language processors, utilities) and application software (end-user programs — MS Office, browsers, payroll systems). System software communicates directly with hardware, is essential at startup, and runs independently; application software depends on it and provides user interfaces. General-purpose application software (word processor, spreadsheet) serves many

users; specific-purpose software (Tally, airline reservation) is customised for one organisation. DOS was command-based; GUI allows interaction via graphical icons and mouse.

3.1.2 Files and Folders

Theory

Files and folders form the foundational framework for organising digital data on computers. A **file** is a digital container storing text, images, audio, video, or programs — identified by name and extension (e.g., `Notes.doc`, `Image.jpg`). A **folder** (directory) groups related files hierarchically; nested folders create subdirectories. The wardrobe analogy: items = files, wardrobe = folder, drawers = subfolders.

Important Points

- File = name + extension (extension identifies type); **Path** = folder hierarchy (e.g., `C:\Users\Name\Documents\file.txt`)
- Examples: `Notes.doc` (Word), `Image.jpg` (JPEG), `document.txt` (text file)
- **File system** = software managing storage organisation on HDD/SSD
- **File operations:** create, open, edit, move, copy, delete, rename
- **Permissions** control access, modify, delete rights per user/group
- **Metadata** = data about data (size, creation date, modification date, author)
- Folders/directories group related files; nested subfolders create hierarchy (music/melody/fast)

For Exam

Files are digital storage units with name and extension identifying type. Folders (directories) group related files in a hierarchical structure with paths showing location. The file system manages organisation on storage devices. Users perform create, open, edit, move, copy, delete, and rename operations. Permissions and metadata (data about data) support security and file management.

3.1.3 Booting and POST

Theory

Booting is the startup sequence that loads the OS when the computer is turned on — like a gym warm-up before exercise. During booting, the system checks all installed hardware/software and loads necessary files for operation. **POST** (Power On Self Test) runs early in booting to verify RAM, hard drives, CD-ROM, keyboard, and other components. **BIOS** (Basic Input Output System) supports booting from HDD, optical drives, and USB.

Important Points

Type	Also Called	Description
Warm booting	Soft reboot	Restart without cutting power; skips full hardware re-init
Cold booting	Hard booting	Start from completely off state; full POST and initialization

Boot Process Steps (in order):

Step	Stage	What Happens
1	Power On	User presses power button; electricity supplied to components
2	BIOS/UEFI Init	Firmware starts; configures basic hardware settings
3	POST	Tests RAM, drives, keyboard, CD-ROM; beeps/error codes on failure
4	Boot Device Selection	BIOS scans boot order (HDD → optical → USB) for bootable media
5	Boot Loader Load	Small program loaded from boot device into memory
6	OS Kernel Load	Core OS files loaded into RAM — first layer of system software
7	Driver & Service Init	OS loads device drivers, starts essential background services
8	Ready State	Login screen or desktop appears; user can interact with system

- Boot devices: hard disks, optical drives, USB (configured in BIOS boot order)
- POST failure halts boot — system displays error codes or audible beeps
- OS is the first major software loaded into memory at startup
- Warm boot reloads OS without full power cycle; cold boot runs complete POST

For Exam

Bootting is the startup sequence loading the OS when the computer turns on. Steps: power on → BIOS init → POST (hardware test) → boot device selection → boot loader → OS kernel → drivers/services → ready state. POST verifies RAM, drives, keyboard, and components. BIOS supports booting from HDD, optical, and USB. Warm booting (soft reboot) restarts without cutting power; cold booting (hard booting) starts from off with full POST and initialization.

3.1.5 Software Layers and Architecture

Theory

Software applications use three layers: **Presentation** (UI/client layer — user interaction), **Application** (business logic — processing between UI and data), and **Data** (storage layer). Gmail login page = presentation; login validation = application; user database = data. These layers deploy in **one-tier** (all on one machine), **two-tier** (client + data server), or **three-tier** (client + application server + data server) architectures.

Important Points

Architecture	Layers Location	Example
One tier	All three in single package	MP3 player, standalone MS Office
Two tier	Client: presentation + application; Server: data	Client-server apps
Three tier	Client: presentation; App server: logic; DB server: data	Scalable web apps

- **Presentation layer** (client layer): topmost; user interacts — Gmail login boxes/buttons
- **Application layer** (business logic): intermediate; processes login, queries database
- **Data layer**: stores and retrieves data for application layer
- **One-tier**: MP3 player, standalone MS Office — all layers in one package on one machine
- **Two-tier**: client handles presentation + application; server handles database; client requests, server responds
- **Three-tier**: client (UI) + application server (logic) + database server (data) — best scalability

For Exam

Three-tier architecture separates Presentation (client UI), Application (business logic on app server), and Data (database server) onto three distinct systems. Client handles only the user interface; application server processes requests and queries the database; database server stores and retrieves data. This provides better scalability than one-tier (all local) or two-tier (client + DB server) because each tier can be upgraded independently. Gmail, banking apps, and e-commerce sites commonly use three-tier design.

Unit Recap (from SLM)

- Software classified into **System Software** and **Application Software**

- **File** = storage unit with name + extension; **Folder/Directory** = organises files
- **Boot** starts OS; types: **Warm** (soft reboot) and **Cold** (hard booting)
- Boot steps: Power on → BIOS → POST → boot device → boot loader → OS kernel → ready
- **POST** = Power On Self Test; **BIOS** selects boot device (HDD, optical, USB)
- Architecture: **One tier**, **Two tier**, **Three tier**
- Layers: **Presentation**, **Application (Business Logic)**, **Data**
- **Metadata** = data about data; **Path** = file location in folder hierarchy

Unit 2: Popular Operating Systems

3.2.1 Operating System — Functions, DOS, Client-Server

Theory

An OS is a collection of software managing computer hardware and providing common services — the interface between user and hardware. Like a company manager, it coordinates all programs. **DOS** is disk-based, single-user, single-task, command-based OS. **Client-server** model: client requests services (like restaurant customer), server responds (like waiter). Open-source software has freely accessible source code; proprietary software is exclusively owned.

Important Points

OS Function	Description
Process Management	Schedules and manages running programs/tasks
Memory Management	Allocates and tracks RAM usage
File System Management	Organises files and folders on storage
Device Management	Controls hardware via drivers
User Interface	Provides CLI (DOS) or GUI (Windows) interaction
Security & Access Control	Protects data; manages user permissions
Networking & Communication	Enables network and Internet connectivity
Error Handling & Logging	Records system errors for diagnosis
Virtualization/Multitasking	Runs multiple programs concurrently
Boot Process	Manages system startup sequence

- **Device drivers** act as translators between OS and hardware (printers, graphics cards, network adapters)

- **Language processors** convert high-level code to machine code (assembler, compiler, interpreter)
- **DOS**: disk-based, single-user, single-task; user types commands; no graphical icons
- **Client** = requests services; **Server** = higher config, responds to requests
- Restaurant analogy: customer = client, waiter = server
- Every computer needs at least one OS; examples: Windows, Linux, Android, iOS

For Exam

An operating system manages hardware and provides common services as the user-hardware interface — like a company manager coordinating workers (applications). DOS is disk-based, single-user, single-task, command-based. In client-server computing, the client requests services and the server responds (restaurant customer/waiter analogy). OS handles process, memory, file, device management, security, networking, error handling, boot process, and time management. Device drivers translate for hardware; language processors convert code. Windows and Linux are common GUI OS; Android (Linux kernel) and iOS are dominant mobile OS.

3.2.1.3 Types of Operating Systems

Theory

Operating systems are broadly classified into seven types based on how they manage users, tasks, processors, and networks. Each type suits different environments — from batch processing of similar jobs to real-time control systems and mobile devices.

Important Points

OS Type	Key Feature	Examples
Batch OS	Similar jobs grouped as batches; offline input; no direct interaction	Punch card systems, payroll batch jobs
Multitasking/Time-sharing	Multiple terminal users simultaneously; quick response; reduces CPU idle time	UNIX, Multics
Multiprocessing	Two+ CPUs with OS copy each; coordinated multi-CPU operations	Windows NT, Windows 2000
Real-Time OS	Immediate data processing; no delay permitted; hard/soft variants	Petroleum monitoring, military/space software
Distributed OS	Multiple systems on shared network; remote access, concurrency, fault tolerance	Cloud computing, Hadoop clusters

OS Type	Key Feature	Examples
Network OS	Runs on servers; centralised user/data management; stable security	Windows Server 2003/2008, Novell NetWare, Mac OS X Server
Mobile OS	Smartphones, tablets, wearables; touch-optimised	Android, iOS, BlackBerry, watchOS

- Real-time types: **Hard** (time constraint critical; delay = drastic consequences) | **Soft** (flexible timing)
- Real-time examples: petroleum temperature monitoring, military systems, space software
- Real-time advantages: maximum device utilisation, near error-free; limitation: complex algorithms
- Batch OS: jobs submitted via punch cards to operator; user does not interact during execution
- Time-sharing advantages: quick response, avoids software duplication, reduces CPU idle time
- Distributed advantages: effective resource sharing, concurrency, scalability, fault tolerance
- Network OS advantages: stable centralised servers, managed security, easy hardware upgradation
- Open-source: source code freely accessible (Linux); Proprietary: exclusively owned (Windows)

For Exam

Seven OS types with examples: **Batch** (punch card payroll jobs — no user interaction), **Multitasking/Time-sharing** (UNIX, Multics — multiple terminal users), **Multiprocessing** (Windows NT/2000 — multiple CPUs), **Real-Time** (petroleum/military/space — immediate response; hard/soft variants), **Distributed** (cloud/Hadoop — networked systems with fault tolerance), **Network OS** (Windows Server, Novell NetWare — server management), **Mobile OS** (Android, iOS — smartphones/tablets). Batch limitation: no interaction during job run. Distributed advantages: resource sharing, concurrency, scalability, fault tolerance.

3.2.1.4 Windows vs Linux and Mobile OS

Theory

Windows and Linux are commonly used GUI operating systems with different management styles. **Android** (Google, 2008) is built on Linux kernel for touch

devices — partially open source, most-used OS overall. iOS (Apple) runs on iPhone, iPod, and iPad.

Important Points

Feature	Linux	Windows
Source	Open source	Licensed/proprietary
Cost	Free	Costly
Filename	Case-sensitive	Case-insensitive
Efficiency	More efficient	Less efficient
Security	More secure	Less secure

- Android (Google, 2008): built on **Linux kernel**; touch-screen tablets/smartphones; partially open source; most-used OS overall
- iOS (Apple Inc.): iPhone, iPod, iPad; proprietary Apple ecosystem
- Other mobile OS: BlackBerry, Web, watchOS
- Windows: licensed, proprietary, case-insensitive filenames, less efficient, less secure
- Linux: open source, free, case-sensitive, more efficient, more secure

For Exam

Linux is open-source, free, case-sensitive, efficient, and more secure. Windows is licensed, costly, case-insensitive, and proprietary. Android (Google, Linux kernel) dominates mobile devices since 2008. iOS (Apple) powers iPhone, iPod, and iPad. Both Windows and Linux act as system managers coordinating application “workers.”

3.2.2 Graphical User Interface (GUI)

Theory

A user interface is required to interact with computers. **GUI** (Graphical User Interface) allows interaction through graphical icons and visual representations instead of typed commands. It offers a visual representation of OS commands and functions, making computers accessible to beginners.

Important Points

- GUI advantages over DOS: easier for beginners, visual icon representation of commands
- Drag-and-drop, cut-and-paste simplify information exchange between applications

- Users can explore options by pointing mouse at icons instead of memorising commands
- GUI offers visual feedback; reduces learning curve for non-technical users
- Windows and Linux both provide GUI; DOS remains command-line only

For Exam

GUI (Graphical User Interface) allows users to interact with electronic devices through graphical icons and visual representations instead of typed commands. It offers a visual representation of OS commands, making computers accessible to beginners. Advantages over DOS include drag-and-drop, cut-and-paste, and easy exploration of options using a mouse. Windows and Linux are widely used GUI operating systems; DOS requires memorising and typing commands.

Unit Recap (from SLM)

- OS manages hardware; acts as interface between user and hardware
- **DOS** = disk-based, single-user, single-task, command-based
- **GUI** = graphical icons; easier than command interface
- Client-server: client requests, server responds (restaurant analogy)
- OS types with examples: Batch, Multitasking (UNIX), Multiprocessing (Win NT), Real-Time (space), Distributed (cloud), Network (Win Server), Mobile (Android/iOS)
- **Windows** = licensed/proprietary; **Linux** = open source/free
- **Android** = Google, Linux kernel, 2008; **iOS** = Apple, iPhone/iPad

Unit 3: System Software and Utilities

3.3.1 Programming Languages

Theory

Programming languages are formal languages programmers use to communicate with computers. **High-level languages** (C, C++, Java, Python) use human-readable syntax and are portable. **Low-level languages** are machine-friendly: **Assembly** uses mnemonics (short codes); **Machine language** uses binary (0 and 1) — the only language computers directly understand.

Important Points

Feature	High Level	Low Level
Nature	Programmer-friendly	Machine-friendly

Feature	High Level	Low Level
Memory	Less efficient	Highly efficient
Understanding	Easy	Tough
Debugging	Simple	Complex
Portability	Portable	Non-portable

- Assembly language uses **mnemonics** (short codes); more user-friendly than machine language
- Machine language = binary using only **0 and 1**; only language computer understands directly
- Writing binary programs is tedious — programmers prefer high-level languages + translators
- System software includes language translators, OS, and utilities making computer functional
- Department analogy: computer = department, HOD = system software, teachers = applications

For Exam

Programming languages are formal languages for programmer-computer communication. High-level languages (C, C++, Java, Python) are programmer-friendly, portable, and easier to debug. Low-level languages are machine-friendly and memory-efficient: assembly uses mnemonics (short codes); machine language uses binary (0 and 1) — the only language computers understand. Translators bridge the gap: compiler (whole program), interpreter (line by line), assembler (assembly to machine). Programmers prefer high-level languages because writing binary is tedious.

3.3.2 Language Translators

Theory

Computers understand only machine language; programmers write in high-level languages. **Language translators** convert programs to machine-understandable form. **Compiler** converts entire program at once; **Interpreter** translates line by line; **Assembler** converts assembly language to machine language.

Important Points

Feature	Compiler	Interpreter
Translation	Entire program at once	Line by line
Analysis time	More	Less

Feature	Compiler	Interpreter
Execution time	Less	More
Debugging	Harder	Easier
Object code	Generated	Not generated
Examples	C, C++, Java	Python, Perl, PHP

- **Assembler** = assembly language → machine language; uses mnemonics (short codes)
- **Compiler**: analyses entire source before execution; generates intermediate object code
- **Interpreter**: analyses less but executes slower overall; no object code file
- C, C++ use compilers; PHP, Python use interpreters
- Hierarchy: High level → (Compiler/Interpreter) → Machine language; Assembly → (Assembler) → Machine language

For Exam

Language translators bridge high-level programs and machine language. A compiler translates the whole program before execution (C, C++ — faster execution, harder debugging). An interpreter translates line by line (Python, PHP — easier debugging, slower execution). An assembler converts assembly language to machine code.

3.3.3 Database and DBMS

Theory

A **database** is a collection of information organised so programs can quickly select desired data — like a spice rack in a kitchen or telephone directory online. **DBMS** (Database Management System) is a set of programs enabling users to access, manipulate, report on, and represent data.

Important Points

- DBMS stores/retrieves data in multiple ways; balances needs of multiple applications
- Consistent data administration processes across organisation
- Sophisticated functions for efficient storage and retrieval
- Provides **Data Integrity** (accuracy) and **Security** (access control)
- Examples: online telephone directory, Facebook member/event/message data
- Kitchen spice rack analogy: organised data easy to select when needed

For Exam

A database is an organised electronic collection of data enabling quick selection — like a kitchen spice rack or online telephone directory. Facebook stores member, event, message, and advertising data in databases. DBMS (Database Management System) manages access, manipulation, reporting, and security. Advantages: stores large volumes, easy access/update, accurate data, consistent administration, balances multiple applications sharing data, and provides data integrity and security through sophisticated storage/retrieval functions.

3.3.4 Utility Software

Theory

Utility software assists the OS in specialised maintenance tasks — analysing, configuring, and maintaining computers. It runs in support of system software and helps keep the system functional and secure.

Important Points

- Utility software = application software assisting OS in specialised tasks
- **Antivirus:** scans for boot viruses, Trojans, worms, spyware; removes/isolates threats
- **Backup software:** creates copies stored in secure location for data recovery
- **Debuggers:** test and debug other programs; fix programming errors
- **Disk checkers:** scan operating hard drives for errors
- **Disk cleaners:** find unnecessary files when hard disk is full; user decides what to delete
- **Disk compression:** transparently compress/decompress disk contents; increases capacity
- **Disk defragmenters:** detect scattered file fragments; rearrange for efficiency

For Exam

Utility software assists the OS in system maintenance and specialised tasks. Examples include antivirus (virus-free environment), backup (data copies), debuggers (fix program errors), disk checkers, disk cleaners, disk compression, and disk defragmenters. Antivirus is a key utility protecting against boot viruses, Trojans, worms, and spyware.

Unit Recap (from SLM)

- System software controls and manages hardware; provides platform for applications

- **High level** (C, Java, Python) = programmer-friendly, portable; **Low level** (Assembly, Machine) = machine-friendly
- **Assembly** uses mnemonics; **Machine language** uses only 0 and 1
- **Compiler:** whole program → machine language (C, C++); **Interpreter:** line-by-line (Python, PHP)
- **Assembler:** assembly → machine language
- **Database:** organised information; **DBMS** manages access, integrity, security
- **Utility software:** antivirus, backup, debuggers, disk checkers/cleaners/compression/defragmenters
- System software analogy: HOD manages department; teachers = applications

Unit 4: Application Software

3.4.1 Application Software Types

Theory

Application software (end-user program) helps users perform specific tasks — takes user input and completes work. Types include word processing, spreadsheet, presentation, multimedia, database, and simulation software. **LaTeX** is a free document preparation system for high-quality scientific/technical documentation, separating content from style.

Important Points

Type	Purpose	Examples
Word processor	Create, manipulate, store text	MS Word, Google Docs
Spreadsheet	Calculations in rows/columns/cells	MS Excel, Google Sheets
Presentation	Display info on slides	MS PowerPoint, Impress
Multimedia	Audio/video handling	Media players
Database	Data management apps	MS Access
Simulation	Model real-world scenarios	Flight simulators

- **Word processor** features: create/save/edit, font/alignment/colour formatting, spell-check, images, headers/footers, watermarks
- **Spreadsheet:** rows × columns = cells; data as text/date/time/number; arithmetic, logical, text ops; charts/graphs
- **Spreadsheet activities:** addition, average, counting, charts, data entry, cell formatting, logical comparisons

- **Presentation:** info broken into slides; add text, graphics, video, images; conveys concept to audience
- **LaTeX features:** journal articles, technical reports, books, slide presentations; complex math formulas (AMS-LaTeX)
- **LaTeX:** cross-references, tables, figures, bibliographies, indexes, multilingual typesetting, artwork; separates content from style
- Specific purpose examples: GIMP, Payroll System, Airline Reservation, Tally, Inventory/HRM systems

For Exam

Application software (end-user program) performs specific tasks for users. Word processors (MS Word, Google Docs) create, format, and store text with spell-check and images. Spreadsheets (Excel) use rows, columns, and cells for calculations, charts, and logical comparisons. Presentation tools (PowerPoint) display information on slides with graphics and video. LaTeX is free scientific documentation software separating content from style, supporting math formulas and bibliographies. Other types: multimedia, database, simulation, and customised software (payroll, Tally).

3.4.2 Computer Virus and Protection

Theory

A **computer virus** is malicious code that changes how a computer works and spreads between machines, damaging devices or stealing data. Entry methods include file sharing, infected webpages, and spam email attachments. Types: worms, Trojan, ransomware. Protection uses antivirus (utility software), firewall, and safe practices.

Important Points

- **Virus types:** **Worms** (self-replicating), **Trojan** (disguised malware), **Ransomware** (locks data for payment)
- **Problems after infection:** slow performance, frequent crashes, data loss/corruption, identity theft
- **Antivirus** = utility software scanning and eliminating viruses
- **Firewall** = network security device filtering traffic per organisation security policy
- **Protection measures:** use antivirus + firewall, update both frequently, update OS regularly
- Increase browser security settings; download only from trusted sites; avoid spam attachments

For Exam

A computer virus is malicious spreading code altering computer operation. Entry via file sharing, infected sites, or spam attachments causes slowdowns, crashes, data loss, and identity theft. Types include worms, Trojan, and ransomware. Protection requires antivirus utility software, firewall (network traffic filter), frequent updates, and downloading from trusted sources only.

Unit Recap (from SLM)

- Application software = end-user program for specific tasks (MS Office, Tally)
- Types: Word Processing, Spreadsheet, Presentation, Multimedia, Database, Simulation
- Word processor: create/edit/format text; Spreadsheet: rows × columns × cells for calculations
- Presentation: slides with text, graphics, video; LaTeX: free scientific/technical documentation
- LaTeX separates content from style; supports math formulas, bibliographies, multilingual typesetting
- Computer virus = malicious spreading code; entry via files, infected sites, spam
- Virus types: Worms, Trojan, Ransomware; protection: antivirus, firewall, updates, trusted downloads

Block 4: Networks and Internet

Unit 1: Basic Concepts and Devices

4.1.1 Computer Networks — Concepts, Need, and Working

Theory

A **computer network** connects computers and networking devices to share resources, data, and applications. Communication involves a **sender**, **transmission medium**, and **receiver** (Fig 4.1.1). **Guided medium** uses physical channels — copper cables, optical cables. **Unguided medium** is wireless — radio waves, microwaves, infrared. Networks enable file sharing, hardware sharing (printers), application sharing, and messaging without duplicating devices on each computer.

Important Points

- Network = connected devices sharing resources; connected device = **node** (PC, server, printer, switch)
- **Server/host** provides data, services, resources, programs to **clients** on request

- **Switch** connects all devices; sends, receives, forwards data across network ports
- **Shared printer workflow (4 steps):** PC-A sends print request → switch acknowledges → switch forwards to printer → printer prints document
- Family network analogy: members share resources; larger network connects to school, workplace, store
- **Advantages:** file-sharing (upload/download), hardware sharing (one printer for all), application sharing, messaging (WhatsApp, SMS)

For Exam

A computer network links computers and devices to share resources, data, and applications. Communication uses sender, medium, and receiver (Fig 4.1.1). Guided media (copper/optical cables) physically guide signals; unguided media (radio, microwaves, infrared) use wireless transmission. Without networks, each computer needs separate printers, scanners, and costly software. A basic network has nodes (PCs, server, printer) connected via a switch; the server provides services to clients on request. Shared printer workflow: PC sends request → switch acknowledges → switch forwards to printer → document prints.

4.1.2 Types of Computer Networks

Theory

Networks are classified by geographical coverage into LAN, MAN, and WAN. Each serves different scales — from a single building to metropolitan areas to global connectivity without distance limitations.

Important Points

Type	Coverage	Example
LAN	Small area: building/campus (few km)	Hospital departments, classroom
MAN	Metropolitan/city area	Cable TV network
WAN	Large geographical area; no distance limit	Inter-hospital national/global networks

- **LAN:** hospital departments sharing info within one building/campus; most common type
- **MAN:** spans metropolitan/city area; example = cable TV connection network
- **WAN:** group of hospitals across country/world; no distance limitation; wireless global reach
- **LAN** limited to few kilometres; **WAN** has no geographic limit

- Classification based on coverage area, service type, and network requirements

For Exam

LAN covers a small geographical area like a hospital building or campus (few kilometres), enabling resource sharing within that site — the most frequently used network type. MAN spans a metropolitan/city area such as a cable TV connection network connecting an entire city. WAN covers large geographical areas with no distance limit — like a group of hospitals across countries sharing information wirelessly worldwide.

4.1.3 Networking Devices

Theory

Networking devices (hardware/equipment) interconnect computers for data transfer and communication. They connect computers, printers, fax machines, and other devices. Five main types: Hub, Switch, Router, Repeater, and Bridge — each with distinct intelligence and function.

Important Points

Device	Function	Key Trait
Hub	Connects multiple devices; broadcasts packets to ALL ports	Not secure; no intelligence
Switch	Forwards packets to designated port only	Intelligent; secure
Router	Connects different networks; chooses optimal route by destination	Links Network A to B
Repeater	Regenerates weakened signals	Extends transmission distance
Bridge	Divides LAN into segments; filters/forwards by address	Reduces traffic

- Hub: data sent as packets to ALL connected ports (Fig 4.1.7 — nodes A–E); insecure
- Switch: receives data on designated port only; forwards to intended recipient (Fig 4.1.8)
- Router: checks destination address → chooses optimal route → forwards between networks (Fig 4.1.9)
- Repeater: strengthens weakened signal and retransmits (petrol pump analogy — Fig 4.1.10)
- Bridge: divides LAN into segments; checks address before forwarding (Fig 4.1.11)

- **Workstation:** dedicated business computer; high-resolution display, faster CPU, more RAM; multitasking; shares resources with clients/servers

For Exam

Hub connects multiple devices but broadcasts data packets to ALL ports — like a post office delivering to everyone at an address (insecure). Switch intelligently forwards packets only to the designated port — like registered post to one addressee (secure; replaces hubs today). Router connects two different networks, checks destination address, and chooses the optimal route (class teacher between two classes analogy). Repeater regenerates weakened signals over long distances (petrol pump refuelling analogy). Bridge divides a LAN into segments, filtering and forwarding by MAC address to reduce traffic.

4.1.5 Network Topologies

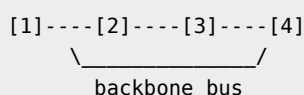
Theory

Topology describes how computers are physically connected and the logical flow of information in a network. Six types: Bus, Ring, Star, Mesh, Tree, and Hybrid — each with distinct layout, fault tolerance, and cost. Physical topology = cable layout; logical topology = data flow path (may differ from physical).

Important Points

Topology	Layout	Key Feature	Weakness
Bus	Single backbone cable	Simple, cheap; all nodes share bus	Backbone failure stops entire network
Ring	Circular cable loop	Token controls access; orderly transmission	One break disrupts ring
Star	All nodes to central hub/switch	Easy to add/remove nodes; fault in one cable isolated	Hub failure disables all nodes
Mesh	Every node to every other	Fault tolerant; multiple paths; $n(n-1)/2$ edges	Expensive; complex wiring
Tree	Tree/branch (Bus + Star)	Scalable hierarchy; segmented traffic	Root/branch failure affects subtree
Hybrid	Mix of two+ topologies	Flexible for specific business needs	Complex to design and manage

Bus Topology — single backbone; data travels both directions:



Ring Topology — token passes clockwise; node transmits only when holding token:

```

[1]-->[2]
  ^      |
  |      v
[4]<--[3]

```

Star Topology — all communication via central hub/switch:

```

      [HUB]
     /  |  \
    [1] [2] [3] [4]

```

Mesh Topology — every node connected to every other; multiple paths:

```

[1]=[2]    (= direct link)
|\  /|
[3]=[4]

```

Tree Topology — bus backbone with star branches via hubs:

```

      [H1]---[H2]
     / |      \
    [1] [2]    [3] [4]
     |
    [H3]--[5]--[6]

```

Hybrid Topology — combines Bus + Star + Ring as needed:

```

[Bus:1]--[Star Hub]--[2][3]
          |
        [Ring:4-5-6-7]

```

- Ring token system: data travels one direction; node transmits only when it holds the token
- Ring example: node 1 → 4 passes through 2, 3 sequentially; no start/end terminators in circle
- Mesh: if n=4 nodes, edges = $4 \times 3 / 2 = 6$; multiple alternate paths if one link fails
- Tree: node 1 → 6 path = node1 → hub1 → hub3 → node6; reduces traffic via segmentation

- Hybrid: node 1→2 uses star hub; node 1→6 uses bus then ring (needs token at node 6)
- **Star weakness:** central hub failure disables entire network; **Bus weakness:** backbone break stops all

For Exam

Topology is the physical connection and logical flow in a network. Bus uses one backbone cable — all nodes share it; cheap but backbone failure stops the network. Ring connects devices in a circle with no terminators; a token system ensures only the node holding the token can transmit (one direction). **Star routes all traffic through a central hub/switch** — easy to manage but hub failure is critical. Mesh connects every node to every other ($n(n-1)/2$ links) — fault tolerant but expensive. Tree combines bus backbone with star branches for scalable hierarchy. **Hybrid mixes topologies (bus + star + ring) for flexible business needs.**

4.1.6 Network Operating Systems (NOS)

Theory

NOS coordinates communication of multiple computers across a network (unlike single-computer OS). Two major types: **Peer-to-Peer** (equal computers share resources, no central file server) and **Client/Server** (dedicated file servers with centralised security and management).

Important Points

Type	Features	Suitability
Peer-to-Peer	No file server; all computers equal; share files/resources	Small-medium LANs
Client/Server	Dedicated file server(s); centralised security; multi-user access	Larger organisations

- **P2P:** share resources/files; access files on other computers; no central file server or centralised management
- **P2P:** all computers equal with same capabilities; designed for small-medium LANs
- **Client/Server:** centralise functions; one or more dedicated file servers with security
- **Client/Server:** integrate all devices; share resources with multiple users regardless of location
- NOS = OS for networks; coordinates multiple computers (unlike single-computer Windows/Linux)

For Exam

Network Operating System (NOS) coordinates multiple computers across a network — unlike Windows/Linux on a single PC. Peer-to-Peer NOS lets equal computers share files and resources without a central file server or centralised management, suitable for small-medium LANs. Client/Server NOS uses dedicated file server(s) providing centralised security, resource management, and multi-user access regardless of user location. P2P: all nodes equal; Client/Server: server has higher configuration and responds to client requests.

Unit Recap (from SLM)

- Communication transfers information; medium may be **guided** (cables) or **unguided** (wireless)
- **Computer network** connects devices to share resources, data, applications
- **Server** provides data/services; **node** = any connected device; **switch** forwards data
- **LAN** = building/campus; **MAN** = city; **WAN** = no geographic limit
- **Hub** = broadcast (no intelligence); **Switch** = intelligent forwarding
- **Router** connects networks; **Repeater** regenerates signals; **Bridge** divides segments
- **Topology** = physical connection and logical flow; six types with distinct ASCII layouts
- **NOS**: Peer-to-Peer (no file server) vs Client/Server (dedicated file server)

Unit 2: World Wide Web and Search Engines

4.2.1 History of the Internet

Theory

The Internet evolved from US defense research. **ARPA** (1958) led to **ARPANET** (1962, J.C.R. Licklider). Packet switching (1965), TCP adoption (1983), **NSFNET** (1985) connected universities. **Tim Berners-Lee** created **HTTP** (1989) — father of WWW. **Mosaic** browser (1993, NCSA). NSFNET decommissioned 1995; commercial ISP model emerged.

Important Points

- 1958: ARPA created for US defense; physical cable networks
- 1962: J.C.R. Licklider proposed ARPANET for nuclear-attack-resistant communication

- 1965: Packet switching introduced; Stanford used first LAN for distant workstations
- 1981: ARPANET extended to national computer science researchers
- 1983: ARPANET adopted TCP; spread to university campuses; acted as early ISPs
- 1985: NSFNET connected university CS departments; linked supercomputing centers
- 1989: Tim Berners-Lee (CERN) created HTTP — father of WWW
- 1990: ARPANET gradually phased out
- 1993: Mosaic browser (NCSA) — key development from NSFNET era
- 1995: NSFNET decommissioned; commercial ISP model and Internet commerce emerged

For Exam

Internet history: ARPA (1958) → ARPANET (1962) → packet switching (1965) → TCP (1983) → NSFNET (1985) → HTTP/WWW by Tim Berners-Lee (1989) → Mosaic browser (1993) → commercial ISPs (post-1995). Tim Berners-Lee is the father of WWW; Mosaic was the key 1993 browser development.

4.2.2 Internet and World Wide Web

Theory

The **Internet** is a global network of millions of computers sharing resources via ISPs, hardware, and wireless/cabling technologies. The **WWW** is a vast collection of linked information on web pages — the most common Internet information system. A **web page** is a single document; a **website** is a collection of pages with a common domain name on a web server.

Important Points

- Internet working (email example): Laptop A → ISP1 → Email Server A → Internet → ISP2 → Server B → Laptop B
- ISPs (MTNL, BSNL, Airtel, Jio) = companies offering Internet access for global data transfer
- WWW = vast linked information on web pages; most common Internet information system
- Encyclopaedia analogy: WWW = whole book; web page = one page you turn to
- Web page built with HTML (structure), CSS (design), JavaScript (interactivity)

- **Website working (4 steps):** browser requests page → Internet forwards to web server → server sends HTML → browser displays page
- **Static website:** fixed content, same for every user, infrequent design changes
- **Dynamic website:** content changes each visit; may differ per user (e.g., Facebook feed)
- **Web server** stores, processes, delivers pages; **Intranet server** = internal, not public

For Exam

Internet is a global network of millions of computers connected via ISPs for resource sharing and communication. ISPs (BSNL, Airtel, Jio) are the essential link between your device and remote servers. WWW is a vast linked collection of web pages — like an encyclopaedia where each page holds text, images, and hyperlinks. A web page is built with HTML (structure), CSS (design), JavaScript (interactivity); a website is a collection with a common domain on a web server. Static sites show fixed content; dynamic sites change per visit or user (e.g., social feeds). Website access: browser requests → Internet → web server → HTML returned → displayed.

4.2.3 Websites — Static vs Dynamic

Theory

A **website** is a collection of web pages with a common domain name published on a web server by an individual, business, or institution. Pages are accessed via Internet using URLs. Websites classify as **static** (fixed content, same for every user) or **dynamic** (content changes each visit, may differ per user).

Important Points

- **Static:** web pages loaded exactly as stored; infrequent content/design changes; built with HTML
- **Dynamic:** portions change without full page reload; HTML + JavaScript; per-user content (e.g., news feed)
- **Web server** stores, processes, delivers pages; internal-only server = **Intranet server**
- Website access flow: browser request → Internet → web server → HTML page → browser display
- Example static: institutional brochure site; example dynamic: Amazon, Facebook

For Exam

A website is a collection of web pages sharing a domain name, published on a web server. Static websites display fixed content identical for every user, created mainly

with HTML. Dynamic websites change content on each visit or per user using HTML and JavaScript. Access flow: browser sends request via Internet, web server returns HTML, browser renders the page. Intranet servers serve internal organisational pages not visible to the public.

4.2.5 URL and Browsers

Theory

URL (Uniform Resource Locator) is a unique web address pointing to one specific page or file — like a postal address. Parts: protocol (https), WWW, domain name (www.google.com), extension (.com), and page name. A **browser** is software to access and view websites (Chrome, Firefox, Safari, IE).

Important Points

- URL example: <https://www.google.com/doodle>
- **https** = protocol rules for browser-computer communication over Internet
- **www** = World Wide Web prefix
- **www.google.com** = domain name (registered to one owner; points to root directory)
- **.com** = extension indicating owner type (.gov, .co.uk also common)
- **doodle** = specific page/file on the website
- Browser features: back/forward navigation, refresh (reload), stop (cancel loading), home (preset page)
- Address bar: enter URLs; dropdown of previously visited sites; tabbed browsing for multiple sites
- Bookmarks save URL addresses; web search tool lets you choose favourite search engine

For Exam

URL is a unique web address directing the browser to a specific page — like a postal address for mail. Example <https://www.google.com/doodle> : **https** = communication protocol; **www** = World Wide Web; **google.com** = registered domain name; **.com** = extension (owner type); **doodle** = specific page. Browsers (Chrome, Firefox, Safari, IE) access websites with navigation buttons, address bar, refresh/stop, home, tabs, bookmarks, and built-in search. Bookmarks save URLs; tabbed browsing opens multiple sites in one window.

4.2.7 Search Engines and Search Tips

Theory

A **search engine** is Internet software querying an indexed database and returning best-matching results. It uses a **web crawler/spider** to systematically index downloaded pages (like a librarian cataloguing books). Tips: specific keywords, quotation marks for exact phrases, Boolean +/- operators, Google Advanced Search, browser history, and Ctrl+F on pages.

Important Points

- Search engine analogy: library = search engine, librarian = browser, user = searcher
- Crawler/spider systematically indexes downloaded pages using algorithms for fast retrieval
- Search does NOT go directly to web — queries pre-built index database first
- Results show page title, text portion size, images, videos
- **Keywords:** be specific (search "lotus" not generic "flower")
- **Quotation marks:** "Maruti Car Service" for exact phrase match
- **Boolean:** + include, exclude — "Maruti Car Service"+Kerala-jobs
- **Advanced Search:** filter by date, country, language, amount
- **Browser History:** revisit previously viewed pages
- **Ctrl+F:** find word/phrase within an already-open web page

For Exam

Search engines (Google, Yahoo, Bing) are Internet software querying a pre-built indexed database — they do not search the live web directly. A web crawler/spider systematically downloads and indexes pages (like a librarian cataloguing books).

Tips: use specific keywords, quotation marks for exact phrases ("Maruti Car Service"), Boolean operators (+Kerala -jobs), Google Advanced Search for date/country filters, browser history for revisiting pages, and Ctrl+F to find text on an open page. Results include page title, text size, images, and videos.

Unit Recap (from SLM)

- Internet launched 1958 (ARPA); **ARPANET** (1962), **TCP** (1983), **NSFNET** (1985), **HTTP** (1989)
- **Tim Berners-Lee** = father of WWW; **Mosaic** = 1993 browser; NSFNET decommissioned 1995
- **Internet** = global network via ISPs; **WWW** = linked web pages (encyclopaedia analogy)

- **Web page** = single document (HTML/CSS/JS); **Website** = collection on web server
- **Static** = fixed content; **Dynamic** = changes per visit/user
- **URL** = unique address (protocol + domain + extension + page)
- **Browser** accesses sites; **Search engine** + **crawler** query indexed database
- Search tips: keywords, quotes, Boolean (+/-), Advanced Search, Ctrl+F

Unit 3: Internet Connectivity

4.3.1 Internet Service Providers (ISP)

Theory

An **ISP** is a company offering Internet access — the gateway between your computer and all Internet servers. Without an ISP subscription, a computer with a modem cannot access the Internet. ISPs enable email, shopping, research, and all online activities.

Important Points

- **ISP** = gateway/link between your computer and all Internet servers
- Without ISP subscription, built-in modem alone cannot access Internet
- **Internet Access:** connecting PDA devices (phones, tablets) online; speed per plan
- **Domain Name Registration:** reserve unique name ~1 year (.com, .gov, .co.in); cannot buy lifetime
- **Web Hosting:** ISP rents space on web server for website files (pictures, audio, video)
- **Usenet:** collection of online discussion newsgroups; Q&A and file sharing forums
- Indian ISPs: BSNL, Airtel, Reliance Jio, Vodafone-Idea, MTNL
- Email route: your PC → ISP servers → destination ISP → recipient (ISP is essential link)

For Exam

ISP is the gateway to the Internet, offering access, domain registration, web hosting, and Usenet discussions. Internet access connects devices online at plan-dependent speeds. Domain registration reserves a unique name annually. Web hosting allocates server space for website files. Examples: BSNL, Airtel, Jio.

4.3.2.1 Dial-up and Cable — Detail

Theory

Dial-up uses PSTN (Public Switched Telephone Network) landline with a modem to connect to an ISP. PSTN uses **circuit switching** — a dedicated physical path between sender and receiver. The modem converts digital computer data to analog for transmission and back to digital at the receiver. Common in remote areas where broadband is unavailable.

Important Points

- Setup: PC A → modem → telephone line → ISP → Internet cloud → ISP → modem → PC B
- Modem = Modulator-Demodulator; converts digital ↔ analog signals
- Circuit switching: connection-oriented; dedicated path for entire session
- Phone line busy during dial-up session — cannot use phone and Internet simultaneously (unlike DSL)
- Cable modem: ISP is cable TV operator; coaxial or fibre to home outlet

For Exam

Dial-up connection uses PSTN landline and modem with circuit switching to reach an ISP. The modem converts digital data to analog for telephone transmission and reconverts at the receiver. It suits remote areas lacking broadband but offers very slow speeds (up to 56 Kbps). Cable modem uses coaxial/fibre from a cable TV operator with a splitter providing separate TV and Internet connections.

4.3.2.3 Wireless Local Loop (WLL)

Theory

WLL connects the subscriber to the nearest telephone exchange via a **radio link** instead of copper telephone lines — making it wireless unlike dial-up. Used where wired infrastructure is difficult. Main components: PSTN, Switch Function, WANU, and WASU.

Important Points

Component	Role
PSTN	Circuit-switched global telephone network
Switch Function	At local exchange; routes PSTN among WANU units
WANU	Wireless Access Network Unit — authentication, routing, transceiving voice/data
WASU	Wireless Access Subscriber Unit — at subscriber home; connects to WANU via antenna

- All local WASU units connect to WANU wirelessly through antenna
- WANU functions: Authentication, Operation & Maintenance, Routing, Transceiving
- WLL speed typically 128 Kbps–2 Mbps; moderate cost; wireless alternative to dial-up

For Exam

WLL (Wireless Local Loop) connects subscribers to the exchange using a radio link instead of telephone copper cables. Components: PSTN (circuit-switched network), Switch Function (at exchange), WANU (network-side unit handling authentication and routing), and WASU (subscriber-side unit with antenna). It provides wireless Internet access where wired lines are unavailable, with speeds typically below broadband but above dial-up.

4.3.2.4 Digital Subscriber Line (DSL)

Theory

DSL uses existing telephone lines to transfer data and connect to the Internet.

Operates on **different frequencies** for telephone voice and data — so phone and Internet work simultaneously without interruption. Two forms: **symmetric** (equal upload/download) and **asymmetric** (faster download, slower upload — most common for home users).

Important Points

- Setup: ISP → phone wall socket → DSL splitter → modem (Internet) + telephone (voice)
- Splitter separates frequencies: one band for voice calls, another for Internet data
- **Symmetric DSL**: equal speeds both directions (business use)
- **Asymmetric DSL (ADSL)**: faster download — suited for browsing, streaming, downloads
- Dedicated continuous connection while line is active; speeds typically 1–100 Mbps

For Exam

DSL is an Internet connection using telephone lines at frequencies separate from voice calls, enabling simultaneous phone and Internet use via a DSL splitter.

Symmetric DSL offers equal upload/download speeds; asymmetric DSL (ADSL) offers faster downloads for typical home use. ISP provides connection to the phone socket; splitter routes voice to telephone and data to DSL modem and computer.

4.3.2.5 Fibre Optics — Detail

Theory

Fibre optics converts electrical signals carrying information into light pulses transmitted through transparent glass fibres about the diameter of a human hair. Speeds exceed DSL and cable — typically tens to hundreds of Mbps or Gbps. Core broadband technology for modern high-speed Internet (FTTH — Fibre To The Home).

Important Points

Layer	Function
Outer jacket	Protects fibre from mechanical/environmental stress
Strength member	Additional mechanical protection
Coating	Extra protection for core and cladding
Cladding	Reduces light scattering; keeps light inside core
Glass core	Innermost; transmits light signals; larger core = more light capacity

- Electrical signals → light → glass fibre → electrical at receiver
- Faster than DSL and cable modem; preferred for high-bandwidth needs
- Used by cable TV operators and ISPs for premium broadband plans

For Exam

Fibre optic technology converts electrical signals to light transmitted through glass fibres. Structure: outer jacket and strength member (protection), coating, cladding (prevents light escape), and glass core (transmits light). It delivers speeds far exceeding DSL and cable, making it the preferred broadband medium for high-speed Internet. Larger core diameter allows more light and higher data capacity.

4.3.2 Internet Connection Types

Theory

ISPs provide connectivity through various technologies — from dial-up over telephone lines to high-speed broadband and dedicated leased lines. **Wired** connections use physical cables (copper, coaxial, optical fibre); **wireless** uses radio waves (Wi-Fi, WiMAX, satellite). Speed and cost vary by technology, coverage area, and whether bandwidth is shared or dedicated.

Important Points

Connection	Technology	Typical Speed	Cost	Best For
Dial-up	PSTN landline + modem; circuit switching	Up to 56 Kbps	Very low	Remote areas without broadband

Connection	Technology	Typical Speed	Cost	Best For
Cable modem	Coaxial/fibre via cable TV operator	10–500+ Mbps	Moderate	Homes with cable TV infrastructure
WLL	Radio link (WANU/WASU) to exchange	128 Kbps–2 Mbps	Low–moderate	Areas lacking wired telephone lines
DSL	Telephone line; symmetric/asymmetric frequencies	1–100 Mbps	Moderate	Homes/offices with existing phone lines
Fibre (FTTH)	Light signals through glass fibre core	100 Mbps–1 Gbps+	Moderate–high	High-speed urban/suburban areas
Wi-Fi	Short-range wireless via router + modem	Depends on ISP plan	Low (uses existing broadband)	Home/office local wireless access
WiMAX	Microwave radio; longer range than Wi-Fi	Up to 1 Gbps	Moderate	Sparsely populated / wide coverage
Satellite	Geostationary satellite + dish + modem	12–100 Mbps (high latency)	High	Remote/rural with no terrestrial lines
BPL	Broadband over electrical power lines	Variable; emerging	Low potential	Limited pilot areas
Leased line	Dedicated symmetric bandwidth	2 Mbps–10 Gbps	Highest	Businesses needing guaranteed SLA

Connection	Technology	Key Detail
Dial-up	PSTN + modem	Analog conversion; circuit switching; phone line busy during use
Cable modem	Coaxial/fibre	Splitter: one to modem (Internet), one to set-top box (TV)
WLL	Radio link	WANU (network unit) + WASU (subscriber unit); wireless to exchange
Broadband	≥25 Mbps down, 3 Mbps up (FCC)	Umbrella: fibre, wireless, cable, DSL, satellite, BPL
DSL	Phone line dual-frequency	Symmetric (equal up/down) or asymmetric (faster download)
Leased line	Dedicated pipe	Fixed bandwidth; no peak-time slowdown

Broadband sub-types:

- **Fibre optics:** electrical → light through glass fibres; core transmits, cladding reduces scattering
- **Wi-Fi:** short-range wireless; router connects modem output to Wi-Fi devices
- **WiMAX:** Worldwide Interoperability for Microwave Access; longer range/speed than Wi-Fi

Feature	Wi-Fi	WiMAX
Range	Short (home/office)	Longer (city/rural coverage)
Speed	Depends on ISP plan	Up to 1 Gbps
Use	Local wireless access	Wide-area wireless broadband
Devices	Router + modem at home	Base station coverage area

- **Satellite:** geostationary satellites; high latency but reaches anywhere with dish view
- **BPL:** broadband over power lines; avoids new cabling; very limited deployment
- WLL components: **PSTN** (circuit-switched telephone network), **Switch Function** (at local exchange), **WANU** (Wireless Access Network Unit — authentication, routing, transceiving), **WASU** (Wireless Access Subscriber Unit — at subscriber home)
- WLL: radio link replaces copper telephone lines; subscriber connects to exchange wirelessly via antenna
- Dial-up: digital → modem → analog → PSTN → ISP → Internet → reverse at receiver; modem converts both ways
- Satellite working: PC A → router → modem → dish → geostationary satellite → provider hub → web server → reverse path
- Satellite uses geostationary satellites (not telephone lines); dish + modem at subscriber; high latency
- BPL (Broadband over Powerline): Internet via existing electrical network; avoids new cabling; limited areas
- Compare cost: Dial-up (cheapest) < WLL < DSL/Cable < Fibre < Satellite < Leased line (most expensive)
- Compare speed: Dial-up (56 Kbps) < WLL < DSL < Cable < Fibre/WiMAX < Leased line (dedicated Gbps possible)

For Exam

Connection types vary in speed, cost, and suitability. Dial-up (56 Kbps, PSTN/modem, very cheap) suits remote areas. Cable modem (high Mbps via coaxial/fibre splitter) and DSL (phone line, symmetric/asymmetric) are common home options. Fibre offers fastest speeds via light in glass fibres. Wi-Fi provides local wireless; WiMAX covers wider areas. Satellite serves remote locations with higher latency and cost. WLL uses radio (WANU/WASU). Leased line gives dedicated symmetric bandwidth at highest cost for businesses. Broadband ($\geq 25/3$ Mbps per FCC) encompasses fibre, wireless, cable, DSL, satellite, and BPL.

4.3.3 Leased Line

Theory

A **leased line** is a dedicated, fixed-bandwidth, symmetric data connection providing reliable high-quality Internet with assured upload and download speeds. Unlike shared connections, bandwidth does not fall at peak times.

Important Points

- Links office locations or computers/servers across sites (e.g., two branch offices)
- Reserved bandwidth for exclusive use — not shared with other ISP customers
- Carries Internet traffic, phone calls, and other data (dedicated pipe analogy)
- Bandwidth does NOT fall at peak times unlike ordinary shared connections
- Symmetric: equal upload and download speeds guaranteed
- Used where reliability and consistent speed are critical (banks, corporates, universities)

For Exam

A leased line is a dedicated, fixed-bandwidth, symmetric connection linking office locations with assured upload/download speeds. It provides reliable high-quality Internet without peak-time bandwidth reduction, unlike shared ISP connections. Used for inter-office linking with reserved bandwidth.

Unit Recap (from SLM)

- **ISP** = gateway to Internet; services: access, domain registration, hosting, Usenet
- **Dial-up**: PSTN, circuit switching, modem; up to 56 Kbps; very low cost
- **Cable modem**: splitter for TV + Internet; moderate speed/cost
- **WLL**: radio link; WANU/WASU components; wireless to exchange
- **Broadband**: fibre, wireless (Wi-Fi/WiMAX), satellite, BPL, DSL; $\geq 25/3$ Mbps

- **Fibre** = fastest; **Satellite** = remote areas; **Leased line** = dedicated symmetric bandwidth
- **Wi-Fi** = short range; **WiMAX** = longer range wireless
- Compare speed/cost: dial-up (cheapest/slowest) → broadband → leased line (costliest/most reliable)

Unit 4: Electronic Mail Systems

4.4.1 Electronic Mail — Address, Components, and Protocols

Theory

Email transmits messages between computers via Internet — one of the most used Internet services. Messages may contain text, images, audio, or attachments.

Address format: `username@domainname` (e.g., sachin@gmail.com) — not case-sensitive, no spaces. Email systems use standard **protocols** for sending and receiving: **SMTP** (send), **POP3** and **IMAP** (receive).

Important Points

Component	Function
User Agent (UA)	Program to compose, send, receive, reply to messages (Outlook, Thunderbird)
Message Transfer Agent (MTA)	Transfers email between systems using SMTP; client MTA + system MTA
Mailbox	File on local hard drive collecting delivered emails
Spool file	Contains outgoing emails; MTA extracts for delivery

Protocol	Full Form	Purpose	Port	Key Trait
SMTP	Simple Mail Transfer Protocol	Sending email between servers	25/587	MTA uses SMTP to push mail to recipient server
POP3	Post Office Protocol v3	Receiving email; downloads to local device	110/995	Mail removed/stored locally; offline access
IMAP	Internet Message Access Protocol	Receiving email; syncs with server	143/993	Mail stays on server; access from multiple devices

- Email services provided by system:

Service	Description
Composition	Creating messages and replies
Transfer	Sending email from sender to recipient
Reporting	Confirming delivery, loss, or rejection
Displaying	Presenting email in readable format
Disposition	Save, delete before/after reading

- **SMTP** = outbound; MTA uses SMTP to push mail to recipient server via spool file
- **POP3** downloads messages to local mailbox; good for single-device offline reading
- **IMAP** keeps messages on server; folders synced across phone, laptop, tablet
- Email services: Composition, Transfer, Reporting, Displaying, Disposition
- Sender and recipient both need valid `username@domainname` addresses

For Exam

Email transfers messages via Internet using `username@domainname` addresses.

Components: User Agent (compose/send/receive), MTA (transfers between systems via SMTP), Mailbox (incoming storage), Spool file (outgoing queue). Protocols: SMTP sends mail between servers; POP3 receives and downloads mail locally; IMAP receives while keeping mail on server for multi-device sync. Services include composition, transfer, delivery reporting, display, and disposition.

4.4.2 Email Software Features

Theory

Email software provides sending, receiving, filtering, attaching, forwarding, CC, and BCC features. Two types: **email clients** (Outlook — manual send/receive, auto-download attachments) and **webmail** (Gmail, Yahoo — refresh page, manual attachment download).

Important Points

- **CC (Carbon Copy)**: all CC recipients visible to each other; like carbon paper between sheets
- **BCC (Blind Carbon Copy)**: only sender sees BCC list; recipients unaware of each other
- **Filtering**: routes mail to folders at server; suspicious → spam/junk; blocks malicious links/code
- **Attach steps**: Compose → Attach Files → Choose File → Open → Send

- **Forward:** select message → enter destination → use forward symbol (no recomposition needed)
- Email clients (Outlook): click Send/Receive; attachments auto-download
- Webmail (Gmail, Yahoo): refresh page; manually download attachments
- Send steps: login → Compose/New → To + Subject + Body → Send

For Exam

Email features include send, receive, filter, attach, forward, CC, and BCC. Email clients (Outlook) run on the computer — click Send/Receive, attachments auto-download. Webmail (Gmail, Yahoo) runs in browser — refresh to check, manually download attachments. CC (Carbon Copy) shows all recipients to each other; BCC (Blind Carbon Copy) hides recipients from one another. Filtering routes mail to folders at the server; suspicious mail goes to spam. Attach: Compose → Attach Files → Choose File → Send. Forward resends a received message without recomposition.

4.4.3 Web-Based Systems

Theory

Web-based systems use web applications over Internet to accomplish tasks (Google Meet, Zoom, online forms, shopping carts). Web applications are coded in JavaScript and HTML, executed via browser. Dynamic apps need server-side processing; static apps do not.

Important Points

- Working (5 steps): browser requests app → Internet forwards to web app server → server processes → returns page via Internet → displays on user screen
- Architecture: web server (receives requests) + application server (processes tasks) + database (stores data)
- Dynamic apps need server-side processing; static apps run entirely in browser
- Benefits: **Data Recovery, Better Security, Competitive Edge, Improved Efficiency, Greater Visibility, 24/7 Accessibility**
- Integratable features: **Mobile Interface, Social Integration, Analytics, Live Chat, Web Payments**
- Examples: Google Meet, Zoom, Google Apps, Microsoft 365, online forms, shopping carts

For Exam

Web-based systems use web applications (JavaScript, HTML) via browsers to accomplish Internet tasks like Google Meet, Zoom, and online shopping.

Architecture: browser → Internet → web application server → database. Five-step flow: request → forward to server → process → return page → display. Benefits: data recovery, security, competitive edge, efficiency, visibility, 24/7 access. Integratable features: mobile interface (desktop + phone), social integration (quick registration), analytics (user insights), live chat (customer support), web payments (one-step checkout).

4.4.4 Web Pages

Theory

A **web page** displays text, figures, audio, video, animations, and links in a browser. Collection of linked pages = **website**. Pages classify as **static** (fixed content, stored as-is) or **dynamic** (content changes without full reload, per-user variation).

Important Points

Page Type	Purpose	Example Use
Home Page	Starting point; website introduction and navigation	www.sgou.ac.in landing tab
Feed Page	Updates content dynamically	Social media following updates
Menu Page	Navigation to content categories	Restaurant-style content list
Search Page	Internal search to jump to results	Site search field
About Page	Company/product/person info; branding	Administration, Academics tabs
Registration Page	Create/login to personalised accounts	Facebook, email signup
404 Page	Error — page not found or broken link	Deleted/dead link response
Portfolio Page	Professional visual presentation	Designs, art, handmade goods
Product Page	E-commerce details, reviews, cart/wishlist	Amazon smartphone listing

- **Static page:** content loaded exactly as stored on server; created with HTML
- **Dynamic page:** portion changes without full reload; HTML + JavaScript; per-user content
- Each page has unique URL; collection of linked pages = website on web server

For Exam

Web pages are browser documents displaying text, figures, audio, video, animations, and links — like encyclopaedia pages on web servers. Static pages load exactly as stored (HTML); dynamic pages change portions without full reload (HTML +

JavaScript). Nine page types: Home (starting point), Feed (content updates), Menu (navigation list), Search (internal results), About (branding info), Registration (account login), 404 (not found error), Portfolio (visual showcase), Product (e-commerce with reviews/cart). Each page has a unique URL; linked pages form a website.

Unit Recap (from SLM)

- **Email** transfers messages; needs sender/recipient addresses (username@domain)
- Components: **UA**, **MTA**, **Mailbox**, **Spool file**
- Protocols: **SMTP** (send), **POP3** (download receive), **IMAP** (server-sync receive)
- Email types: **clients** (Outlook) and **webmail** (Gmail)
- Features: send, receive, filter, attach, forward, **CC**, **BCC**
- **Web-based systems** use web applications (JavaScript, HTML)
- Web pages: **Static** (fixed) and **Dynamic** (changing)
- Page types: Home, Feed, Menu, Search, About, Registration, 404, Portfolio, Product

Block 5: Hypertext Markup Language

Course Code: B21CA01DC | **Programme:** BCA | **University:** SGOU

Coverage: Block 5 — Units 1–4 (HTML) | **Source:** SGOU SLM B21CA01DC

Unit 1: HTML — Basic Tags and Divisions

5.1.1 HTML Structure, Document Types and Rules

Theory

A **website** is a collection of web pages accessible through the Internet; each page is built with **HTML** (Hypertext + Markup Language). Hypertext links documents; markup tags define structure. **CSS** handles presentation; **JavaScript** adds interactivity.

HTML5 is the latest version (video streaming, location support). An **HTML element** has opening tag, content, and closing tag; syntax is `<tagname> Content </tagname>`. Tags are **paired** (opening + closing) or **singular/empty** (e.g. `<br>`). Attributes in the opening tag use `name="value"` in quotation marks.

The document structure (Fig 5.1.2): `<!DOCTYPE html>` → `<html>` (root) → `<head>` (metadata, not visible) → `<title>` (browser title bar) → `<body>` (visible content). Save as `.html` or `.htm` using Notepad/TextEdit; browser renders the file top-to-bottom,

left-to-right. HTML versions evolved from 1.0 (basic text/images) through 2.0, 3.0 (CSS support), 4.1 (external CSS) to HTML5.

HTML document types: (1) **Transitional** — most used, flexible syntax, browsers use “best effort” without reporting errors; (2) **Strict** — enforces rules, all opened tags need closing tags, faster on mobile; (3) **Frameset** — multiple documents on one screen (menu systems).

Version	Year	Key Features
HTML 1.0	1991	Basic text controls and images only
HTML 2.0	—	Common rules; text boxes, buttons
HTML 3.0	—	Improved tags; CSS support
HTML 4.1	—	External CSS file inclusion
HTML5	Current	Video streaming, location support, modern APIs

HTML rules: sketch layout on paper first; tags in angle brackets `<tag>`; opening on / closing off (`<BR>` exception); closing uses forward slash `</tag>`; nested tags closed inner-first; optional attributes modify behaviour (e.g. `<P ALIGN=CENTER>`). Older browsers ignore unknown tags without breaking recognised content.

Important Points

- HTML = Hypertext (links) + Markup Language (tags); invented 1991 by Tim Berners-Lee
- HTML versions: 1.0 → 2.0 → 3.0 → 4.1 → **HTML5**
- **DOCTYPE** on first line tells browser the HTML version
- `<title>` = browser tab title; `<h1>` = page content heading (different purposes)
- **Three document types:** Transitional, Strict, Frameset
- **Five coding rules:** angle brackets, on/off switch, forward slash, correct nesting, attributes
- Paired tags have opening and closing; singular/empty tags (br, img, meta) need no closing
- Attribute values must always be enclosed in quotation marks
- Design on paper before coding to avoid rework with HTML tags

Common Global Attributes (apply to most HTML elements):

Attribute	Example	Purpose
id	<code>id="section1"</code>	Unique identifier; target for internal links
class	<code>class="header-box"</code>	Groups elements for CSS styling

Attribute	Example	Purpose
style	<code>style="color:red;"</code>	Inline CSS styling on any element
title	<code>title="Tooltip text"</code>	Tooltip shown on mouse hover
lang	<code><html lang="en"></code>	Declares document language

For Exam

HTML is a structured markup language combining hypertext links and tags to define web page structure. A basic document contains DOCTYPE, html, head (title, meta, link, style), and body. Documents are classified as Transitional (flexible), Strict (rule-enforced), or Frameset (multiple frames). Key rules: tags in angle brackets, paired tags with forward-slash closing, correct nesting, and attributes in quotes. HTML5 is the current standard supporting multimedia and modern web features. Tim Berners-Lee invented HTML in 1991; each HTML file is plain text with a .html extension read by browsers top-to-bottom.

```
<!DOCTYPE html>
<html>
<head><title>Web Page</title></head>
<body>
<h1>Distance Learning Computer Science</h1>
<p>A portal for Students</p>
</body>
</html>
```

5.1.4–5.1.9 HTML Tags — Head, Body, Headings, Div and Center

Theory

HTML tags are keywords helping browsers format content; each tag has opening tag, content, and closing tag (some have no closing). The **head tag** (`<head>`) holds metadata — title, character set, styles, links, scripts — not displayed on the page. Head elements: **title** (one per document, browser title bar), **style** (inline CSS in head), **base** (absolute URL, one only, empty tag), **link** (external CSS via `rel` and `href`, empty tag), **meta** (keywords, description, viewport for search engines, empty tag).

The **body tag** defines visible content (text, images, links); placed inside `<html>` after `<head>`; only one body per document. **Headings** `<h1>` – `<h6>`: h1 = most important, h6 = least; help search engines index structure. **Division tag** `<div>` groups elements for CSS styling; block-level, no layout effect alone. **Center tag** `<center>` aligns text, graphics, and tables to centre. Browsers read HTML top-to-bottom and distinguish tagged content from plain text.

Head Element	Syntax Example	Purpose
title	<code><title>Page Title</title></code>	Browser title bar; one per document
style	<code><style>h1{color:#1c87c9}</style></code>	Inline CSS styling inside head
base	<code><base href="https://site.com/" target="_blank"></code>	Absolute base URL; one only; empty tag
link	<code><link rel="stylesheet" href="style.css"></code>	Links external CSS file; empty tag
meta	<code><meta name="keywords" content="HTML, CSS"></code>	Keywords, description, viewport; empty tag
meta viewport	<code><meta name="viewport" content="width=device-width, initial-scale=1.0"></code>	Responsive design on all devices

Body Tag	Syntax Example	Purpose
h1–h6	<code><h1>Main Heading</h1></code>	Six heading levels by importance
div	<code><div>Grouped content</div></code>	Block-level container for CSS grouping
center	<code><center>Centred text</center></code>	Aligns text, graphics, tables to centre
p	<code><p>Paragraph text</p></code>	Paragraph of text
body	<code><body>Visible content</body></code>	All visible page content; one per document

Important Points

- Head metadata: title, style, base, link, meta — invisible to user
- Title tag ≠ h1: title = entire document title in browser bar; h1 = page heading
- Link tag: `<link rel="stylesheet" href="style.css">` loads external CSS
- Meta viewport: `<meta name="viewport" content="width=device-width, initial-scale=1.0">`
- Six heading levels: h1 (largest/most important) to h6 (smallest)
- `<div>` = container for grouping; `<center>` = centre alignment
- Base tag sets default URL for all relative links on the page
- Meta description helps search engines index the page content
- Only one `<title>` and one `<base>` tag allowed per document

For Exam

HTML tags format browser display. The head section contains metadata elements: title (browser title bar, one only), style (CSS in head), base (absolute URL), link

(external resources), and meta (keywords, description, viewport). The body tag holds all visible content. Six heading levels (h1–h6) structure content by importance. The div tag groups elements for CSS styling as a block-level container. The center tag aligns content to the centre of the page. Title tag content appears in the browser title bar but not on the page itself, whereas h1 appears as visible page content. Meta keywords and description help search engines index the page; meta viewport ensures responsive display on mobile devices.

Unit Recap — Unit 1

- HTML = Hypertext + Markup Language; HTML5 is the latest version
- CSS = presentation; JavaScript = interactivity
- Element syntax: `<tagname> Content </tagname>`; attributes in opening tag
- Document structure: DOCTYPE → html → head → title → body
- Document types: Transitional (flexible), Strict (enforced), Frameset (frames)
- Head elements: title, style, base, link, meta
- Body = visible content; h1–h6 headings; div = grouping; center = alignment

Unit 2: Managing List and Table

5.2.1 HTML Lists

Theory

HTML lists present information in a well-formed, semantic manner. Three types: **Unordered list** (`<ul>`) — items with no numerical order, shown as bullets; each item uses `<li>`. **Ordered list** (`<ol>`) — items numbered 1, 2, 3...; each item uses `<li>`. **Description list** (`<dl>`) — terms with descriptions like a dictionary; **dt** = term, **dd** = description (at least one dt followed by one dd per group). Lists help organise content the way a programme coordinator arranges festival events. All three list types use paired opening/closing tags.

List Type	Opening Tag	Item Tag	Description Tag	Display Style
Unordered	<code></code>	<code></code>	—	Bullet points
Ordered	<code></code>	<code></code>	—	Numbers 1, 2, 3...
Description	<code><dl></code>	—	<code><dt></code> + <code><dd></code>	Term + description pairs

Important Points

- **Unordered:** `<ul><li>item</li></ul>` — bullet points

- **Ordered:** `<ol><li>item</li></ol>` — numbered list
- **Description:** `<dl><dt>term</dt><dd>desc</dd></dl>`
- `<li>` must be inside ul, ol, or menu parent
- Lists introduce information semantically
- Changing order in unordered list does not change meaning
- Each element in unordered/ordered list declared inside parent tag
- Description list arranged like a dictionary with terms and meanings

For Exam

HTML provides three list types. Unordered lists (`<ul>`) group items without numerical order using bullet points; each item is an `<li>` element. Ordered lists (`<ol>`) number items sequentially starting 1, 2, 3. Description lists (`<dl>`) pair terms (`<dt>`) with descriptions (`<dd>`), similar to a dictionary. The `<li>` tag represents an individual list item within its parent list. All three list types come in paired opening/closing tags. Changing the order of items in an unordered list does not change the meaning, whereas order matters in ordered lists.

5.2.2 HTML Tables

Theory

HTML **tables** create rows and columns on a web page using `<table>`, `<tr>` (table row), `<td>` (standard data cell), and `<th>` (header cell). All tr tags are declared inside table. td/th are child elements of tr. Table size adjusts to content. **colspan** spans cells across columns; **rowspan** spans across rows. Tables are analogous to inserting a table in Microsoft Word using rows and columns.

Tag	Attribute Examples	Purpose
table	<code><table border="1" width="80%"></code>	Defines the table container
tr	<code><tr align="center"></code>	Defines a row inside table
td	<code><td colspan="2">Merged cell</td></code>	Standard data cell; left-aligned default
th	<code><th rowspan="2">Header</th></code>	Header cell; bold/centred by default
colspan	<code>colspan="2"</code>	Merges cell across 2 columns
rowspan	<code>rowspan="3"</code>	Merges cell across 3 rows

Important Points

- Structure: `<table>` → `<tr>` → `<td>` or `<th>`
- `<th>` = header cell (bold/centred by default)
- `<td>` = data cell (left-aligned by default)

- colspan and rowspan control cell spanning
- tr must be inside table tag
- Tables organise tabular data on web pages
- td and th can contain text, images, and other HTML content
- Table size automatically adjusts based on content size

Example table with colspan:

```
<table>
<tr><th>Name</th><th colspan="2">Contact</th></tr>
<tr><td>Ali</td><td>Email</td><td>Phone</td></tr>
</table>
```

For Exam

HTML tables are built with the table tag containing rows (`<tr>`), which contain data cells (`<td>`) or header cells (`<th>`). Header cells label columns or rows and appear bold/centred by default; data cells hold content and are left-aligned. The colspan attribute merges cells horizontally across columns; rowspan merges vertically across rows. All tr tags must be declared inside the table tag. Tables provide structured display of tabular information on web pages, similar to tables in Microsoft Word.

5.2.3 HTML Frames

Theory

An HTML **frame** defines a window loading another web page; uses **src** attribute for URL; empty tag (no closing). **Frameset** divides the browser window into frames using **rows** (horizontal) or **cols** (vertical) attributes. Frameset comes in pairs; can contain one or more frame tags and nest framesets for smaller divisions. Used for menu systems where one section reloads while menu stays fixed. Clicking a menu item reloads only the content frame, not the entire page. Percentage values in cols/rows (e.g. `cols="50%,50%"`) define proportional frame sizes.

Tag	Attribute Examples	Purpose
frameset	<code><frameset cols="50%,50%"></code>	Divides window into vertical frames
frameset	<code><frameset rows="30%,70%"></code>	Divides window into horizontal frames
frame	<code><frame src="page.html"></code>	Loads external page; empty/singular tag
src	<code>src="https://example.com"</code>	URL of the web page to load in frame

Important Points

- `<frame src="url">` — empty/singular tag

- `<frameset cols="50%,50%">` divides window vertically
- `<frameset rows="...">` divides horizontally
- Frameset can nest for complex layouts
- Frame tag used with frameset element
- Common use: fixed menu + changing content area
- src attribute defines address of web page in frame
- Nested framesets divide windows into smaller sections

For Exam

HTML frames display multiple web pages in one browser window. The frameset tag divides the window using rows or cols attributes and contains frame tags. Each frame tag loads an external page via the src attribute. Framesets can be nested for complex layouts. This technique is commonly used for menu systems where navigation remains static while content reloads. The frame tag is empty (no closing tag) and must be used inside a frameset element. Percentage values in cols/rows define proportional frame sizes.

```
<!DOCTYPE html>
<html>
<head><title>Lists Example</title></head>
<body>
<h1>Unordered List</h1>
<ul><li>Item 1</li><li>Item 2</li></ul>
<h1>Ordered List</h1>
<ol><li>First</li><li>Second</li></ol>
</body>
</html>
```

Unit Recap — Unit 2

- Three list types: unordered (ul), ordered (ol), description (dl/dt/dd)
- `<li>` represents a list item inside ul or ol
- Tables: table → tr → td/th
- colspan and rowspan span multiple columns/rows
- Frame tag loads external pages; frameset divides window
- rows = horizontal frames; cols = vertical frames

Unit 3: Presenting Multimedia

5.3.1–5.3.2 Image Tag and HTML Colors

Theory

Multimedia is an interactive mass communication medium (text, graphics, audio, video, animation). The **image tag** `<img>` embeds images; image is not inserted directly — browser loads from source. Required attributes: **src** (image URL) and **alt** (alternate text). Also specify **width** and **height** to prevent flicker. Empty tag — no closing tag. Image formats include JPEG, PNG, GIF, BMP, TIFF.

HTML colors three methods: (1) **Hex codes** — six-digit number with `#` (e.g. `#FF0000` = red; min `#000000`, max `#FFFFFF`); (2) **Color names** (e.g. `blue`, `white`); (3) **RGB values** — `rgb(255,0,0)`; values 0–255 per channel; black = `rgb(0,0,0)`, white = `rgb(255,255,255)`. Hex uses base-16 (0–9, A–F); letters not case-sensitive.

Tag/Method	Attribute / Syntax	Example	Purpose
img	<code>src</code>	<code>src="/images/photo.jpg"</code>	Image file URL (required)
img	<code>alt</code>	<code>alt="Baby Photo"</code>	Alternate text if image fails (required)
img	<code>width</code> , <code>height</code>	<code>width="200" height="185"</code>	Prevents page flicker while loading
Hex color	<code>#RRGGBB</code>	<code>#FF0000</code> = red; <code>#0000FF</code> = blue	Six-digit hex colour code
Color name	name	<code>background-color: blue</code>	Named colour in CSS/style
RGB	<code>rgb(R,G,B)</code>	<code>rgb(255,0,0)</code> = red	Red/Green/Blue 0–255 each

Colour	Hex Value	RGB Value
Black	<code>#000000</code>	<code>rgb(0,0,0)</code>
White	<code>#FFFFFF</code>	<code>rgb(255,255,255)</code>
Red	<code>#FF0000</code>	<code>rgb(255,0,0)</code>
Blue	<code>#0000FF</code>	<code>rgb(0,0,255)</code>
Cyan	<code>#00FFFF</code>	<code>rgb(0,255,255)</code>
Magenta	<code>#FF00FF</code>	<code>rgb(255,0,255)</code>

Important Points

- ``
- Image formats: JPEG, PNG, GIF, BMP, TIFF

- Hex: base-16; 6 digits after # ; not case-sensitive
- RGB: red, green, blue each 0–255
- style attribute adds colour to elements
- Multimedia elements: text, image, audio, video, animation
- src and alt are the two required attributes of img tag
- Minimum hex/RGB = black; maximum = white

For Exam

The img tag inserts images using src (URL) and alt (alternate text) attributes; width and height prevent layout flicker while the image loads. Colours are specified three ways: hexadecimal codes (#RRGGBB, six digits starting with #), HTML colour names (blue, red, white), or RGB values rgb(R,G,B) with each component 0–255. Hex black is #000000 and white is #FFFFFF; RGB black is rgb(0,0,0) and white is rgb(255,255,255). Hex uses base-16 (0–9, A–F). The img tag is empty — no closing tag required. Image formats include JPEG, PNG, GIF, BMP, and TIFF.

5.3.3–5.3.4 Marquee and Multimedia Tags

Theory

The **marquee tag** scrolls text or images horizontally or vertically: `<marquee> content </marquee>`. **Multimedia tags** embed audio/video: **audio** — plays audio files; uses src or `<source>`; **controls** adds play/pause/volume; formats MP3, WAV. **video** — displays video; formats MP4/MPEG-4, WebM; attributes: controls, muted, width, height. **source** — defines multiple media formats (empty tag). **embed** — container for external applications/plugins (empty tag). **object** — embeds multimedia or another HTML document (paired tag). Browser chooses supported format from source tag options.

Tag	Key Attributes	Example	Purpose
marquee	direction, behaviour	<code><marquee>Scrolling text</marquee></code>	Scrolls text/image horizontally or vertically
audio	controls, src	<code><audio controls src="song.mp3"></code>	Embeds audio; MP3, WAV formats
source	src, type	<code><source src="file.mp3" type="audio/mpeg"></code>	Multiple format fallback; empty tag
video	controls, muted, width, height	<code><video controls width="300"></code>	Embeds video; MP4, WebM formats
embed	type, src, width, height	<code><embed type="audio/mpeg" src="audio.mp3"></code>	External plugin content; empty tag

Tag	Key Attributes	Example	Purpose
object	width, height, data	<code><object data="video.swf"></code>	Embedded object/document; paired tag

Important Points

- Marquee: paired tag; scrolls text left-right or top-bottom
- Audio: `<audio controls><source src="file.mp3" type="audio/mpeg"></audio>`
- Video: `<video controls src="file.mp4"></video>`
- source tag enables multiple format fallbacks
- embed = external plugin content (no closing tag)
- object = embedded object with opening/closing tags
- controls attribute adds play, pause, and volume buttons
- muted attribute silences video on load

For Exam

The marquee tag creates scrolling text or images on a web page horizontally or vertically. Multimedia is embedded using audio tag (plays audio with controls and source), video tag (displays video with controls, muted, width, height), embed tag (external plugin content, empty tag), and object tag (embedded objects/documents, paired tag). The source tag specifies alternative media formats so browsers choose supported formats. Supported audio formats include MP3 and WAV; video formats include MP4/MPEG-4 and WebM. The controls attribute adds play, pause, and volume buttons to audio and video players.

```
<!DOCTYPE html>
<html>
<head><title>Multimedia</title></head>
<body>

<audio controls>
<source src="song.mp3" type="audio/mpeg">
</audio>
</body>
</html>
```

Unit Recap — Unit 3

- Multimedia = interactive digital medium (text, image, audio, video)

- `img`: `src` + `alt` required; empty tag
- Colours: hex (`#RRGGBB`), names, RGB `rgb(0–255)`
- Marquee scrolls text/images
- `audio`, `video`, `embed`, `object` = multimedia tags
- `source` defines multiple media formats

Unit 4: Linking in HTML

5.4.1–5.4.4 HTML Links, Anchor, Internal and External Linking

Theory

An HTML **hyperlink** connects one web page to another; has two ends — **anchor** (text marking start/end) and **direction**. Link starts at source anchor, ends at destination anchor. Created with **anchor tag** `<a href="URL">content</a>`. Default colours: **unvisited = blue**, **visited = purple**, **active = red** (all underlined). Links apply to text, images, and other elements. Hyperlinks can connect to HTML documents, images, video, audio, or elements within a document.

Internal linking — links within same page using **hash (#)** and **id**: `<a`

`href="#section1">Go</a>` ... `<a id="section1">Section</a>`. Path may be absolute or relative.

External linking — connects to other web pages via href URL; no hash symbol needed. Example: `<a href="https://example.com">Visit</a>`. Relative paths like

`../tutorial.php` navigate within site structure.

Tag/Concept	Attribute / Syntax	Example	Purpose
a (anchor)	<code>href</code>	<code>Shop</code>	Creates hyperlink to URL
a (internal)	<code>href="#id"</code>	<code>Lesson 1</code>	Scrolls to section on same page
a (external)	<code>href="url"</code>	<code>CSS</code>	Links to external/relative page
id	<code>id="name"</code>	<code>Introduction</code>	Target anchor for internal links
linked image	<code>a + img</code>	<code></code>	Image acts as clickable link

Link State	Default Colour	Description
Unvisited	Blue (underlined)	Link not yet clicked

Link State	Default Colour	Description
Visited	Purple (underlined)	Link previously clicked
Active	Red (underlined)	Link currently being clicked

Important Points

- **Anchor tag:** `<a href="url">text</a>` — **paired tag**
- **href attribute** = **destination URL**
- Internal: `#id` scrolls to section on same page
- External: full or relative URL to other pages
- Default link colours: blue (unvisited), purple (visited), red (active)
- Images as links: `<a href="url"></a>`
- Source anchor = where link starts; destination anchor = where it ends
- Absolute path = full original URL; relative path = path within site

For Exam

HTML hyperlinks connect pages using the anchor tag with href attribute. A link has source anchor (where it starts) and destination anchor (where it ends). Default colours: blue (unvisited), purple (visited), red (active) — all underlined. Internal links use hash (#) with id attributes to scroll to sections on the same page. External links use href with absolute URLs (full web address) or relative URLs (path within site). Hyperlinks can be applied to text, images, and other HTML elements. Clicking a linked image redirects to the href URL.

5.4.5 Creating a Website

Theory

Creating a website requires **HTML** (structure/content) and **CSS** (styling/presentation). Prerequisites: web hosting service and unique domain name (e.g. www.example.org). **Eleven steps:** (1) Study HTML basics; (2) Learn document structure; (3) Save as .html/.htm (UTF-8 encoding); (4) Learn CSS selectors (e.g. `p { font-size: 18px; }`); (5) Combine CSS documents in order; (6) Install **Bootstrap** (open-source HTML/CSS framework); (7) Pick a design/template; (8) Customize with HTML and CSS; (9) Add content and images; (10) Fine-tune colours and fonts; (11) Create additional pages (about, contact, portfolio, products, team, policies) and link to homepage. Sublime Text editor provides syntax colour highlighting for easier coding.

Important Points

- HTML = structure; CSS = presentation; Bootstrap = responsive framework
- Save HTML with .html extension; UTF-8 encoding preferred

- CSS class selector: `.classname { property: value; }`
- Bootstrap provides pre-built templates and grid system
- Additional pages: about, contact, portfolio, services, policies
- Upload .html files to web server main directory
- CSS tag selector: `p { font-size: 18px; }` styles all paragraph tags
- Bootstrap viewport meta tag: `width=device-width, initial-scale=1`
- Copy .html file to web server directory to publish online

Eleven Website Creation Steps (Detailed):

Step	Action	Detail
1	Study HTML basics	Recall tags: <code><i></code> , <code><p></code> , <code></code> , headings from previous units
2	Learn document structure	DOCTYPE → html → head → title → body
3	Save HTML document	Notepad/TextEdit; save as index.html; UTF-8 encoding
4	Learn CSS selectors	Tag selector <code>p{}</code> and class selector <code>.name{}</code>
5	Combine CSS documents	Arrange CSS in order matching HTML structure
6	Install Bootstrap	Open-source framework; download from getbootstrap.com
7	Pick a design	Choose Bootstrap template; copy to web server directory
8	Customize with HTML/CSS	Edit homepage head section, meta viewport, link CSS
9	Add content and images	Use header, footer, section tags; assign CSS classes
10	Fine-tune colours/fonts	Hex values e.g. <code>#FF0000</code> for red text via style attribute
11	Create additional pages	About, contact, portfolio, products, team, policies — link all to homepage

For Exam

Website creation combines HTML for structure and CSS for styling. Steps include learning HTML/CSS, saving documents as .html files with UTF-8 encoding, selecting CSS selectors (tag and class), installing Bootstrap framework, choosing a template, customizing design, adding content/images, fine-tuning colours/fonts, and creating linked additional pages. Prerequisites include web hosting (space on web server) and

a domain name (unique website identifier). Bootstrap is open-source software providing templates and responsive design tools. Sublime Text editor provides syntax highlighting for easier HTML coding.

```
<!DOCTYPE html>
<html>
<head><title>Links</title></head>
<body>
<a href="https://www.example.com">External Link</a>
<a href="#top">Internal Link</a>
<a id="top">Section Top</a>
</body>
</html>
```

Unit Recap — Unit 4

- Hyperlink connects pages; anchor tag + href attribute
- Link colours: blue (unvisited), purple (visited), red (active)
- Internal links: #id on same page
- External links: URL to other pages
- Website needs hosting + domain name
- Bootstrap + HTML + CSS for website creation

HTML Quick Reference — Key Attributes Summary

Tag	Key Attributes	Example Usage
<html>	lang	<html lang="en"> — root element
<head>	—	Contains metadata; not visible on page
<title>	—	<title>Page Title</title> — browser tab
<meta>	name, content	<meta name="description" content="About page">
<link>	rel, href, type	<link rel="stylesheet" href="style.css">
<body>	bgcolor, background	Contains all visible page content
<h1> — <h6>	align	Heading
<p>	align	Centred paragraph
<div>	id, class, style	<div id="main" class="box"> — grouping
/	type, start	<ol start="5"> — ordered from 5

Tag	Key Attributes	Example Usage
<code></code>	value	<code><li value="3">Third item</code>
<code><table></code>	border, width, cellpadding	<code><table border="1" width="100%"></code>
<code><tr></code>	align, valign	Row inside table
<code><td></code>	colspan, rowspan, align	<code><td colspan="2">Merged</td></code>
<code><th></code>	colspan, rowspan, scope	Header cell; bold by default
<code></code>	src, alt, width, height	<code></code>
<code><a></code>	href, target, id	<code></code>
<code><audio></code>	controls, src, autoplay	<code><audio controls src="song.mp3"></code>
<code><video></code>	controls, width, height, muted	<code><video controls width="300"></code>
<code><source></code>	src, type	<code><source src="file.mp3" type="audio/mpeg"></code>
<code><marquee></code>	direction, behaviour	<code><marquee direction="left">Text</marquee></code>
<code><frameset></code>	rows, cols	<code><frameset cols="50%,50%"></code>
<code><frame></code>	src, name, noresize	<code><frame src="page.html"></code> — empty tag
<code><embed></code>	type, src, width, height	External plugin content
<code><object></code>	width, height, data	Embedded object; paired tag

Block 6: Trends in Information Technology

Course Code: B21CA01DC | **Coverage:** Block 6 — Units 1–4 | **Source:** SGOU SLM B21CA01DC

Unit 1: Applications of IT

6.1.1 Applications of IT

Theory

Information Technology (IT) includes everything associated with computer technology — networking, hardware, software, the Internet, and the people who use them. IT is a wide discipline dealing with **processing, management, transfer, storage, protection, and retrieval** of information. Key associated terms: hardware, software, database, communication, server, network, Internet, applications. Major IT applications: business, banking, education, medical field, science, and mobile application development. Most organisations maintain IT departments. IT enables online advertising through Facebook, WhatsApp, and Twitter — saving time and money compared to offline media.

IT Term	Definition
Hardware	Physical parts: keyboard, mouse, monitor, speakers
Software	Programs that tell the computer how to work
Database	Organised collection of information for easy access/update
Communication	Transfer of information between computers (email, WhatsApp)
Server	Computer providing data, files, programs to other computers
Network	Collection of connected computers sharing resources
Internet	Global network of millions of computers worldwide
Applications	Practical implementation of IT in various fields

Important Points

- IT = processing, management, transfer, storage, protection, retrieval of information
- Hardware = physical parts; Software = programs; Database = organised data
- Server provides data; Network connects computers; Internet = global network
- Applications: business, banking, education, medical, mobile apps
- IT enables online advertising via social media (saves time and money)
- Organisations have dedicated IT departments
- Communication transfers information via email, WhatsApp, etc.
- Database helps organise, access, manage, and update data easily

For Exam

Information Technology encompasses computer technology including networking, hardware, software, and the Internet. It deals with processing, management, transfer, storage, protection, and retrieval of information. Associated terms include hardware (physical parts), software (programs), database (organised data), server (provides data to other computers), network (connected computers sharing resources), and Internet (global network). Major applications span business, banking, education, healthcare, science, and mobile development. IT has become integral to modern organisational operations. Online advertising via social media saves time and money compared to offline newspaper/radio/TV ads.

6.1.2 IT in Business

Theory

IT supports business through: **Organisational Communication** — email, Google Meet, virtual meetings worldwide. **Inventory Management** — software tracks stock

levels, monitors and reorders automatically. **E-commerce** — online buying/selling of goods and services via social media and shopping websites; saves time, labour, advertising costs. **CRM (Customer Relationship Management)** — tracks purchases, provides 24/7 customer service online. **Decision-Making** — decision support systems show real-time performance (capital, sales, trends) for promotion/expense decisions. **Internet Enabled Systems** — improve security; e.g. CCTV reduces theft and data loss. Employees can share work reports regardless of location.

Business IT Application	Tool / Method	Benefit
Communication	Email, Google Meet	Virtual meetings across locations
Inventory Management	Inventory software	Auto stock tracking and reordering
E-commerce	Facebook, WhatsApp, shopping sites	Saves time, labour, advertising cost
CRM	CRM systems	24/7 customer service; tracks purchases
Decision-Making	Decision support systems	Real-time performance monitoring
Security	CCTV, Internet Enabled Systems	Reduces theft and data loss

Important Points

- Communication: email, video conferencing for virtual meetings
- Inventory Management software tracks stock and auto-reorders
- E-commerce = online buying/selling; saves costs
- CRM improves customer experience 24/7
- Decision support systems enable real-time performance monitoring
- Internet Enabled Systems (CCTV) improve business security
- CRM tracks customer purchases and assists via website/social media
- E-commerce uses Facebook, WhatsApp, YouTube, and shopping websites

For Exam

IT plays a vital role in business through organisational communication (email, Google Meet, virtual meetings with staff/clients worldwide), inventory management software (tracks stock, auto-reorders raw materials), e-commerce (online buying/selling via Facebook, WhatsApp, shopping websites — saves time, labour, advertising costs), CRM systems (tracks customer purchases, provides 24/7 service after business hours), decision support systems (real-time monitoring of capital, sales, marketing

trends for promotion decisions), and internet-enabled security systems like CCTV (reduces theft and confidential data loss).

6.1.3 IT in Banking — Electronic Services

Theory

IT revolutionised banking enabling **Any Time, Any Where Banking**. Multiple electronic mechanisms connect bank branches and customers digitally. **ECS** handles bulk salary payments to many accounts. **MICR** uses magnetic ink for electronic cheque clearing with a **9-digit code** (3 city + 3 bank + 3 branch digits). **NEFT** is RBI-maintained nationwide transfer in batches (not instant). **RTGS** settles funds in real time — the fastest channel. **CBS** networks all branches for any-branch banking. **ATM** dispenses cash via magnetic strip cards. Tele banking provides 24-hour voice service; internet banking via website; mobile banking via installed app.

Important Points

- ECS = bulk electronic fund transfer between accounts
- MICR = 9-digit code for electronic cheque clearing
- NEFT = RBI nationwide one-to-one transfer (batch, not real-time)
- RTGS = real-time gross settlement; fastest banking channel
- CBS = all branches networked; any-branch banking
- ATM = cash dispensing; magnetic strip cards
- NEFT requires beneficiary addition before transfer
- RTGS has no waiting period — processed immediately

For Exam

IT in banking includes multiple electronic services enabling Any Time, Any Where Banking. ECS handles bulk salary/fund transfers (one-to-many). MICR electronically clears cheques using 9-digit magnetic ink codes (3 city + 3 bank + 3 branch). NEFT (RBI-maintained) transfers funds one-to-one between NEFT-enabled accounts in batches — add beneficiary first via Internet banking. RTGS settles in real time without delay — the fastest banking channel for high-value transfers. CBS networks all branches for any-branch operations via centralized data. ATMs dispense cash using debit/credit card magnetic strips. Tele banking (24-hour voice), Internet banking (website + username/password), and Mobile banking (installed app) extend round-the-clock access.

Service	Full Form	Timing / Settlement	Key Features
ECS	Electronic Clearing Services	Batch processing	Bulk transfer: one account to many or vice versa; used for salary payments
MICR	Magnetic Ink Character Recognition	Cheque clearing	9-digit code (3 city + 3 bank + 3 branch); magnetic ink at cheque bottom
NEFT	National Electronic Fund Transfer	Batch (not real-time)	RBI-maintained; one-to-one; add beneficiary first; all NEFT-enabled banks
RTGS	Real-Time Gross Settlement	Real-time (instant)	Fastest channel; no delay; high-value transfers; processed immediately
CBS	Core Banking Solutions	Real-time (centralised)	All branches interconnected; operate account from any branch; single platform
ATM	Automated Teller Machine	Instant	Cash dispensing via debit/credit card; magnetic strip; Any Time Banking
Tele Banking	—	24-hour voice service	Customer care phone; block cards, account info, report loss anytime
Internet Banking	—	24/7 online access	Bank website; username + password; fund transfer, FD info on laptop/PC
Mobile Banking	—	24/7 on mobile app	Requires installed app; transactions, fund transfer, account info on phone

6.1.4 IT in Education

Theory

IT transforms education through seven areas. **Access to Study Material** — e-books, guides, previous question papers online as supplements. **Continuous Learning** — teachers send assignments online; students submit without physical classroom attendance. **Sharing of Knowledge** — online forums and social media for distance-independent collaboration. **Audio and Visual Learning Aids** — voice clips, YouTube, classroom videos for better understanding and practical demos. **Distance Learning** — online courses for all ages; overseas university courses from home. **Proper Record Keeping** — secure digital student records replacing manual files. **Video**

Conferencing — virtual classes, seminars, meetings, training sessions from anywhere.

Education IT Area	Description	Example
Access to study material	E-books, guides, question papers online	SGOU SLM e-content
Continuous learning	Online assignments and submissions	Learning beyond classroom
Sharing of knowledge	Forums and social media collaboration	Distance-independent discussion
AV learning aids	Voice clips, YouTube, classroom videos	Practical demos of theory
Distance learning	Online courses for all ages	Overseas courses from home
Record keeping	Secure digital student records	Eliminates manual file loss
Video conferencing	Virtual classes and seminars	Quality education from anywhere

Important Points

- E-books, guides, question papers accessible via Internet
- Online assignments enable learning anywhere, anytime
- AV aids: YouTube, voice clips, classroom videos
- Distance learning reaches populations unable to attend offline
- Digital records eliminate manual file loss
- Video conferencing delivers quality education remotely
- Online forums enable knowledge sharing across distances
- Overseas university courses accessible from home country

For Exam

IT in education enables access to electronic study materials (e-books, guides, previous question papers as supplements), continuous learning (teachers send assignments online; students submit without physical classroom attendance), knowledge sharing (online forums and social media for distance-independent collaboration), audio-visual learning aids (voice clips, YouTube, classroom videos for better understanding and practical demos), distance learning (online courses for all ages; overseas university courses from home), secure digital record keeping (eliminates manual file loss), and video conferencing (virtual classes, seminars, meetings, training sessions from anywhere).

6.1.5 IT in the Medical Field

Theory

Health Information Technology (HIT) applies IT in healthcare using electronic devices to store, share, and analyse health information. Improves care quality, diagnostic accuracy; reduces costs and errors. **MPM (Medical Practice Management)** software manages administrative and clinical practice — appointments, insurance verification. **EHR/EMR** digitally documents patient medical records; eliminates manual charting errors; enables sharing when patients move hospitals. **RPM (Remote Patient Monitoring)** uses medical sensors (blood sugar, pressure, heartbeat) sending data to doctor's device for remote treatment of chronic patients. HIT defines management of information between doctors and patients.

Medical IT System	Full Form	Function
HIT	Health Information Technology	IT applied to healthcare overall
MPM	Medical Practice Management	Admin/clinical practice; appointments, insurance
EHR	Electronic Health Records	Digital patient records; shareable across hospitals
EMR	Electronic Medical Records	Documentation and storage of patient medical info
RPM	Remote Patient Monitoring	Sensors send vitals to doctor's device remotely

Important Points

- HIT = IT applied to healthcare
- MPM automates appointments and insurance tasks
- EHR/EMR = digital patient records; shareable across hospitals
- RPM = sensors monitor vitals remotely
- HIT improves quality, accuracy; reduces costs and errors
- Telemedicine and online patient monitoring supported by IT
- EHR allows doctors to record on computer or mobile device
- RPM reduces costs and saves lives of chronic disease patients

For Exam

Health Information Technology applies IT in healthcare using electronic devices to store, share, and analyse health information by professionals. MPM software

manages administrative and clinical practice — automates appointments and insurance verification. EHR/EMR digitally stores patient medical records on computer/mobile, eliminating manual charting errors and enabling sharing when patients move between hospitals. RPM uses medical sensors (blood sugar, pressure, heartbeat) to send data to doctor's mobile/laptop for remote treatment. HIT improves healthcare quality and diagnostic accuracy while reducing costs, errors, and enabling telemedicine for chronic patients.

6.1.6 IT in Science

Theory

IT enables collection, storage, and processing of vast scientific data. Computations done in seconds (e.g. digits of Pi) that humans cannot complete in a lifetime. Detects particular data from millions of records. **Automated machines** (robotic arms) handle dangerous radioactive samples safely. Enables global collaboration — scientists share ideas worldwide. **Computer simulation** creates models and analyses results before physical laboratory experiments. Scientists can collaborate on projects with colleagues anywhere in the world without travel.

Important Points

- Fast computation and large-scale data processing
- Data detection from millions of records
- Robotic automation for dangerous sample handling
- Global scientific collaboration via networks
- Computer simulation before physical experiments
- IT stores and organises vast scientific information
- Robotic arms handle radioactive samples safely
- Simulation models tested before costly lab experiments

For Exam

IT in science enables fast computation (e.g. digits of Pi in seconds — work exceeding human lifetimes), detection of specific data from millions of records, automated machines (robotic arms handling dangerous radioactive samples safely), global collaboration among scientists sharing ideas worldwide, and computer simulation to create models and analyse results before costly physical laboratory experiments. IT stores and organises vast scientific information for different research purposes.

6.1.7 Mobile Applications Development

Theory

Mobile application development creates software for mobile platforms. Three types: **Native Applications** — built for specific platform (e.g. Android) using platform tools; uses full device potential; examples: WhatsApp, Facebook. **HTML5 Applications** — web technologies (HTML5, JavaScript, CSS); write-once-run-anywhere; minimal changes per OS. **Hybrid Applications** — native container embedding HTML5 app; combines both approaches; examples: Twitter, Gmail, Instagram. Developers consider existing apps before creating new ones for greater impact.

Important Points

- Native = platform-specific; full device capability
- HTML5 = web-based; cross-platform compatibility
- Hybrid = native shell + HTML5 inside
- IT supports all three development approaches
- Native preferred for device integration (smart home apps)
- Hybrid uses native container with embedded web app
- HTML5 requires minimal changes for each operating system
- Native apps integrate with smart home devices for personalised experiences

For Exam

Mobile application development creates software for mobile platforms. Three types exist: Native applications are built for specific platforms (e.g. Android) using platform-specific tools, utilizing full device potential (WhatsApp, Facebook). HTML5 applications use web technologies (HTML5, JavaScript, CSS) with write-once-run-anywhere approach requiring minimal changes per OS. Hybrid applications combine native containers with embedded HTML5 apps (Twitter, Gmail, Instagram). IT advancement supports all three approaches for entertainment, business, and education apps.

App Type	Platform	Technology	Advantages	Examples
Native	Specific (Android/iOS)	Platform SDK and languages	Full device potential; best performance	WhatsApp, Facebook
HTML5	Cross-platform	HTML5, JavaScript, CSS	Write-once-run-anywhere; minimal OS changes	Web-based mobile apps
Hybrid	Cross-platform	Native container + HTML5	Best of both; unique native elements	Twitter, Gmail, Instagram

Unit Recap — Unit 1

- IT = processing, management, transfer, storage, protection, retrieval of information
- Business: communication, inventory, e-commerce, CRM, decision support, CCTV
- Banking: ECS, MICR, NEFT, RTGS, CBS, ATM, tele/internet/mobile banking
- Education: e-materials, distance learning, AV aids, video conferencing
- Medical: HIT, MPM, EHR/EMR, RPM
- Science: computation, simulation, robotic automation, global collaboration
- Mobile apps: Native, HTML5, Hybrid

Unit 2: IT in Other Disciplines

6.2.1 Bioinformatics

Theory

Bioinformatics combines biology and information technology. Term coined by **Paulien Hogeweg** in 1979. Uses computational tools to manage, analyse, and manipulate large biological datasets. Three components: (1) database creation for biological data storage; (2) algorithms and statistics for relationship analysis; (3) analysis of DNA, RNA, protein sequences, structures, gene expression. **Human Genome Project** (1990) advanced bioinformatics — ~3 billion base pairs across 60,000–100,000 genes. Applications: pharmaceuticals, preventive medicine, gene therapy, drug discovery, crop improvement.

Bioinformatics Application	Description
Pharmaceuticals	Drug discovery; personalized medicine for infectious diseases
Preventive medicine	Combined with epidemiology; understand disease patterns
Gene therapy	Incorporate genetic materials into diseased/unhealthy cells
Drug discovery	Computational biology validates cost-effective drugs
Crop improvement	Drought-resistant and insect-resistant crop development

Important Points

- Bioinformatics = IT + biology
- Three components: databases, algorithms/statistics, data analysis
- Coined by Paulien Hogeweg (1979)
- Human Genome Project drove genome-stage development
- Applications: pharma, gene therapy, drug discovery, crop improvement

- Computers essential for processing billions of base pairs
- Gene therapy incorporates genetic materials into diseased cells
- Crop improvement: drought-resistant and insect-resistant varieties

Bioinformatics Component	Description
Database creation	Storage and management of large biological data sets
Algorithms & statistics	Determine relationships among members of large data sets
Data analysis tools	Analyse DNA, RNA, protein sequences, structures, gene expression

For Exam

Bioinformatics is the combination of biology and information technology, coined by Paulien Hogeweg in 1979. It encompasses database creation for biological data storage, algorithm/statistics development for relationship analysis, and analysis of biological data (DNA, RNA, protein sequences, structures, gene expression profiles). The Human Genome Project (1990) advanced bioinformatics — 23 chromosomes, 60,000–100,000 genes, ~3 billion base pairs. Applications include pharmaceuticals (personalized medicine), preventive medicine (epidemiology), gene therapy (incorporating genetic materials into diseased cells), drug discovery (computational biology), and crop improvement (drought/insect-resistant crops).

6.2.2 IT in Medicine

Theory

IT in medicine includes: **Hospital Information System (HIS)** — manages registration, billing, medical records, pharmacy, payroll across hospital departments. **Data Analysis** — statistical calculations on large medical research data in seconds. **Laboratory Computing** — accurate blood chemistry, microbiology results linked to patient IDs. **CADM (Computer-Assisted Decision Making)** — interactive system assisting doctors with clinical decisions using computer memory and processing. **Critical Care** — computerised vital sign monitoring in ICU. **Computer-Assisted Therapy** — dosage planning for toxic drugs. **Medical Imaging** — CT, MRI, ultrasound, 3D anatomy via dedicated hardware/software. **Computer-Aided Diagnostics (CAD)** — assists interpretation of X-ray, MRI images; detects breast cancer, lung tumors.

Medicine IT Application	Description
HIS	Hospital registration, billing, records, pharmacy, payroll
Data Analysis	Statistical calculations on large medical research datasets
Laboratory Computing	Blood chemistry, microbiology results linked to patient IDs

Medicine IT Application	Description
CADM	Interactive system assisting doctors with clinical decisions
Critical Care (ICU)	Computerised vital sign, medication, lab value monitoring
Computer-Assisted Therapy	Dosage planning for powerful/toxic drugs
Medical Imaging	CT, MRI, ultrasound, 3D anatomy generation
CAD / CADe	Assists medical image interpretation; detects cancer/tumors

Important Points

- HIS covers registration, billing, records, pharmacy, payroll
- CADM assists clinical decision-making tasks
- Medical imaging: CT, MRI, ultrasound, gamma cameras
- CAD/CADe highlights suspicious areas in medical images
- Laboratory computing delivers fast, accurate results
- ICU data management: vitals, medications, lab values
- CAD detects breast cancer on mammography, lung tumors on CT
- Closed-loop systems control drug infusion based on patient vitals

For Exam

IT in medicine includes Hospital Information Systems (registration, admission/discharge, billing, medical records, pharmacy, payroll), computer-assisted data analysis (statistical calculations on large research data in seconds), laboratory computing (blood chemistry, microbiology linked to patient IDs), CADM (interactive system assisting doctors with clinical decisions using computer memory/reliability), computerised ICU vital sign monitoring, computer-assisted therapy (dosage planning for toxic drugs), medical imaging (CT, MRI, ultrasound, 3D anatomy), and computer-aided diagnostics/CAD (assists X-ray/MRI interpretation; detects breast cancer on mammography, lung tumors on CT scans).

6.2.3 Computational Economics and Finance

Theory

Computational Economics is interdisciplinary (computer science, economics, management science) involving computational modelling of economic systems.

Computational Finance combines mathematical science, economic theory, statistics, and computer simulation for investment planning and risk management. **Economic**

Forecasting uses computer simulations to predict market changes. **Online**

Trading/E-commerce enables instant digital transactions. **Stock Markets** — online trading apps, stock prediction robots, faster transactions, real-time monitoring,

enhanced security, **Blockchain** for secure trading (SEBI exploring adoption). Blockchain eliminates third-party authorities using smart contracts.

Important Points

- Computational Economics = computational modelling of economic systems
- Computational Finance = simulation for investment and risk management
- Forecasting models predict market changes rapidly
- Online trading apps eliminate brokers and paperwork
- Blockchain enables secure, transparent stock transactions
- Real-time monitoring gives accurate prices instantly
- Automated robots analyse thousands of data points for stock prediction
- SEBI exploring blockchain for Indian stock market infrastructure

For Exam

Computational economics uses computer science, economics, and management science to computationally model economic systems and develop strategies. Computational finance combines mathematical science, economic theory, statistics, and simulation for investment planning and risk management. Economic forecasting uses computer models to predict market changes rapidly. IT revolutionised stock markets through online trading apps (no broker/paperwork), automated stock prediction robots, faster secure transactions, real-time price monitoring, enhanced security, and blockchain technology (SEBI exploring adoption for transparent, smart-contract-based trading).

Stock Market IT Application	Description
Online trading apps	Direct trading without broker; hassle-free, lower costs
Stock prediction robots	Analyse thousands of data points; execute at minimal prices
Faster transactions	Eliminates manual records, audits, verification delays
Real-time monitoring	Accurate trusted prices; investors react quickly to market changes
Enhanced security	Complete transaction records; trust and transparency
Blockchain	Secure trading via smart contracts; SEBI exploring adoption in India

6.2.4 Cognitive Science and AI Applications

Theory

Cognitive Science and AI integrates AI with human cognition study — language, reasoning, learning, vision, human-technology interaction, AI ethics. **Speech-to-text** converts spoken audio to text; **text-to-speech** converts text to speech (Google Assistant, Voice Aloud Reader). **Personaliser** (Microsoft Azure) uses reinforcement learning for personalised content recommendations. AI lifestyle applications: autonomous vehicles (Tesla, Toyota, Volvo), spam filters (Gmail 99.9%), facial recognition, recommendation systems (YouTube, e-commerce), GPS navigation (Uber route optimisation using CNN and GNN).

Cognitive AI Application	Description	Real-World Example
Speech-to-text	Converts spoken audio into text; real-time streaming	Conversation transcription APIs
Text-to-speech	Converts written text into spoken audio	Google Assistant, Voice Aloud Reader
Personaliser	Reinforcement learning for content recommendations	Microsoft Azure Personaliser Preview
Autonomous vehicles	Machine learning for driving and object detection	Toyota, Audi, Volvo, Tesla
Spam filters	AI filters unwanted email automatically	Gmail ~99.9% filtration capacity
Facial recognition	Detects and identifies faces for secure access	Phone unlock, high-security areas
Recommendation systems	Analyses user behaviour for customised content	YouTube, e-commerce platforms
GPS navigation	CNN + GNN for route detection and optimisation	Uber, logistics companies

Important Points

- Cognitive science AI = AI + human cognition study
- Speech-to-text and text-to-speech are key applications
- Personaliser delivers customised user experiences
- Autonomous vehicles use machine learning
- Spam filters and facial recognition use AI daily
- Recommendation systems analyse user behaviour
- Personaliser uses reinforcement learning (Microsoft Azure)
- GPS combines Convolutional Neural Network and Graph Neural Network

For Exam

Cognitive science integrates artificial intelligence with the study of human cognition covering language, reasoning, learning, vision, and human-technology interaction. Key applications include speech-to-text (converts spoken audio; real-time streaming transcription), text-to-speech (Google Assistant, Voice Aloud Reader), and Personaliser (Microsoft Azure reinforcement learning for customised content). AI lifestyle applications include autonomous vehicles (Toyota, Audi, Volvo, Tesla — machine learning for driving and object detection), spam filters (Gmail 99.9% filtration), facial recognition (phones, high-security areas), recommendation systems (YouTube, e-commerce analyse browsing history), and GPS navigation (Uber uses CNN and GNN for route optimisation).

6.2.5 Quantum Computing

Theory

Quantum computing harnesses quantum mechanics to solve problems too complex for classical computers. Uses **qubits** — quantum bits existing in multiple states simultaneously (unlike classical bits 0 or 1). Emerged in 1980s; companies: IBM, Google, Microsoft, Intel, Alibaba, Nokia. **Silq** is a high-level programming language for quantum computing developed at ETH Zürich. AI, machine learning, and big data search combine with quantum computing. Applications: complex problem solving, secure data encryption, detecting intruders via light signals. QIST (Quantum Information Science and Technology) combines quantum mechanics with information technology.

Concept	Classical Computing	Quantum Computing
Basic unit	Bit (0 or 1 only)	Qubit (multiple states simultaneously)
Programming language	C, Python, Java, etc.	Silq (ETH Zürich)
Emergence	1940s onwards	1980s
Key companies	Intel, AMD	IBM, Google, Microsoft, D-Wave
Applications	General computing	Complex problems, encryption, AI/ML integration

Important Points

- Qubits exist in multiple states simultaneously
- Classical bits = 0 or 1; qubits = multidimensional quantum state
- Silq = programming language for quantum computing
- Field emerged in 1980s
- Used with AI, ML, and big data search

- Companies: IBM, Google, Microsoft, Intel
- QIST = Quantum Information Science and Technology
- Quantum algorithms tackle problems faster than classical counterparts

For Exam

Quantum computing uses quantum mechanics and qubits (quantum bits that exist in multiple states simultaneously) to solve complex problems beyond classical computers. Unlike classical bits (0 or 1 only), qubits engage 0 and 1 multidimensionally. The field emerged in the 1980s when quantum algorithms were found more efficient than classical ones for certain problems. Silq is a high-level programming language for quantum computing developed at ETH Zürich with a strong static type system. It combines with AI, machine learning, big data search, and secure data encryption. Companies investing in quantum computing include IBM, Google, Microsoft, Intel, and Alibaba.

6.2.6 Nanotechnology

Theory

Nanotechnology is science/engineering at the **nanoscale** (1–100 nanometers; 1 nm = 10^{-9} meter). Involves seeing and controlling individual atoms and molecules. Tools: STM (Scanning Tunneling Microscope), AFM (Atomic Force Microscope) invented in early 1980s. IT applications: **data exploration and visualization** (real-time interactive environments), **computational geometry** (specialized algorithms for nanoscale geometries), **software usage** (open-source, object-oriented codes for simulation and analysis on multiple platforms). Medieval stained glass used alternate-sized gold/silver nanoparticles for colour.

Important Points

- Nanoscale = 1 to 100 nanometers
- STM and AFM enable nanoscale observation
- IT provides data visualization and exploration tools
- Computational geometry algorithms for nanoscale structures
- Open-source modular software essential for nanotech simulation
- Enhanced properties: strength, lightness, chemical reactivity
- Nanoscale materials used for centuries in stained glass windows
- Real-time data interactivity hides complexity from the user

For Exam

Nanotechnology operates at 1–100 nanometers (1 nm = 10^{-9} meter), studying and manipulating individual atoms and molecules. Tools like STM (Scanning Tunneling Microscope) and AFM (Atomic Force Microscope) invented in the early 1980s enabled the nanotechnology age. Medieval stained glass used gold/silver nanoparticles for colour centuries ago. IT contributes through data exploration and visualization (real-time interactive environments hiding data complexity), computational geometry algorithms for nanoscale structures, and open-source object-oriented modular software for simulation and analysis across platforms.

Unit Recap — Unit 2

- Bioinformatics = IT + biology; databases, algorithms, analysis
- Medicine: HIS, CADM, medical imaging, computer-aided diagnostics
- Computational economics/finance: modelling, forecasting, online trading
- Cognitive science AI: speech-to-text, text-to-speech, personaliser
- Quantum computing: qubits, Silq programming language
- Nanotechnology: nanoscale; IT for visualization and computational geometry

Unit 3: Computer Security and Malware

6.3.1 Computer Security and Threats

Theory

Computer Security protects data and computer systems from harm, theft, and unauthorized use. Four major **threats**: (1) **Theft of data** — stealing secret/sensitive data from organisations; (2) **Computer Vandalism** — extracting passwords, erasing hard disk via malicious programs; (3) **Fraud** — unauthorized fund/data transfers; (4) **Invasion of privacy** — illegal access to personal/financial/medical databases. Countermeasures: security software (firewall, antivirus, antispyware), strong passwords, data backup on separate media, encryption. Six security measures: firewall, human aspects, data backup, cryptography, antivirus, anti-spyware.

Security Measure	Description	Example
Firewall	Hardware/software filtering network traffic per security policy	Blocks prohibited inbound/outbound communications
Human Aspects	Protecting against unwanted user/intruder actions	Hardest task; training users against social engineering
Data Backup	File duplication for emergency recovery	WhatsApp backup; daily/weekly backup to separate server

Security Measure	Description	Example
Cryptography	Hides data by converting to coded representation	Encrypt "attack" as "@2&7?+"; decrypt with secret key
Antivirus	Identifies and removes viruses from memory/storage/email	Kaspersky, K7, Avira scan boot/OS modifications
Anti-Spyware	Finds and removes spyware secretly collecting data	Spybot Search and Destroy, Ad-aware

Important Points

- Computer security = protection from harm, theft, unauthorized use
- Four threats: data theft, vandalism, fraud, privacy invasion
- Firewall filters network traffic per security policy
- Cryptography: encrypt/decrypt with keys
- Data backup = file duplication for recovery
- Antivirus identifies and removes viruses
- Intruder enters system without permission (e.g. hacker)
- WhatsApp data backup saves data on servers for recovery

For Exam

Computer security is the protection of data and computer systems from harm, theft, and unauthorized use. Four major threats are theft of data (stealing sensitive organisational data), computer vandalism (extracting passwords or erasing hard disk), fraud (unauthorized fund transfers), and invasion of privacy (illegal access to personal/financial/medical databases). Six security measures are firewall (filters network traffic), human aspects (user/intruder protection), data backup (file duplication for recovery), cryptography (encryption/decryption with keys), antivirus (removes viruses), and anti-spyware (removes Spybot). Strong passwords and encryption protect against unauthorized access.

6.3.2 CIA Triad — Confidentiality, Integrity, Availability

Theory

Three elements of computer security: **Confidentiality** — concealment of information; only authorized persons access it (encrypted email with secret key; "Hello!" → encrypted as f7#E+). **Integrity** — trustworthiness of data; prevents unauthorized changes; sub-elements: data integrity (content) and authentication (originality); hashing converts data to unique text strings. **Availability** — ability to access data when needed by authorized users at the right time; server attacks cause

unavailability. Together these form the **CIA triad** — the foundation of cybersecurity policy.

CIA Element	Definition	Mechanism	Example
Confidentiality	Conceal information from unauthorized access	Encryption with secret key	Encrypted email readable only with key
Integrity	Data trustworthiness; prevent tampering	Hashing (# function)	Rs 500 payment protected from change to Rs 5000
Availability	Authorized access when needed	Server uptime, backup systems	Bank server accessible for loan repayment

Important Points

- Confidentiality = information secrecy; encryption with keys
- Integrity = data trustworthiness; prevents tampering
- Integrity sub-elements: data integrity + authentication
- Hashing converts data to unique string (# function)
- Availability = access when needed by authorized users
- CIA triad = Confidentiality, Integrity, Availability
- Example: Rs 500 payment changed to Rs 5000 = integrity loss
- Server attack preventing loan payment = availability loss

For Exam

The three elements of computer security are Confidentiality (concealing information so only authorized users access it, using encryption — e.g. "Hello!" encrypted as f7#E+ with secret key), Integrity (maintaining data trustworthiness and preventing unauthorized changes like payment amount tampering, using hashing for authentication), and Availability (ensuring authorized users can access data when needed — server attacks cause unavailability). Together they form the CIA triad fundamental to cybersecurity. Computer security aims to prevent disruption of information and services.

6.3.3 Security Terminology and Malware

Theory

Key terminologies: **Unauthorized access** — accessing server/data without owner knowledge. **Hacker** — exploits systems for money, cause, or fun. **Threat** — action/event compromising security. **Vulnerability** — flaw allowing attacker manipulation. **Attack** — assault violating security. **Social Engineering** —

psychological manipulation to steal data. **Malware** — collective name for viruses, ransomware, worms, Trojans, adware, spyware; delivered via links/files over Internet requiring user click. **Spybot/Ad-aware** remove spyware; **Kaspersky, K7, Avira** are antivirus examples.

Malware Type	Behaviour	Delivery Method	Key Damage
Virus	Inserts into standalone program; spreads on execution	Email attachment, file download	Corrupts files; steals passwords
Worm	Self-replicating standalone software	Network, email without user action	Spreads automatically computer-to-computer
Trojan	Disguised as legitimate software	Tricked user installation	Takes control of system after install
Ransomware	Encrypts hard drive files	Link or file over Internet	Demands payment for decryption key
Adware	Forces browser redirect to advertisements	Bundled with free downloads	Downloads further malicious software
Spyware	Secretly collects sensitive user data	Installed without user knowledge	Sends data to external third parties

Term	Definition
Unauthorized access	Accessing server/website/data without owner's knowledge
Hacker	Person exploiting computer system for money, cause, or fun
Threat	Action or event that might compromise computer security
Vulnerability	Flaw/weakness in system allowing attacker manipulation
Attack	Assault on computer system security violating policies
Social Engineering	Stealing data via psychological manipulation, not hacking

Important Points

- Six security measures: firewall, human aspects, backup, cryptography, antivirus, antispyware
- **Malware** = malicious software collective term
- **Virus** attaches to programs; **worm** self-replicates independently
- **Trojan** disguises as legitimate software
- **Ransomware** encrypts and demands payment
- **Spyware/Adware** steal data or force ad redirects
- Social engineering uses psychological tricks, not technical hacks

- Malware delivered as link or file; user must click to execute

For Exam

Computer security terminologies include unauthorized access (accessing data without owner knowledge), hacker (exploits systems for money or fun), threat (action compromising security), vulnerability (system flaw allowing manipulation), attack (assault on security), and social engineering (psychological manipulation to steal data). Malware is malicious software including viruses (attach to programs, spread on execution), worms (self-replicating standalone), Trojans (disguised as legitimate software), ransomware (encrypts files demanding payment), adware (forces browser ad redirects), and spyware (secret data collection). Malware is delivered via Internet links or files requiring user click to execute.

6.3.4 Types of Computer Viruses

Theory

A **computer virus** attaches to program files, stays inactive until execution, then spreads across networks corrupting files, stealing passwords, and damaging systems. Spreads via email attachments, downloads, social media links. General signs: pop-ups, blue screen, homepage changes, bulk emails, crashes, missing files, slow performance, unknown startup programs, password changes, disabled security software. Like coronavirus spreading person-to-person, a computer virus infects one machine then spreads across the network.

Important Points

- Virus spreads on execution of infected program/file
- Signs: pop-ups, crashes, slow performance, missing files
- Boot sector virus infects boot sector via infected USB
- Polymorphic virus changes code to evade antivirus
- Multipartite attacks boot sector AND executable files
- Macro virus written in Word/Excel macro language
- Virus stays dormant until infected program is executed
- Spreads via email attachments and social media scam links

General Virus Infection Signs:

Sign	Description
Repeated pop-up windows	Unexpected pop-ups appear frequently
Blue screen errors	System crashes with blue error screen
Homepage changes	Browser homepage altered without permission

Sign	Description
Bulk emails sent	Emails sent from your account to contacts without your action
Repeated system crashes	Computer freezes or crashes frequently
Missing files	Files disappear from folders
Slow performance	Unusual slowness in normal operations
Unknown startup programs	Unfamiliar programs launch at boot
Password changes	Passwords changed without your knowledge
Disabled security software	Antivirus or firewall turned off by virus

For Exam

Computer viruses attach to programs and spread on execution, causing file corruption, password theft, spam emails, and system damage via email attachments, Internet downloads, and social media links. Eight types exist: boot sector (boot-time infection via USB), web scripting (browser/webpage exploit), browser hijacker (modifies browser settings/redirects), resident (stays in memory after program closes), direct action (activates on .exe/.com execution), polymorphic (changes code each execution to evade antivirus), multipartite (attacks boot sector AND files — most dangerous), and macro virus (embedded in Word/Excel documents, runs on open). General signs include pop-ups, blue screen, homepage changes, bulk emails, crashes, missing files, slow performance, and disabled security software.

Virus Type	Characteristic	Target / Behaviour	Symptoms / Signs
Boot Sector	Copies code to boot sector of disk	Activates on startup; spreads via infected USB	System fails to boot; damage during startup from pen drive
Web Scripting	Exploits web browser/page code	Infects when visiting affected webpage	Browser malfunctions after visiting infected site
Browser Hijacker	Modifies browser settings without permission	Redirects to unwanted sites; replaces homepage/search	Unwanted ads; redirects to money-making scam sites
Resident	Stays in memory after program closes	Continues running after software terminated	Damage persists even after uninstalling infected program
Direct Action	Activates only when infected file executes	Attaches to .exe and .com files	Files become inaccessible; no performance delay
Polymorphic	Changes its code each execution	Evades traditional antivirus detection	Antivirus fails to detect; virus reappears after scan

Virus Type	Characteristic	Target / Behaviour	Symptoms / Signs
Multipartite	Attacks boot sector AND program files	Most dangerous; contaminates multiple times	Combined boot failure and file corruption simultaneously
Macro	Written in macro language (Word/Excel)	Embedded in documents; runs on document open	Infection spreads when opening Word/Excel attachments

6.3.5 Antivirus Software

Theory

Antivirus is software designed to prevent, search, detect, and remove malicious software. Essential because hackers release new viruses daily; unprotected computers get infected on Internet connection. Functions: scan files/directories/external devices, schedule automatic scans, remove detected malware, show computer health, parental controls, act as firewall, warn about dangerous websites, protect online accounts. Scans programs attempting to modify boot program, OS, or other protected programs.

Important Points

- Antivirus: prevent, search, detect, remove malware
- Scans files, directories, external devices
- Scheduled automatic scans
- Examples: Kaspersky, K7, Avira
- Anti-spyware (Spybot, Ad-aware) removes spyware
- New viruses daily require updated antivirus
- Warns about dangerous websites and links
- Can act as firewall and provide parental controls

For Exam

Antivirus software prevents, searches, detects, and removes malicious software from files and computer systems. It scans programs attempting to modify boot program, operating system, or other protected programs, schedules automatic scans, removes detected threats, monitors computer health, provides parental controls, acts as firewall, and warns about dangerous websites. Examples include Kaspersky, K7, and Avira. Anti-spyware programs like Spybot Search and Destroy and Ad-aware remove spyware secretly collecting user data. Updated antivirus is essential as hackers release new viruses daily; unprotected computers get infected immediately on Internet connection.

Antivirus Function	Description
Scan files/directories	Checks files, folders, external devices for malware
Scheduled automatic scans	Regular scans without manual initiation
Remove detected malware	Deletes or quarantines malicious programs
Monitor computer health	Shows overall security status of system
Parental controls	Restricts children's online activities
Warn dangerous websites	Alerts before visiting harmful links
Protect online accounts	Guards credentials and sensitive login data

Unit Recap — Unit 3

- Six security measures: firewall, human aspects, backup, cryptography, antivirus, antispyware
- Four threats: data theft, vandalism, fraud, privacy invasion
- CIA triad: Confidentiality, Integrity, Availability
- Malware types: virus, worm, Trojan, ransomware, adware, spyware
- Eight virus types: boot sector, web scripting, browser hijacker, resident, direct action, polymorphic, multipartite, macro
- Antivirus prevents, detects, and removes malware

Unit 4: Latest Trends in Computer Science

6.4.1 Virtual Reality (VR)

Theory

Virtual Reality (VR) creates a simulated virtual environment — an alternate world of reality using IT. User feels immersed in imaginary surroundings via VR headsets, gloves, and sensory equipment tracking head/eye movements. Setup: VR headset (A), battery pack (B), controller box (C), laptop (D), USB/HDMI cables (E, F). Three types: **Fully-Immersive** (most realistic; gaming/entertainment; HTC Vive, PlayStation VR), **Semi-Immersive** (partial virtual; education/training with projectors), **Non-Immersive** (common video games). Benefits: immersive learning, interactive atmosphere, realistic exploration, skill enhancement. Real Reality = actual world; Virtual Reality = imaginary world created with IT.

VR Type	Description	Use Case	Examples
Fully-Immersive	Most realistic; high-resolution; wide field of view	Gaming, entertainment	HTC Vive, PlayStation VR

VR Type	Description	Use Case	Examples
Semi-Immersive	Partially virtual; computer graphics + projectors	Education, training	Large projector systems
Non-Immersive	Least immersive; standard screen-based	Everyday video games	Average PC/console games

Important Points

- VR = simulated virtual environment; user feels immersed
- Equipment: VR headset, gloves, controller box, laptop
- Three types: Fully-Immersive, Semi-Immersive, Non-Immersive
- Examples: HTC Vive, PlayStation VR
- Applications: 3D movies, gaming, education, science
- Tracks head and eye movements for immersion
- Two lenses between screen adjust to eye movement
- Uses video, audio, and haptic (touch) simulations

For Exam

Virtual Reality creates a simulated virtual environment using computers and sensory equipment (VR headsets, gloves, controller box, laptop). The user experiences an alternate world with vision, hearing, and touch senses — e.g. experiencing Mars, swimming with sharks at home. Three types: Fully-Immersive (most realistic; gaming/entertainment; HTC Vive, PlayStation VR), Semi-Immersive (partial virtual; education/training with projectors), and Non-Immersive (standard video games). Setup: VR headset blocks real world; HDMI connects laptop; lenses adjust to eye movement. Benefits: immersive learning, interactive atmosphere, realistic exploration, skill enhancement, comfortable education.

6.4.2 Augmented Reality (AR)

Theory

Augmented Reality (AR) enhances real-world objects with digital content (images, sounds, text) creating interactive experiences. Unlike VR (fully virtual), AR exists in the real world — like using a magnifying glass or smartphone zoom to enhance viewing. Setup: camera captures real world → video merger combines with virtual objects from graphics system → augmented stream to user. Devices: **HUD** (Head Up Display — complex info in small area), Smart Glasses, Holographic Display, Smartphones. Four types: **Marker Based** (QR codes), **Superimposed/Markerless** (virtual objects on physical world), **Location Based** (GPS/compass; Google Maps), **Projection Based** (interactive hologram).

AR Type	Mechanism	Example
Marker Based	Visual markers (QR codes) connect digital to real world	QR code scanning
Superimposed / Markerless	Virtual objects superimposed on physical world	Move virtual objects in real space
Location Based	GPS, compass provide location-specific data	Google Maps navigation
Projection Based	Light projected on surfaces; senses human interaction	Interactive hologram

Important Points

- AR = real world enhanced with digital objects
- VR = fully virtual; AR = real world augmented
- Devices: HUD, Smart Glasses, Holographic Display, Smartphones
- Four types: Marker Based, Superimposed, Location Based, Projection Based
- Camera + graphics system + video merger = AR pipeline
- Benefits: education, healthcare, gaming, accuracy, remote sharing
- HUD shows complex info while interacting with real environment
- Holographic display needs no wearable device

For Exam

Augmented Reality generates interactive experiences by enhancing real-world objects with digital content (images, sounds, text). Unlike VR which creates fully virtual environments, AR superimposes digital elements on the real world — like a magnifying glass enhancing a document. Setup: camera captures real world → video merger combines with virtual objects from graphics system → augmented stream to user. Devices: HUD (complex info in small area), Smart Glasses, Holographic Display (3D space, no wearable), Smartphones. Four types: Marker Based (QR codes connect digital to real), Superimposed/Markerless (virtual objects on physical world), Location Based (GPS/compass; Google Maps), Projection Based (interactive hologram with light projection).

Feature	Virtual Reality (VR)	Augmented Reality (AR)
Environment	Fully simulated/virtual world	Real world enhanced with digital objects
User experience	Immersed in imaginary surroundings	Sees real world + overlaid digital content
Equipment	VR headset, gloves, controller box, laptop	HUD, Smart Glasses, Smartphone, Holographic display

Feature	Virtual Reality (VR)	Augmented Reality (AR)
Setup	Headset blocks real world; HDMI to laptop	Camera → video merger → augmented stream
Types	Fully-Immersive, Semi-Immersive, Non-Immersive	Marker Based, Superimposed, Location Based, Projection Based
Examples	HTC Vive, PlayStation VR, 3D gaming	QR codes, Google Maps, interactive hologram
Applications	Gaming, 3D movies, education, science	Education, healthcare, gaming, navigation
Key difference	Creates alternate reality	Augments existing reality

6.4.3 Artificial Intelligence (AI) and Automation

Theory

AI was coined by **John McCarthy in 1956** — “Science and Engineering of making intelligent machines.” Machines work and behave like humans using data and algorithms. **Automation** = production/delivery of goods/services with minimal human intervention. AI works by: perceive environment (data) → detect patterns → learn from patterns → repeat until confident predictions (Fig 6.4.4). **Three types:** **Narrow/Weak AI** (specific tasks — Alexa, face verification, Google Maps), **General/Strong AI** (any intellectual task like humans; not yet achieved), **Super AI** (exceeds human ability; hypothetical/science fiction). **Applications:** healthcare (Fitbit), automobiles (self-driving), surveillance, social media, entertainment, education.

AI Application Area	Example	How AI Is Used
Healthcare	Fitbit, iWatch	Early disease detection; doctor decision support
Automobiles	Tesla, Toyota, Volvo	Self-driving; object detection; route optimisation
Surveillance	Face recognition tools	Observation and security in public/industrial areas
Social Media	Facebook, Twitter, Google	Personalised feeds; hate speech detection
Entertainment	YouTube, Netflix	Recommendations based on viewing history
Education	AI monitoring systems	Track academic, psychological, physical student wellbeing

Important Points

- AI coined by John McCarthy (1956)
- **Three types:** **Narrow (Weak)**, **General (Strong)**, **Super AI**

- Working: perceive → detect patterns → learn → predict
- **Narrow AI** examples: Alexa, Google Maps, face verification
- **Automation** = minimal human intervention in production
- **Applications:** healthcare, automobiles, surveillance, social media, education
- Google AI eye doctor detects diabetic retinopathy from retina scans
- Twitter AI identifies hate speech and terrorist language

For Exam

Artificial Intelligence, coined by John McCarthy in 1956 ("Science and Engineering of making intelligent machines"), makes machines work like humans by processing data through algorithms adjusted by past experiences. Three types: Narrow AI (specific tasks — Alexa operates in limited predefined range, no self-awareness; Google Maps, face verification), General AI (any intellectual task like humans; high computation but cannot yet think/reason like humans), and Super AI (exceeds human ability; hypothetical as in science fiction). AI workflow: perceive environment (data) → detect patterns → learn from patterns → repeat until confident predictions. Automation delivers goods/services with minimal human intervention. Applications: healthcare (Fitbit early disease detection), self-driving cars, surveillance (face recognition), social media (Facebook feeds, Twitter hate speech detection), entertainment (streaming recommendations), education (student wellbeing monitoring).

AI Type	Also Called	Capability	Status	Examples
Artificial Narrow Intelligence	Weak AI	Specific tasks only; predefined range	Currently exists	Alexa, Google Maps, face verification, Siri
Artificial General Intelligence	Strong AI	Any intellectual task like a human	Not yet achieved	Hypothetical human-level reasoning machines
Artificial Super Intelligence	Super AI	Exceeds human cognitive ability	Hypothetical/sci-fi	Machines surpassing humans in all tasks

6.4.4 Smart Technology

Theory

Smart Technology — "SMART" = **Self-Monitoring, Analysis, and Reporting**

Technology using AI and Machine Learning for intellectual awareness. Three types:

IoT Devices — sensors, gateway, software, servers, UI bringing physical objects to life (smart cities, smart homes). **Smart Connected Devices** — controlled via remote/smartphone over Internet/Bluetooth/WiFi; customized but less adaptive (smart cameras, smart bulbs). **Smart Devices** — limited automation, no Internet required, programmable (smart coffee makers). Amazon Echo enables voice control of smart fan. Benefits: accessibility, sustainability, security, efficiency, saves time and money.

Smart Technology Type	Connectivity	Adaptability	Examples
IoT Devices	Internet; sensors + gateway + server	Most adaptive to user behaviour	Smart cities, smart homes
Smart Connected Devices	Internet / Bluetooth / WiFi	Customized but less adaptive	Smart cameras, smart bulbs
Smart Devices	No Internet required	Limited automation; programmable	Smart coffee makers, Amazon Echo

Important Points

- SMART = Self-Monitoring, Analysis, and Reporting Technology
- Uses AI and Machine Learning
- Three types: IoT devices, Smart Connected Devices, Smart Devices
- IoT most adaptive; Smart Connected via remote/app
- Smart Devices need no Internet
- Examples: Amazon Echo (voice control), smart TV, smartwatch
- Smart Connected uses Internet, Bluetooth, or WiFi
- IoT devices most capable of adapting to user behaviour

For Exam

Smart Technology (SMART = Self-Monitoring, Analysis, and Reporting Technology) uses AI and Machine Learning to give objects intellectual awareness so they perform smartly. Three types: IoT devices (network of sensors, gateway, software, servers, UI — most adaptive; smart cities/homes), Smart Connected Devices (remote/smartphone controlled via Internet/Bluetooth/WiFi — smart cameras/bulbs; customized but less adaptive), and Smart Devices (limited automation, no Internet, programmable — smart coffee makers). Amazon Echo enables voice control of appliances. Benefits: accessibility, sustainability, security, efficiency, saves time and money.

6.4.5 Internet of Things (IoT)

Theory

IoT = system of interconnected computing devices, machines, animals, or people with **UIDs (Unique Identifiers)** transferring data over a network without human-to-human or human-to-computer interaction. A “thing” can be a heartbeat monitor implant, biochip animal, automobile tyre sensor, etc. **Four components:** Sensors, Gateway, Server, Mobile/Laptop. Example: soil moisture sensor detects dry soil → gateway sends data → server triggers sprinkler → notification sent to phone. IoT enables man-to-machine and machine-to-machine interactions simultaneously.

Cloud computing (Amazon Web Services — AWS) delivers on-demand servers, storage, and databases over the Internet — a related trend in modern IT alongside IoT, VR, AR, and AI.

IoT Workflow (5 Steps):

1. **Collect Data** — Sensors/IoT devices gather environmental data (temperature, moisture, heartbeat, pressure)
2. **Transport Data** — Gateway receives sensor data and sends via WiFi, Bluetooth, mobile, or satellite network
3. **Process Data** — Server runs data processing software on the acquired information
4. **Trigger Action** — Server triggers automated response (e.g. turn on sprinkler when soil is dry)
5. **Notify User** — Processed information delivered to mobile/laptop via alarms, texts, or email notifications

IoT Component	Role	Examples
Sensors	Collect environmental/physical data	Temperature, moisture, heartbeat, tyre pressure sensors
Gateway	Transport data to server	WiFi, Bluetooth, mobile network, satellite
Server	Process data; trigger automated actions	Sprinkler activation when soil moisture drops
Mobile/Laptop	Display results; notify user	Alarms, SMS, email notifications on smartphone

IoT Application Area	Use Case
Smart Homes	Control lights, fan, AC, TV, washing machine via smartphone
Wearables	Smartwatch tracks heartbeat, oxygen, pressure; public safety routing
Healthcare	Remote patient monitoring; pharmaceutical inventory management
Agriculture	Soil moisture sensors trigger automated irrigation

IoT Application Area	Use Case
Smart Cities	Smart streetlights; occupancy-based AC temperature adjustment
Automobile	Tyre pressure sensors alert driver via network

Important Points

- IoT = interconnected things with UUIDs transferring data automatically
- Four components: Sensors, Gateway, Server, Mobile/Laptop
- Gateway options: WiFi, Bluetooth, mobile/satellite networks
- No human-to-human interaction required
- Applications: smart homes, wearables, healthcare, agriculture, smart cities
- Benefits: remote access, automation, time/money savings
- UUID = unique identification number for each IoT device
- Choose gateway based on power consumption, range, and bandwidth

For Exam

Internet of Things (IoT) is a system of interconnected devices with unique identifiers (UUIDs) that collect and transfer data over networks without human-to-human or human-to-computer interaction. Four components: Sensors (collect environmental data — temperature, heartbeat, moisture), Gateway (transports via WiFi/Bluetooth/mobile/satellite; choose based on power, range, bandwidth), Server (processes data and triggers automated actions), and Mobile/Laptop (delivers results via alarms/notifications). Workflow: (1) Collect → (2) Transport → (3) Process → (4) Trigger Action → (5) Notify User. Example: soil moisture sensor → gateway → server turns on sprinkler → report to phone. Applications: smart homes, wearables, healthcare, agriculture, smart cities. IoT enables automation, remote access, and improved service quality. **Cloud computing** (e.g. Amazon Web Services — AWS) delivers on-demand IT resources (servers, storage, databases) over the Internet without physical hardware — another key trend alongside VR, AR, AI, and IoT.

Unit Recap — Unit 4

- **VR** = fully virtual simulated environment; three immersion levels
- **AR** = real world enhanced with digital objects; four types
- **AI** (McCarthy 1956): Narrow, General, Super AI; perceive-learn-predict
- Smart Technology: IoT devices, Smart Connected, Smart Devices
- IoT: Sensors → Gateway → Server → Mobile; UUIDs identify things
- Cloud computing (AWS) = on-demand IT resources over Internet

Block 6 Quick Reference — Key Terms Summary

Topic	Key Terms / Facts
IT Definition	Processing, management, transfer, storage, protection, retrieval of information
Banking	ECS (bulk), MICR (9-digit cheque code), NEFT (batch), RTGS (real-time/fastest), CBS (any-branch), ATM
HIT / Medical IT	MPM (practice management), EHR/EMR (digital records), RPM (remote sensors)
Mobile Apps	Native (platform-specific), HTML5 (cross-platform), Hybrid (native + HTML5)
Bioinformatics	IT + biology; Paulien Hogeweg 1979; databases + algorithms + analysis
Security Measures	Firewall, human aspects, backup, cryptography, antivirus, anti-spyware
CIA Triad	Confidentiality (encryption), Integrity (hashing), Availability (access when needed)
Malware Types	Virus, Worm, Trojan, Ransomware, Adware, Spyware
8 Virus Types	Boot sector, Web scripting, Browser hijacker, Resident, Direct action, Polymorphic, Multipartite, Macro
VR Types	Fully-Immersive, Semi-Immersive, Non-Immersive
AR Types	Marker Based, Superimposed, Location Based, Projection Based
AI Types	Narrow/Weak (Alexa), General/Strong (not yet), Super (hypothetical)
AI Coined By	John McCarthy, 1956
Smart Technology	IoT devices, Smart Connected, Smart Devices
IoT Components	Sensors → Gateway → Server → Mobile/Laptop
IoT Workflow	Collect → Transport → Process → Trigger → Notify
Quantum Computing	Qubits; Silq language; emerged 1980s
Nanotechnology	1–100 nm; STM/AFM microscopes
Cloud Computing	AWS; on-demand servers/storage over Internet

Block 1 Revision Checklist

- ☐ Four functions of a computer: Input, Processing, Output, Storage
- ☐ Hardware, Software, Firmware (BIOS/POST), Liveware
- ☐ Block diagram: Input unit → CPU (CU, ALU, Memory) → Output unit
- ☐ Von-Neumann architecture and stored program concept (1945)
- ☐ Processor, Motherboard, Memory (RAM/ROM types), Registers, SMPS

- ☐ Binary (base-2), Decimal (base-10), Octal (base-8), Hex (base-16)
- ☐ Bit, Nibble (4 bits), Byte (8 bits), MSB, LSD
- ☐ BCD, Gray code, Excess-3, ASCII (A=65, a=97)
- ☐ Input devices: Keyboard, Mouse, Touchpad, Scanner, Barcode reader
- ☐ Output devices: Monitor (soft), Printer/Plotter (hard), Speakers (audio)
- ☐ Impact vs Non-impact printers; Laser vs Dot matrix
- ☐ Evolution: Abacus → Pascaline → Babbage → ABC → Transistors → IC → Microcomputer
- ☐ Analog vs Digital vs Hybrid computers
- ☐ Types by size: Laptop, PC, Workstation, Mainframe, Supercomputer, Handheld, Wearable

Block 2 Revision Checklist

- ☐ Memory structure: cells (1 bit), addresses, memory words
- ☐ Units: KB, MB, GB, TB, PB ($\times 1024$ each)
- ☐ Read/Write operations; access time
- ☐ Primary memory: RAM (volatile), ROM (non-volatile)
- ☐ SRAM vs DRAM comparison table
- ☐ Cache: L1/L2/L3, hit/miss/penalty, hit ratio formula
- ☐ Memory hierarchy: Registers → Cache → Main → Secondary → Offline
- ☐ Instruction set and ISA; opcode + address fields
- ☐ Instruction cycle: Fetch → Decode → Operand Fetch → Execute
- ☐ RISC vs CISC comparison (simple/fast vs complex/compact)
- ☐ CPU registers: MAR, MDR, PC, IR, Accumulator, GPR, Flags
- ☐ Virtual memory: virtual vs physical/logical addresses
- ☐ Secondary storage characteristics: non-volatile, large, cheap
- ☐ Magnetic tape vs disk; HDD specs (rpm, capacity)
- ☐ Optical media: CD, DVD, Blu-ray capacities
- ☐ USB, pen drive, external HDD, memory stick, SSD
- ☐ Number conversions: decimal ↔ binary, binary ↔ hex, decimal ↔ octal
- ☐ Digital codes comparison: BCD, Gray, Excess-3, ASCII

- ☐ Memory hierarchy access times (registers ns → tape seconds)
- ☐ Instruction cycle phases with register roles (PC, IR, MAR, MDR)
- ☐ Secondary storage capacity table (CD 700MB → Blu-ray 50GB → SSD TB)

Block 3 Revision Checklist

- ☐ Define software; distinguish instruction, program, hardware, software
- ☐ Classify software: System vs Application with comparison table
- ☐ Distinguish static vs dynamic websites with examples
- ☐ Explain website access flow (browser → Internet → web server → display)
- ☐ List OS functions with descriptions (process, memory, file, device management)
- ☐ Explain device drivers and language processors as system software
- ☐ Describe files (name, extension, path, metadata) and folders (directories, subfolders)
- ☐ List boot process steps in order (power on through ready state)
- ☐ Explain BIOS role and boot device priority
- ☐ Define booting; compare warm (soft reboot) vs cold (hard booting)
- ☐ Explain POST and what hardware it checks
- ☐ Describe three software layers: Presentation, Application, Data
- ☐ Compare one-tier, two-tier, three-tier architecture
- ☐ Define OS; explain DOS (command-based, single-user, single-task)
- ☐ Explain client-server model with restaurant analogy
- ☐ List and describe 7 OS types: Batch, Time-sharing, Multiprocessing, Real-time, Distributed, Network, Mobile
- ☐ Compare Linux vs Windows (open source, cost, case sensitivity, security)
- ☐ Describe Android (Linux kernel) and iOS (Apple)
- ☐ Define GUI and its advantages over command interface
- ☐ Classify programming languages: High level vs Low level (Assembly, Machine)
- ☐ Compare Compiler vs Interpreter vs Assembler
- ☐ Define database and DBMS with advantages
- ☐ List utility software examples (antivirus, backup, disk tools)

- ☐ Describe application software types: word processor, spreadsheet, presentation, LaTeX
- ☐ Define computer virus, entry methods, types, and protection measures

Block 4 Revision Checklist

- ☐ Define computer network; explain communication model (sender, medium, receiver)
- ☐ Distinguish guided vs unguided transmission media
- ☐ List network advantages: file sharing, hardware sharing, application sharing, messaging
- ☐ Explain basic network working with switch and shared printer
- ☐ Compare LAN, MAN, WAN with examples
- ☐ Compare Hub (broadcast) vs Switch (intelligent forward) vs Router (connects networks)
- ☐ Explain Repeater (signal boost) and Bridge (segment division)
- ☐ Draw/describe ASCII for all six topologies (Bus, Ring, Star, Mesh, Tree, Hybrid)
- ☐ Explain ring topology token system
- ☐ Compare Peer-to-Peer vs Client/Server NOS
- ☐ Trace Internet history: ARPA, ARPANET, TCP, NSFNET, HTTP, Mosaic
- ☐ Define Internet, WWW, web page, website, URL, browser, search engine
- ☐ Explain URL parts: protocol, domain, extension, page
- ☐ Describe search engine working with web crawler/spider
- ☐ List Internet search tips: keywords, quotes, Boolean, Ctrl+F
- ☐ Define ISP and its services (access, domain, hosting, Usenet)
- ☐ Compare connection types with speed/cost table: Dial-up, Cable, WLL, Broadband, Leased Line
- ☐ Explain Wi-Fi vs WiMAX; DSL symmetric vs asymmetric; fibre vs satellite
- ☐ Describe email components: UA, MTA, Mailbox, Spool file
- ☐ Compare email protocols: SMTP (send), POP3 (local receive), IMAP (server sync)
- ☐ Explain DSL splitter and symmetric vs asymmetric DSL
- ☐ Describe dial-up circuit switching and modem analog conversion

- ☐ Compare SMTP vs POP3 vs IMAP (send vs local receive vs server sync)
- ☐ Explain email address format username@domainname
- ☐ List email system services: Composition, Transfer, Reporting, Displaying, Disposition
- ☐ Describe WLL components: PSTN, Switch Function, WANU, WASU
- ☐ Compare fibre optics structure: jacket, cladding, core
- ☐ Explain CC vs BCC; email filtering, attach, and forward
- ☐ List web application benefits and characteristics (mobile, social, analytics)
- ☐ Name nine types of web pages

Block 5 Revision Checklist

- ☐ Define HTML, Hypertext, Markup Language; list versions to HTML5
- ☐ Draw HTML structure: DOCTYPE, html, head, title, body
- ☐ Compare Transitional, Strict, Frameset document types
- ☐ State five HTML coding rules and element syntax with attributes
- ☐ List head elements: title, style, base, link, meta — with purpose
- ☐ Differentiate title tag vs h1; explain div and center tags
- ☐ Create ul, ol, dl lists with li, dt, dd tags
- ☐ Build table with tr, td, th; explain colspan/rowspan
- ☐ Explain frameset/frame with cols/rows attributes
- ☐ Use img (src, alt), hex/RGB colours, marquee, audio/video tags
- ☐ Write anchor links: internal (#id) and external (href URL)
- ☐ List 11 website creation steps including Bootstrap
- ☐ Know HTML global attributes: id, class, style, title, lang
- ☐ Reference HTML Quick Reference table for tag attributes

Block 6 Revision Checklist

- ☐ Define IT and list associated terms (hardware, software, database, server, network)
- ☐ Explain IT in business: communication, inventory, e-commerce, CRM, decision support, CCTV

- ☐ Describe banking IT table: ECS, MICR, NEFT, RTGS, CBS, ATM, tele/internet/mobile banking
- ☐ List IT in education: study materials, continuous learning, distance learning, video conferencing
- ☐ Explain HIT, MPM, EHR/EMR, RPM in medical field
- ☐ Differentiate Native, HTML5, and Hybrid mobile applications
- ☐ Define bioinformatics (3 components) and IT in medicine (HIS, CADM, CAD)
- ☐ Explain computational economics, finance, forecasting, stock markets, blockchain
- ☐ Describe cognitive science AI, quantum computing (qubits, Silq), nanotechnology
- ☐ List six security measures and four computer security threats
- ☐ Explain CIA triad: Confidentiality, Integrity, Availability
- ☐ Define malware types and eight computer virus types (table)
- ☐ Describe antivirus functions; compare VR and AR types
- ☐ Explain AI types (Narrow, General, Super) and how AI works
- ☐ Define Smart Technology types and IoT components/workflow/applications
- ☐ Explain cloud computing (AWS) as on-demand IT resource delivery
- ☐ Use Block 6 Quick Reference table for last-minute revision

Quick Reference — Exam Essentials

Topic	Must-Know
Computer functions	Input, Processing, Output, Storage
Hardware/Software/Firmware/Liveware	Physical parts / Programs / ROM (BIOS) / Users
Von-Neumann	Stored program; CU, ALU, memory, I/O
Number systems	Binary, Decimal, Octal, Hex; BCD, Gray, Excess-3, ASCII
Memory hierarchy	Registers → Cache → RAM → Secondary
Instruction cycle	Fetch → Decode → Execute → Store
RISC vs CISC	Simple/fast vs Complex/compact
Boot process	POST → Boot loader → OS (warm vs cold boot)
Translators	Compiler, Interpreter, Assembler
LAN/MAN/WAN	Building / City / Worldwide

Topic	Must-Know
Topologies	Bus, Ring, Star, Mesh, Tree, Hybrid
Email	SMTP send; POP3/IMAP receive
HTML skeleton	<code><!DOCTYPE html><html><head><body></code>
Banking IT	ECS, MICR, NEFT, RTGS, CBS, ATM
Security (CIA)	Confidentiality, Integrity, Availability
Emerging tech	VR, AR, AI, IoT, Quantum

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Duaon mein Yaad Rakhna.

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